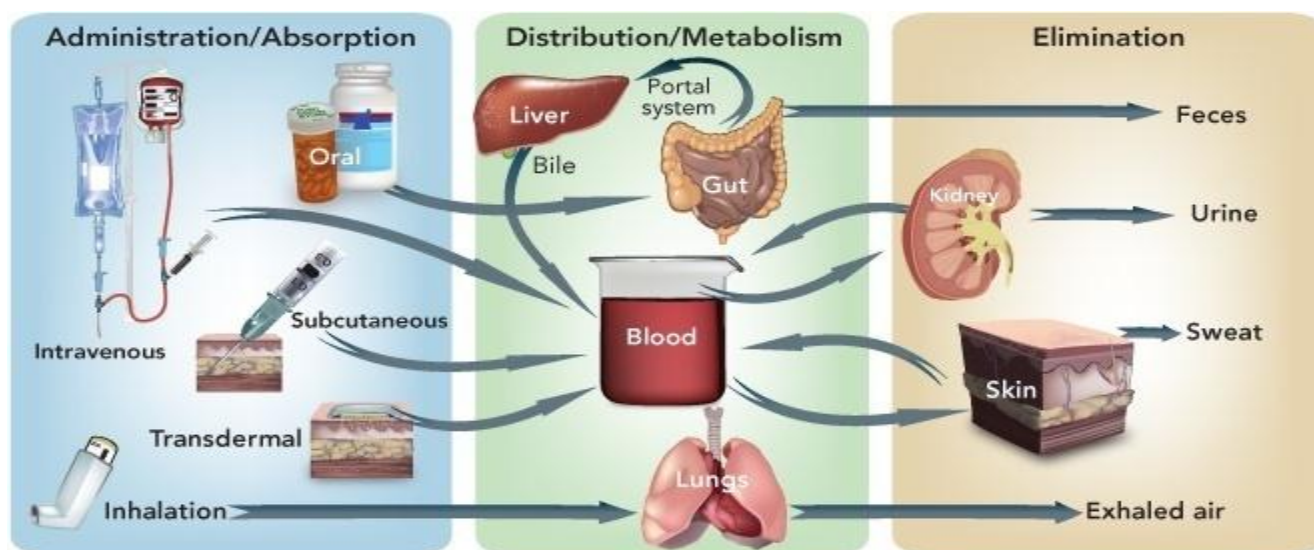




Biopharmaceutics and Pharmacokinetics

PT- 312

Theoretical notes

Prepared By

Staff Members of Pharmaceutical Technology Department

2026-2027

Faculty Vision and Mission

رؤية البرنامج

التميز العلمي والتطوير المستمر لخدمة المنظومة الصحية العلاجية والوصول لمكانة مرموقة عالميا في مجال الصيدلة الإكلينيكية.

رسالة البرنامج:

إعداد صيادلة مؤهلين بأحدث المفاهيم الصيدلية والطبية يمكنهم المساهمة في رفع كفاءة المنظومة العلاجية على المستوى المحلي والإقليمي من خلال التعامل مع الفريق الصحي في المستشفيات وتقديم الخدمات الصيدلية بمستوى مهاري محترف بالصيدليات العامة والخاصة وشركات الأدوية ومعامل الرقابة الدوائية وتحليل الأغذية بالإضافة إلى العمل في مجال الإعلام والتسويق الدوائي والمشاركة بفاعلية في البحث العلمي من خلال مراكز البحوث والجامعات لخدمة المجتمع.

Program Mission

Preparing qualified pharmacists with the latest pharmaceutical and medical concepts who can contribute to raising the efficiency of the treatment system at the local and regional levels through dealing with the health team in hospitals and providing pharmaceutical services at a professional skill level in public and private pharmacies, pharmaceutical companies, drug control laboratories and food analysis centers, in addition to working in the field of media and pharmaceutical marketing and actively participate in scientific research through research centers and universities to serve the community.

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Pharmacokinetics

Kinetic study (study of the rate process) of ADME (absorption, distribution, metabolism and elimination)

Biopharmaceutics

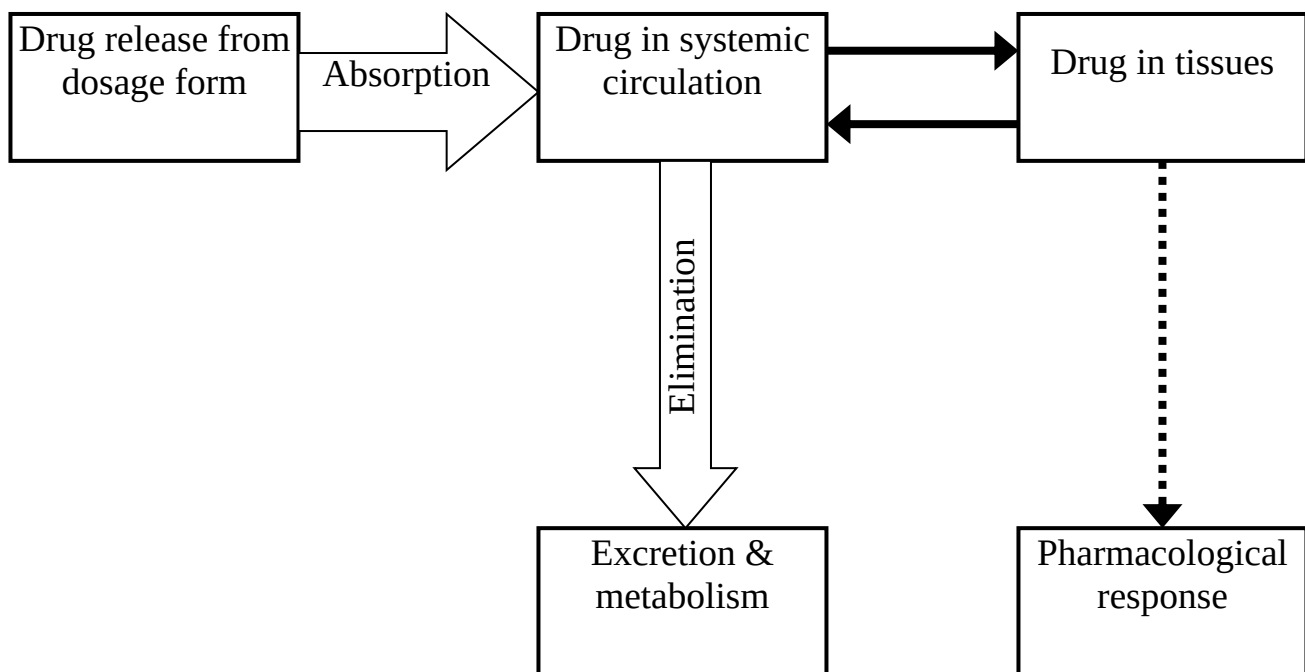
Science that study relationship between the physicochemical properties of the drug (active pharmaceutical ingredient API), dosage form and route of administration on the rate and extent of drug absorption

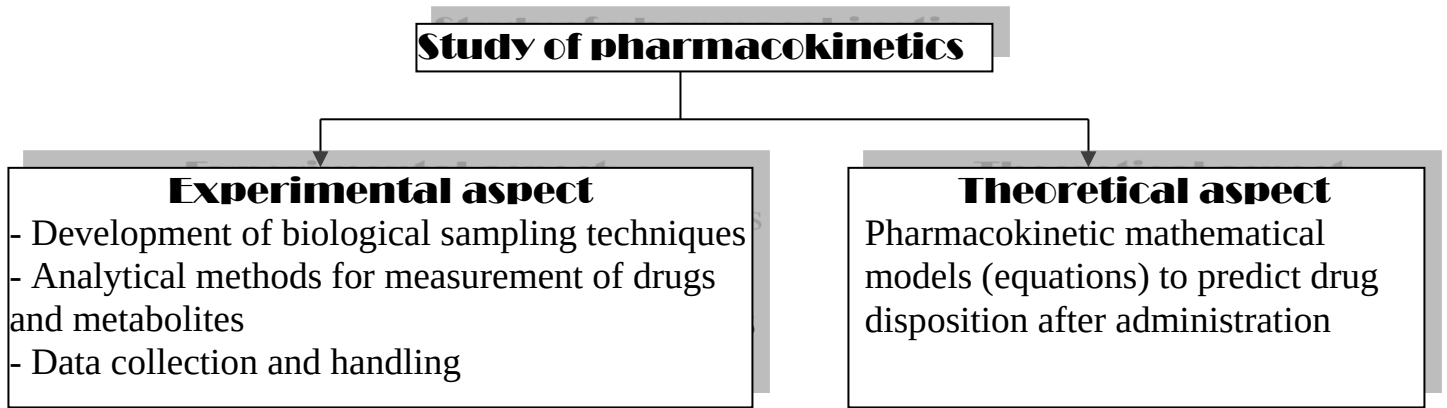
Biopharmaceutics study factors that affect

- 1) Stability of drug in the dosage form product
- 2) Drug release from the product
- 3) Rate of drug release from the product
- 4) Systemic absorption of drug

Bioavailability = physiological availability = biological availability:

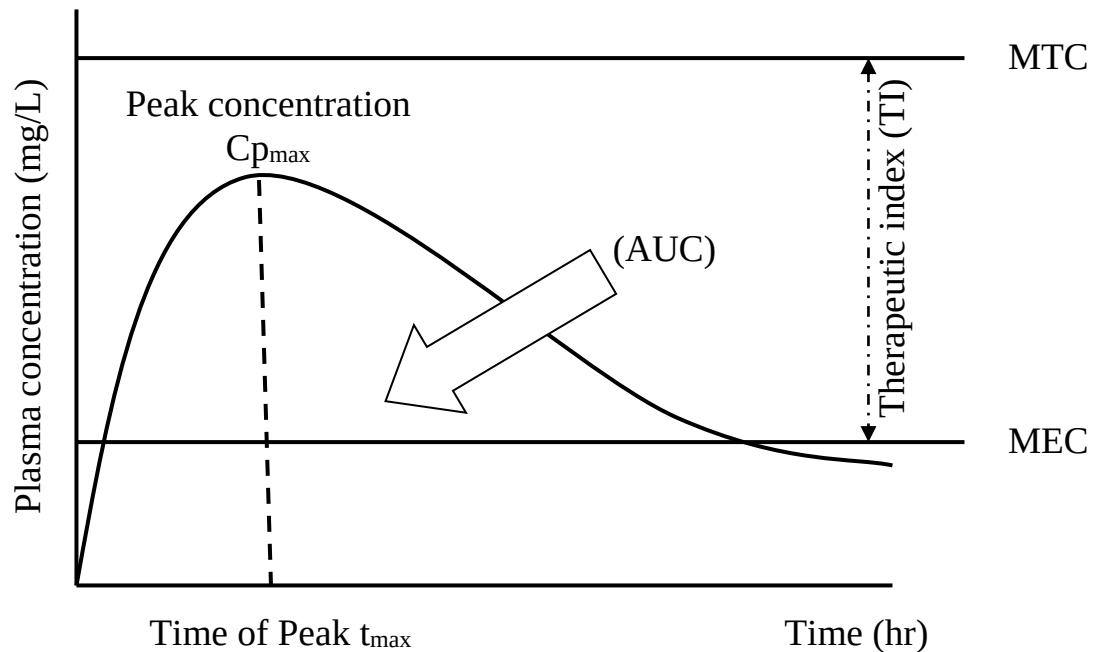
the rate and extent of drug absorption from dosage form and become available at the site of action. For drug products intended not to reach blood; their rate and extent will be measured by assessing effect of drug at site of action e.g. antacids.





Drug concentrations can be measured in plasma (most accurate and most direct method), urine, saliva or milk

Plasma concentration – time curve



$C_{p_{max}}$: maximum concentration of drug in blood; related to dose, K_a and K

t_{max} : time of the maximum drug concentration in blood; it is a measure of the rate of drug absorption

AUC: area under the plasma concentration – time curve; it is related to amount or extent of drug absorption

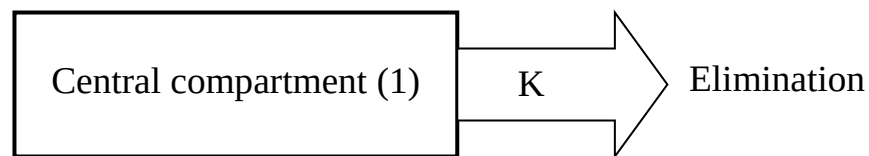
Pharmacokinetic models

- ▶ A mathematical model (equation) simulating the rate process of ADME and predict concentration of drug in the body at any time.

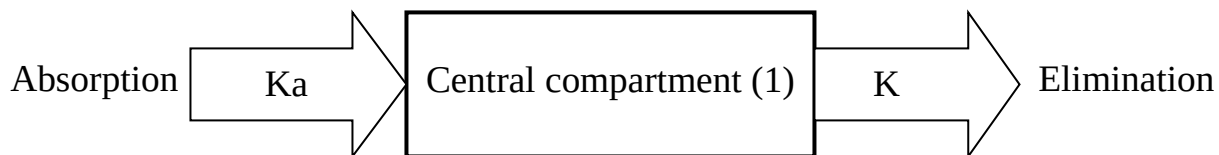
Applications of pharmacokinetic models

1. Used to estimate possible accumulation of drug inside the body
 2. Allow maintaining drug concentration within therapeutic range (TI) thus optimize the therapy
 3. Used to correlate concentration of drug with therapeutic response
- ▶ Simple models are compartmentally based models where group of organs or tissues are grouped to one or more compartments where drug move to or from central compartments (blood)
 - ▶ They are divided based on number of compartments into

1) One compartment open model



One compartment open model following IV injection



One compartment open model following first order

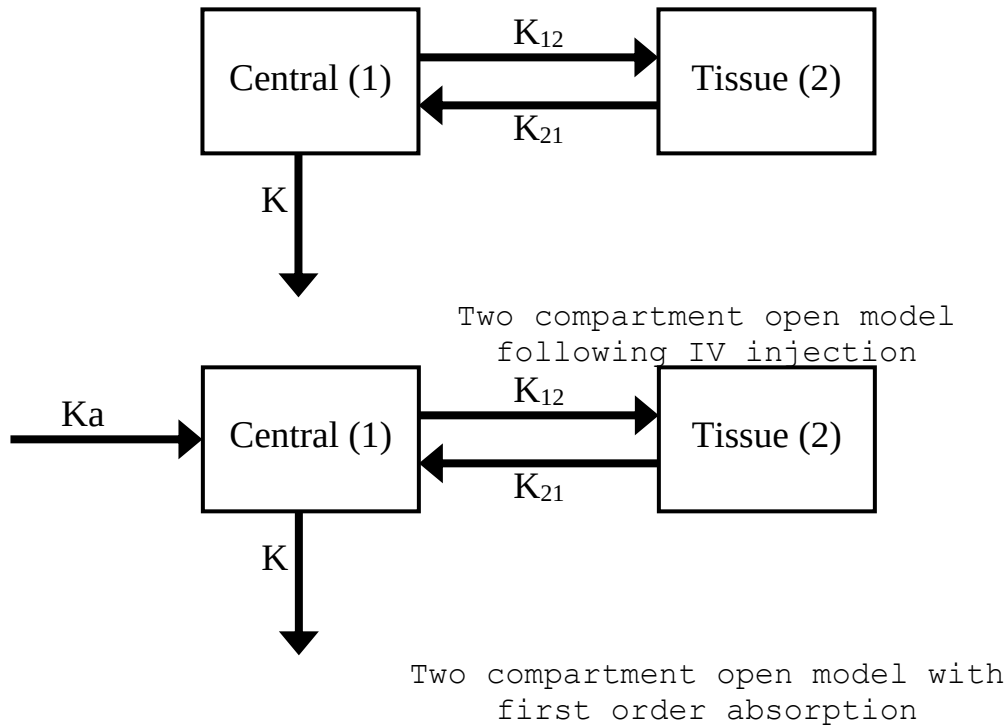
Assumptions

- a) Body is one central compartment (plasma and highly perfused organs) to which drug is added and eliminated
- b) Once drug is injected into blood; it undergo instantaneous equilibrium and distribution into the surrounding tissues
- c) Concentration of drug is proportional to dose (linear relationship)
- d) Elimination of drug is first order (rate of drug elimination depend on concentration of drug in blood)
- e) Changes in plasma level of the drug will result in proportional changes in tissue and urine drug levels

f) Amount of drug inside the body can not be measured directly but plasma drug level can be measured accurately

- Open means that the drug undergoes continuous elimination

2) Two compartment open model

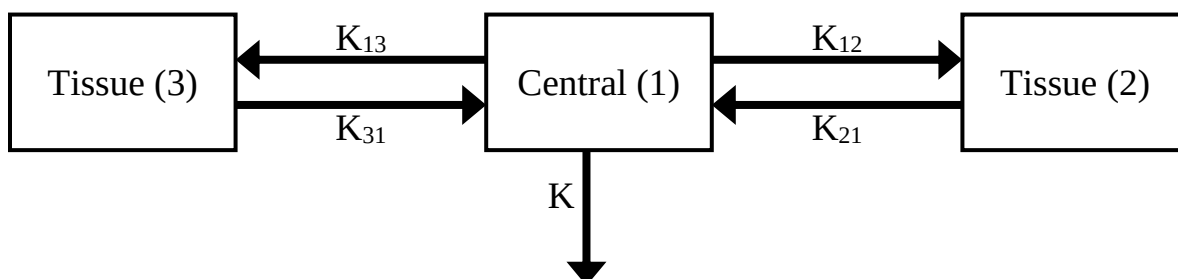


- ▶▶ Central compartment: blood with other highly perfused and highly equilibrating organs
- ▶▶ Peripheral (tissue) compartment: slowly perfused slowly distributing tissues
- ▶▶ Total amount of drug = drug in central compartment + amount in tissue compartment
- ▶▶ Elimination of drug from central compartment

Drawing of pharmacokinetic models functions to

- 1) Enable to write differential equations to describe concentration in each compartment
- 2) Give a representation of the rate process
- 3) Show different pharmacokinetic constants

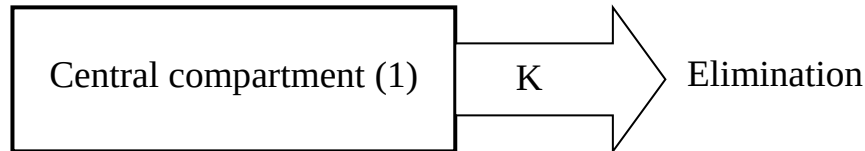
3) Three compartment open model



Three compartment open model following IV injection

Peripheral compartments 2 and 3 are not directly connected but each is connected to central compartment 1 from which drug elimination takes place

1) One compartment open model following IV injection



One compartment open model following IV injection

Since elimination of the drug follow first order kinetics

$$\frac{dD_B}{dt} = -KD_B$$

$$\log D_B = \log D_B^0 - \frac{Kt}{2.303}$$

or $DB = DB^0 e^{-Kt}$

$$\frac{dC_p}{dt} = -KC_p$$

$$\log C_p = \log C_p^0 - \frac{Kt}{2.303}$$

or $C_p = C_p^0 e^{-Kt}$

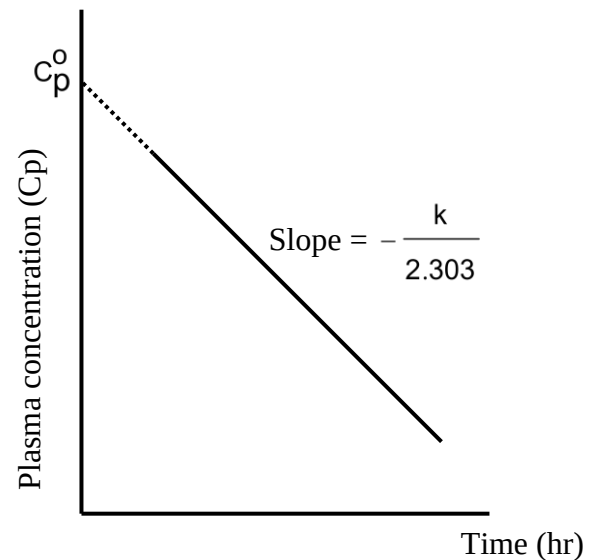
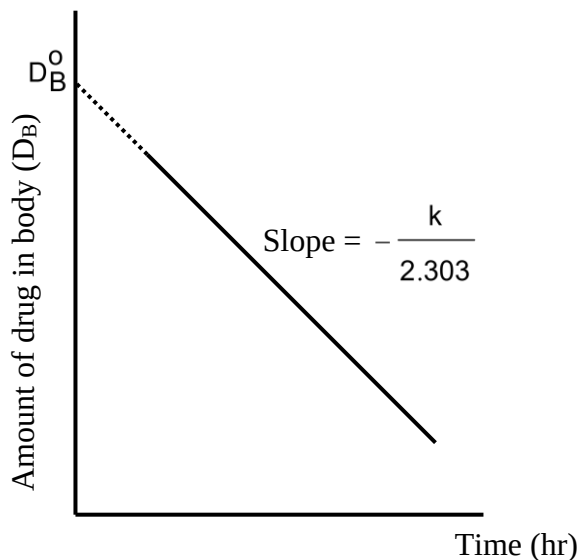
D_B = total amount of drug in body

D_B^0 = I V dose (initial amount of drug)

K: 1st order elimination rate constant

C_p = plasma concentration of the drug

C_p^0 = initial plasma concentration of the drug



Elimination half life

- ▶▶ Time required for 50% of the drug to be eliminated from the body $t_{\frac{1}{2}} = \frac{0.693}{k}$
- ▶▶ $t_{\frac{1}{2}}$ is constant for each drug using 1st order kinetics; it is independent on C_p^0
- ▶▶ Both k and $t_{\frac{1}{2}}$ are inherent constant properties of the drug
- ▶▶ $t_{\frac{1}{2}}$ differ only in sustained release formulation (have longer $t_{\frac{1}{2}}$)

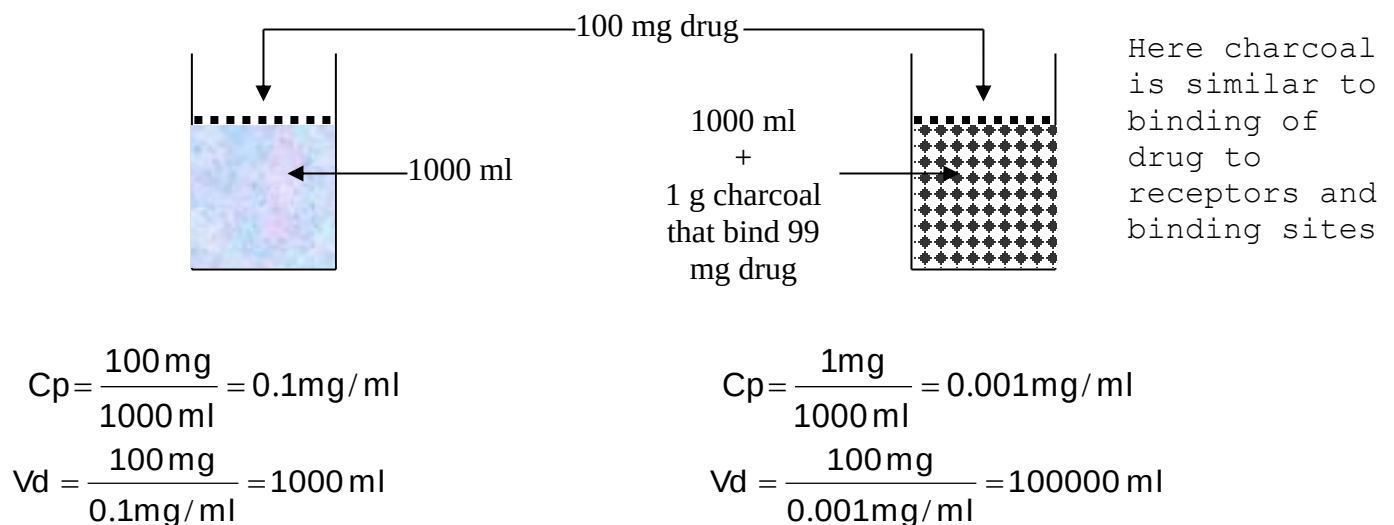
Vd (apparent volume of distribution)

A proportionality constant relating amount of drug in the body to drug concentration according to the following equation

$$\text{Total amount of drug in body } D_B = C_p \times V_d \Rightarrow V_d = \frac{D_B (\text{Amount of drug in the body})}{C_p (\text{plasma concentration of the drug})}$$

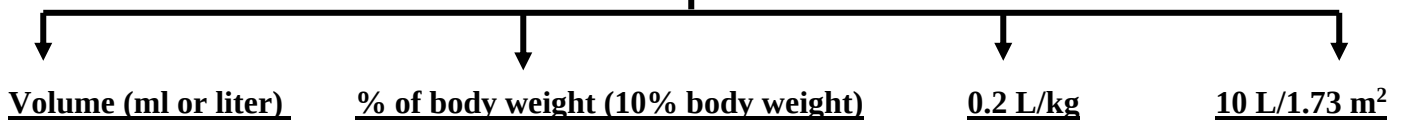
Significance of Vd

1) It is a **hypothetical** volume (not a true physiological volume) of the fluid at which the drug is distributed i.e. reflect degree of distribution of the drug. It may be smaller, equal or larger than the body mass

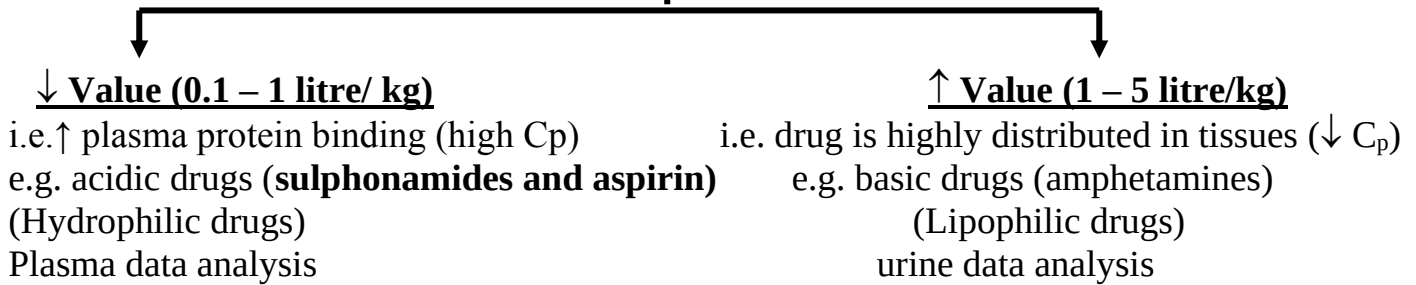


2) Vd is a **proportionality constant** for each drug (under normal healthy condition) but change only in **pathological conditions** e.g. renal dysfunction, liver cirrhosis, CHF and edematous conditions ($\uparrow V_d$ due to $\uparrow$ of total body fluids) but in hemorrhage diarrhea and vomiting ($\downarrow V_d$ due to loss of body fluids)

3) Units of Vd



4) Vd may be low or high (for different drugs)



$$\frac{dD_B}{dt} = -K D_B \Rightarrow \frac{dD_B}{dt} = -K \cdot V_d \cdot C_p \Rightarrow dD_B = -K \cdot V_d \cdot C_p dt \text{ By integration}$$

$$\int_0^{D^0} dD_B = -k \cdot V_d \int_0^{\infty} C_p dt \Rightarrow D^0 = k \cdot V_d \cdot [AUC]_0^{\infty} \Rightarrow V_d = \frac{D^0}{k \cdot [AUC]_0^{\infty}} \text{ And } V_d = \frac{D_B^0}{C_p^0}$$

So Vd may be determined from IV dose and initial concentration and from IV dose, elimination rate constant and area under the curve

Clearance

- ▶▶ The volume of the fluid (plasma or blood) containing the drug which is cleared of the drug per unit time, so if clearance of penicillin is 15 ml / min in a patient with Vd = 12 liters, this means that 15 ml of the 12 liters is cleared of the drug per minute. Its unit is volume per time (ml/min or L/hr)
- ▶▶ It is a measure of drug elimination from the body without identifying the mechanism or the process (may be through kidney excretion, liver metabolism or both)
- ▶▶ It is a constant for a first order process regardless of drug concentration
- ▶▶ $Cl_T = K \cdot V_d = \frac{0.693}{t_{1/2}} \cdot V_d$, **Clearance is inversely proportional to $t_{1/2}$**

Example 1 A drug has an elimination $t_{1/2}$ of 6 hrs and follows first-order kinetics. If a single 200 mg dose is given to an adult male patient (68 kg) by IV bolus injection, what percent of the dose is lost in 24 hrs

Answer

Example 2

A 50 kg volunteer is given an intravenous dose of an antibiotic, and serum concentrations were determined at 2 and 5 hours after administration. The concentrations were 1.2 and 0.3 $\mu\text{g/ml}$, respectively. What is the biologic half-life for this drug, assuming first-order elimination kinetics?

Example 3

A new drug was given in a single IV dose of 200 mg to an 80 kg male. After 6 hrs, the blood concentration of drug was found to be 1.5 mg/100 ml of blood. Assuming the apparent V_d is 10% of body weight, compute the total amount of drug in the body fluids after 6 hrs. What is the half life of this drug?

Example 4

A drug has an elimination half life of 8 hr and follows first-order elimination kinetics. If a single 600 mg dose is given to an adult female patient (62) by rapid IV injection, what percent of the dose is eliminated (lost) in 24 hr assuming the apparent V_D is 400 ml/kg. What is the expected plasma drug concentration (C_p) at 24 hr post dose?

Example 5

A single I.V dose of a new antibiotic was given to a **50 kg** woman at a dose level of **20 mg / kg**, blood samples were removed periodically and examined for parent drug, the data are tabulated in the following table

Time (hour)	Cp ($\mu\text{g/ml}$)
0.25	4.2
0.5	3.5
1	2.5
2	1.25
4	0.31
6	0.08

- Plot **Cp** versus **time** on semilog paper; determine **K**, **$t_{1/2}$** , **Vd** and **total clearance**

Example 6

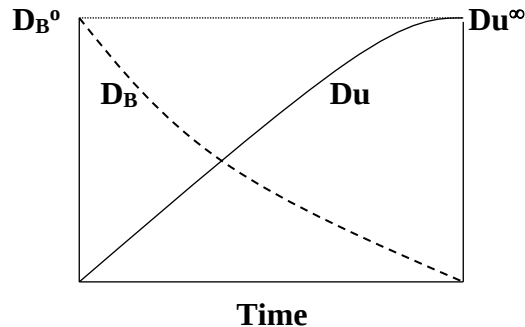
A 50 kg woman was given a single IV dose of an antibacterial drug at dose level of 6 mg/kg. blood samples were taken at various time intervals. The concentration of the drug (Cp) was determined in plasma fraction of each blood sample and the following data were obtained

t (hr)	Cp ($\mu\text{g/ml}$)
0.25	8.21
0.50	7.87
1.0	7.23
3.0	5.15
6.0	3.09
12.0	1.11
18.0	0.40

- What are the values for Vd, K and $t_{1/2}$ for this drug
- This antibacterial agent is not effective at a plasma concentration of less than 2 $\mu\text{g/ml}$. what is the duration of activity for this drug
- How long would it take for 99.9% of this drug to be eliminated
- If the dose of the antibiotic were doubled exactly, what would be the increase in duration of activity

Calculation of K using urine data

- Urine data is a **mirror image of plasma data** especially for drugs which are excreted unchanged in urine.



Du: cumulative amount of the drug excreted in urine

Du[∞]: total cumulative amount of drug Excreted in urine

DB: amount of the drug in blood

(1) Urinary excretion method

By plotting the **rate of urinary Excretion** $\frac{dDu}{dt}$ versus **time*** (midpoint)

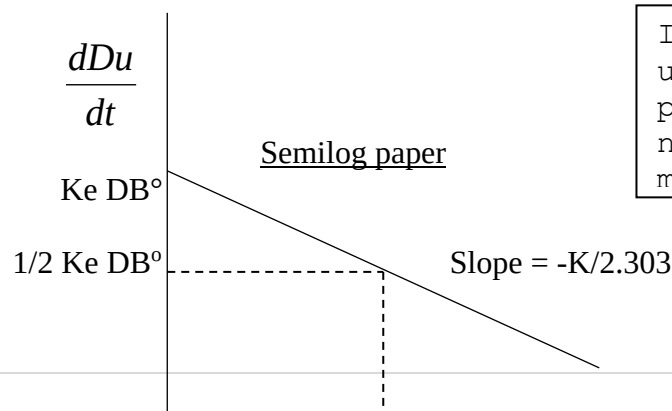
$$\frac{dDu}{dt} = K_e D_B \quad \text{but} \quad D_B = D_B^0 e^{-Kt} \quad \text{So} \quad \frac{dDu}{dt} = K_e D_B^0 e^{-Kt}$$

Taking **log** we get $\rightarrow \log \frac{dDu}{dt} = \log K_e D_B^0 - \frac{Kt}{2.303}$

K : Total elimination rate constant

Ke: Renal elimination rate constant

Knr = (K – Ke) i.e. non renal rate constant of drug elimination.



If the rate of urinary excretion is plotted against normal time not midpoint time it

$t_{1/2}$ **Time (midpoint)****Time* (mid point), why?? !**

Because if we use time at end of each interval, this will give the urinary elimination rate at the end of this interval so, to be representative to the total interval, use time at the midpoint of each time interval

(2) Sigma – minus method

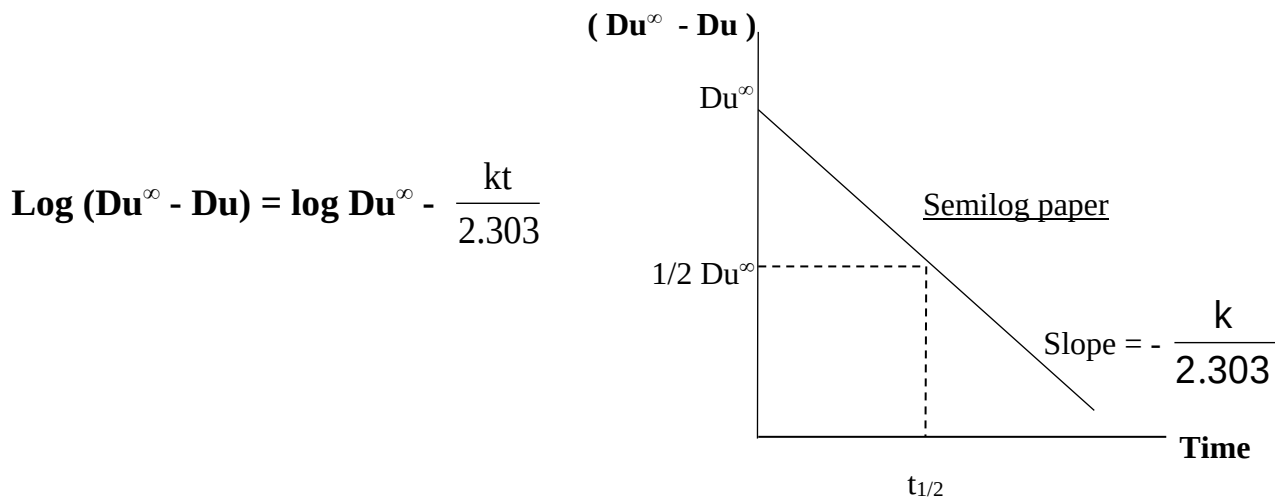
By plotting log ARE (amount Remained to be excreted) ($Du^\infty - Du$) versus time

The amount of unchanged drug in urine can be given from $Du = \frac{K_e D^0}{K} (1 - e^{-kt})$

At time $t = \infty \Rightarrow Du$ become Du^∞ and $e^{-k\infty} = \text{zero}$

So $Du^\infty = \frac{K_e D^0}{K}$ and by substituting for Du^∞ in the above equation

$$Du = Du^\infty (1 - e^{-kt}) \Rightarrow Du = Du^\infty - Du^\infty e^{-kt} \Rightarrow \mathbf{Du^\infty - Du = Du^\infty e^{-kt}}$$

**Example 7**

In a study on an antibiotic, urine data was collected and the equation that fits to the curve of sigma minus method was found to be $Du^\infty - Du = 50 e^{-0.2t}$. Determine Du^∞ and $t_{1/2}$. Units of Du^∞ mg and $t_{1/2}$ hours?

Comparison between the urinary excretion rate method and sigma minus method

	Urinary excretion rate method	Sigma minus method
knowledge of Du^∞	Not required and loss of one urine sample will not invalidate the method	Required and small error in its determination will cause error in all results as all results depends on it $Du^\infty - Du$
Fluctuations in rate of drug elimination and experimental error e.g. incomplete evacuation of bladder	Changes in linearity of the plot	Accuracy is less affected
Order of elimination	Applicable to zero order and first order elimination	Applicable to first order elimination only
k_e	Can be determined	Can not be determined

Validity of urine data

- 1- At least **20 %** of the drug should be excreted unchanged (as intact drug).
- 2- **Collection period**: Urine samples are collected till most of the drug is excreted i.e. to about **7 t $\frac{1}{2}$ where 99 % of drug is eliminated or 10 t $\frac{1}{2}$ where 99.9% of drug is eliminated.**
- 3- **Sampling interval**: Frequent sampling to get a good curve description.
- 4- Subjects should be instructed for **complete evacuation of the bladder** at each collection period to collect the entire drug present in bladder.
- 5- Considering alteration in urinary **PH** and volume which may cause variation in urinary excretion rate.
- 6- Assay technique should be **accurate and specific for the drug (HPLC)**.
- 7- **Statistical analysis** should be done to compensate the individual variation (2 way ANOVA is used not student "t" test)

Example 8

A single I.V dose of a new antibiotic was given to a **50 kg** woman at a dose level of **20 mg / kg**, urine samples were removed periodically and examined for parent drug, the data are tabulated in the following table

Time (hour)	Du (mg)
0.25	160
0.5	140
1	200
2	250
4	188
6	46

Determine **K**, **t_{1/2}**, using **urinary excretion method** and **Sigma – minus method**

Answer**(1) Urinary Excretion Method**

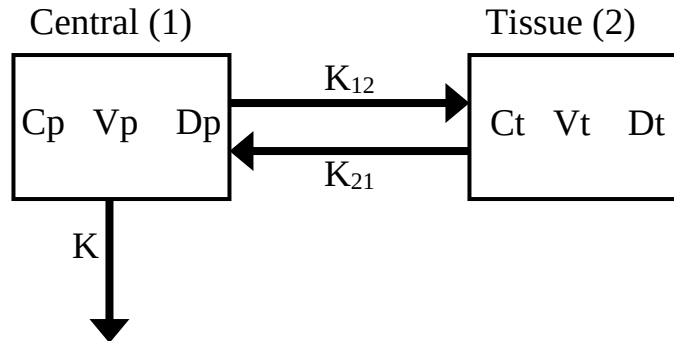
Time (hour)	Du (mg)	t* (hr) midpoint time	Du/t or (dDu/dt)	mg/hr
0.25	160			
0.5	140			
1	200			
2	250			
4	188			
6	46			

(2) Sigma – Minus Method

Time (hour)	Du (mg)	Cumulative Du	Du [∞] - Du
0.25	160		
0.5	140		
1	200		
2	250		
4	188		

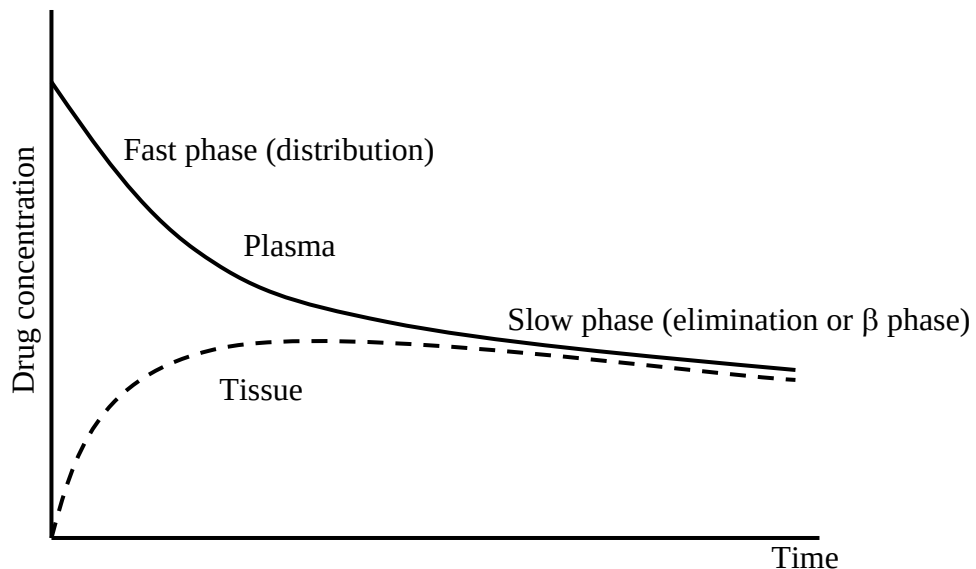
6	46		
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Two compartment open model following I V route (single dose)



Two compartment open model
following IV injection

- ▶▶ The drug equilibrate and distribute in 2 compartments, the central compartment (rapidly and uniformly) and peripheral compartment (slowly)
- ▶▶ Drug transfer between the 2 compartments follow 1st order kinetics
- ▶▶ At time $t = 0$ no drug in tissue compartment, while it is peak in plasma (central compartment)
- ▶▶ After drug being distributed into tissue compartment, rapid decline of plasma conc. while tissue conc. starts to peak
- ▶▶ Then tissue conc. starts to decline as the conc. gradient between the 2 compartments does



- ▶▶ In this model K_{12} and K_{21} are the 1st order rate constants of drug transfer from central to tissue compartment and from tissue to central compartment respectively

- ▶▶ Rate of drug change in tissue is given from $\frac{dC_t}{dt} = K_{12}.C_p - K_{21}.C_t$
- ▶▶ Concentration of drug in central compartment is $C_p = \frac{D_p \text{ (amount of drug in plasma)}}{V_p \text{ (volume of central compartment)}}$
- ▶▶ And concentration in peripheral compartment is $C_t = \frac{D_t \text{ (amount of drug in tissue)}}{V_t \text{ (volume of tissue compartment)}}$
- ▶▶ a and b are hybrid 1st order constants for drug distribution and elimination phase respectively
- $a + b = K_{12} + K_{21} + K$ and $ab = K_{21}.K$
- ▶▶ The general equation that describe the plasma concentration of drug at any time is

$$C_p = Ae^{-at} + Be^{-bt}$$

Calculation of pharmacokinetic parameters in two – compartment Model using method of residual (Feathering method)

Method of residuals (feathering method): a graphical method for separation of exponents e.g. separation of the absorption rate constant or Distribution rate constant from elimination rate constant.

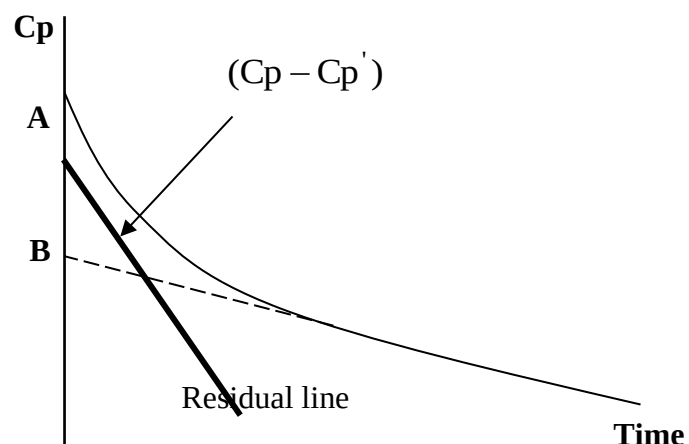
The biexponential equation for C_p is $C_p = Ae^{-at} + Be^{-bt}$

- ▶▶ In the beginning distribution is faster than elimination so $a > b$, but after a while Ae^{-at} will approach zero, So $C_p = Be^{-bt}$ (drug undergo complete elimination) and by in log form it become $\log C_p = \log B - \frac{bt}{2.303}$ (**logarithmic expression of elimination or β - phase**)
- and b can be obtained from slope of the terminal straight portion (slope = $-b/2.303$)
- And $t_{1/2}$ of elimination phase equal $0.693/b$

- ▶▶ To convert distribution phase to straight line, use feathering method by plotting the residual concentration ($C_p - C_p'$) versus time.

- So distribution phase is expressed by the following **equation of residual line**

$$* \log(C_p - C_p') = \log A - \frac{at}{2.303}$$



- Slope of residual line – $a / 2.303$ from which a can be determined
- Some pharmacokinetic parameters may be derived from the proceeding equation.

$$K = \frac{ab(A + B)}{bA + Ba} \quad K_{12} = \frac{AB(b - a)^2}{(A + B)(Ab + Ba)} \quad K_{21} = \frac{Ab + Ba}{A + B}$$

$$A = \frac{D^0(a - K_{21})}{V_p(a - b)} \quad B = \frac{D^0(K_{21} - b)}{V_p(a - b)}$$

$$t_{\frac{1}{2}} = \frac{0.693}{K} \quad C_p^0 = A + B \quad V_p = \frac{D^0 \text{ (IV dose)}}{A + B}$$

$$(AUC)_0^\infty = \frac{D^0}{KV_p} \quad V_t = V_p \frac{K_{12}}{K_{21}}$$

K is the true elimination rate constant while b is a hybrid constant that is affected by the rate of drug transfer in and out of tissue compartment so b is smaller than K

Determination of the compartment model

- Compartmental models should always use the fewest number of compartments necessary to describe the experimental data adequately.

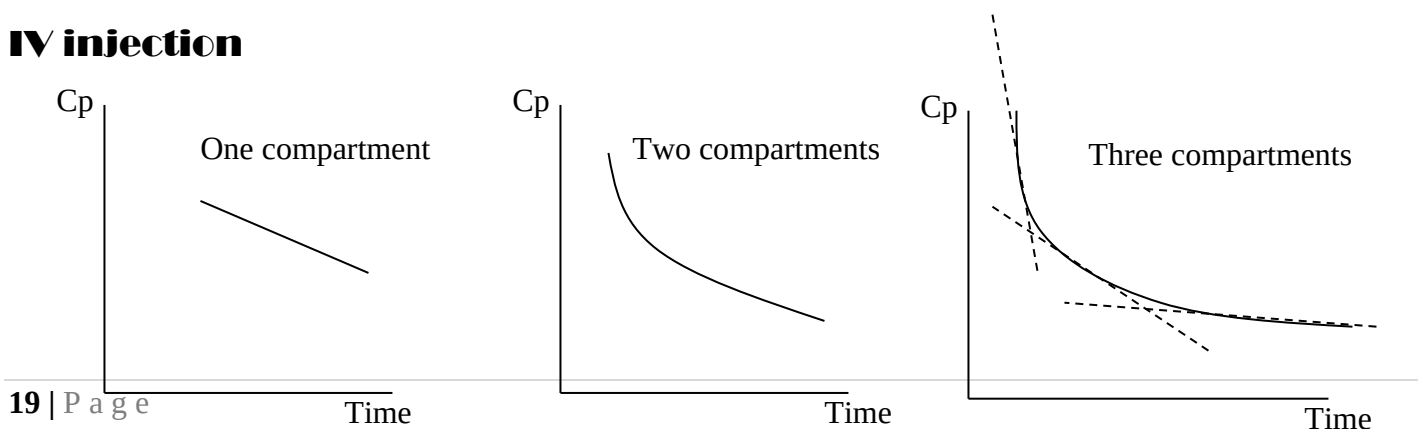
►► The observed number of compartments will depend on

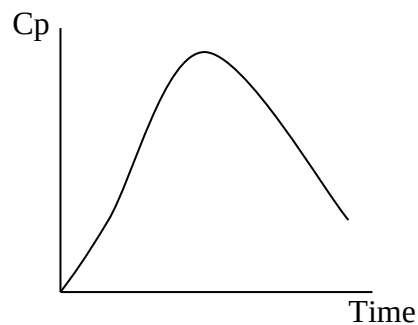
- 1) The route of drug administration
- 2) The rate of drug absorption
- 3) The total time of sampling
- 4) The number of samples taken during the collection period
- 5) The assay sensitivity

- Plot C_p versus time, provided that sampling is done at short intervals to show distribution phase (if present).

- (1) If straight line is obtained $\Rightarrow$ one compartment open model.
- (2) Curved line is obtained $\Rightarrow$ two compartments or three compartments open model.

IV injection



Oral absorption**Example 9**

An Antibiotic was given IV in dose of 300 mg to healthy volunteers (75 kg). The drug was found to follow **2 compartment** open model & the plasma drug concentration versus time curve was constructed on a semilog paper, the equation that fit to the obtained curve was found to be

$$C_p = 45 e^{-1.8 t} + 15 e^{-0.21 t}$$

♥ Determine K (hr^{-1}), $t_{1/2}$ and V_p

Answer

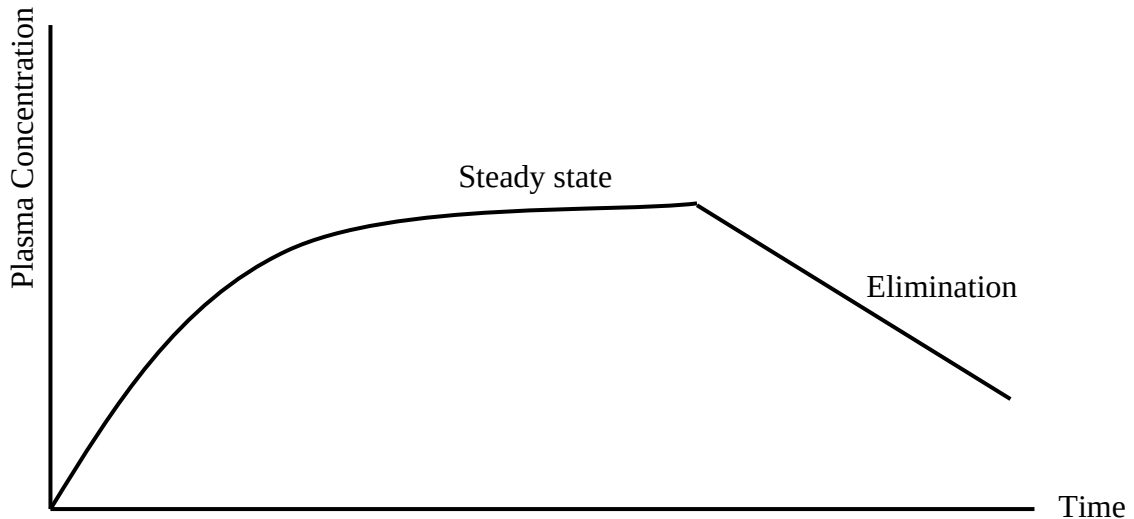
Example 10: plot the following data on semilog paper; determine a , b , A , B , K , $t_{1/2}$, K_{12} and K_{21}

Time (hr)	0.25	0.5	1	1.5	2	4	8	12	16
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Cp (µg/ml)	43	32	20	14	11	6.5	2.8	1.2	0.52
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Intravenous Infusion

- ▶▶ Drugs are given by constant IV infusion at a constant zero order rate to maintain constant plasma drug concentration at steady state
- ▶▶ At steady state; rate of drug elimination from the body equal rate of drug entering the body (rate of infusion)



- ▶▶ **At steady state** $\frac{dC_p}{dt} = 0 \Rightarrow$ typical zero order kinetics through injection
- ▶▶ And rate of drug input (zero order infusion) = rate of drug output (first order elimination)

$$\frac{dD_B}{dt} = R - k \cdot D_B$$

Where R: is rate of zero order infusion

By integration the above equation and substitution for $D_B = K \cdot V_d$

$$C_p = \frac{R}{k \cdot V_d} (1 - e^{-kt})$$

- ▶▶ At $t = \infty \Rightarrow e^{-kt} = \text{zero}$ and $C_{ss} = \frac{R}{k \cdot V_d} = \frac{R}{Cl}$ thus $\uparrow R$ will $\uparrow$ steady state but $\uparrow Cl$ will $\downarrow$

steady state i.e. $\frac{R_1}{R_2} = \frac{C_{ss1}}{C_{ss2}}$ thus doubling R will increase (double) steady state but time required to reach steady state will remain constant as it depends on drug elimination only for a constant V_d

- ▶▶ Steady state is reached after $5 t_{1/2}$ ($4.32 t_{1/2}$) where more than 95% of steady state is reached, by infusion alone we can never reach 100% of steady state required

- » As theophylline has a $t_{1/2} = 7$ hrs thus to take a sample for drug monitoring, take it after $5 \times 7 \text{ hr} = 35 \text{ hrs}$
- » After $3.32 t_{1/2} \Rightarrow 90\%$ of C_{ss} is reached and after $6.65 (\approx 7) t_{1/2} \Rightarrow 99\%$ of C_{ss} is reached
- » Thus following any infusion rate (R); steady state will be reached only after $5 t_{1/2}$

» **Loading dose (D_L) with IV infusion**

The dose of the drug that should be given initially by IV bolus injection to reach steady state as rapidly as possible (**immediately**)

Concentration of drug following loading dose is $C_1 = C^0 e^{-kt} \Rightarrow C_1 = \frac{D_L}{V_d} e^{-kt}$

But concentration of drug following infusion with rate R is $C_2 = \frac{R}{k \cdot V_d} (1 - e^{-kt})$

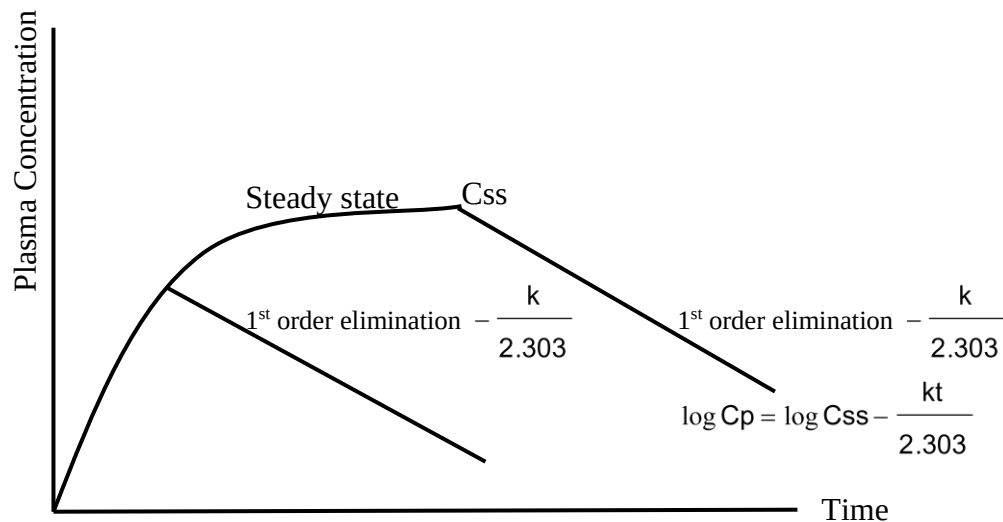
So plasma level following both IV bolus injection and IV infusion become $C_p = C_1 + C_2$

$$\therefore C_p = \frac{D_L}{V_d} e^{-kt} + \frac{R}{k \cdot V_d} (1 - e^{-kt}) \Rightarrow C_p = \frac{D_L}{V_d} e^{-kt} + \frac{R}{k \cdot V_d} - \frac{R}{k \cdot V_d} e^{-kt}$$

$$C_p = \frac{R}{k \cdot V_d} + \left(\frac{D_L}{V_d} e^{-kt} - \frac{R}{k \cdot V_d} e^{-kt} \right) \Rightarrow \text{At } t = \infty \Rightarrow C_{ss} = \frac{R}{k \cdot V_d} \Rightarrow \text{but } D_L = C_{ss} V_d$$

$$\text{So } D_L = \frac{R}{K}$$

- » **After steady state (when infusion is stopped);** drug level undergo elimination by first order kinetics



» **Calculation of half life from infusion**

$$C_p = \frac{R}{k \cdot V_d} (1 - e^{-kt}) \text{ but } C_{ss} = \frac{R}{k \cdot V_d} \text{ by substitution } C_p = C_{ss} (1 - e^{-kt})$$

$$C_p = C_{ss} - C_{ss} e^{-kt} \Rightarrow C_{ss} - C_p = C_{ss} e^{-kt} \text{ by taking log for the whole equation}$$

$$\log(C_{ss} - C_p) = \log C_{ss} - \frac{kt}{2.303}$$

From the above equation K can be determined by knowing C_{ss} and any concentration before steady state C_p and time for that concentration t and then determine $t_{1/2} = \frac{0.693}{K}$

Example 11

An antibiotic has $V_d = 10$ liters, $k = 0.2 \text{ hr}^{-1}$, a steady state plasma concentration of 10 ug / ml , calculate rate of infusion.

Example 12

If $R = 2 \text{ mg / hr}$ $k = 0.1 \text{ hr}^{-1}$ $V_d = 10 \text{ L}$

What dose should be recommended if the drug level is desired to be 2 ug / ml immediately?!

Example 13

If an antibiotic has elimination half life of 3.6 in the general population, a patient was given an I V infusion of that antibiotic (at an infusion rate of 15 mg / hr). Blood samples were taken at 8 and 24 hrs, plasma drug concentration were 5.5 and 6.5 respectively in mg / L .

Estimate elimination half life of the drug in this patient? If the desired C_p is 8 mg / L for the above patient, what is suitable infusion rate for the patient?

Example 14

A drug was given to a 75 kg patient by single IV dose and by constant IV infusion. The serum concentration were

Time (hour)	Plasma concentration ($\mu\text{g/ml}$)	
	Single IV dose (1 mg/kg)	Constant IV infusion (0.2 mg/kg/hr)
0	10	0
2	6.7	3.3
4	4.5	5.5
6	3.0	7.0
8	2.0	8.0
10	1.35	8.6
12		9.1
18		9.7
24		9.9

- 1) What is the **steady-state** plasma drug level?
- 2) What is the **time for 95 % steady-state** plasma drug level?
- 3) What is the drug **clearance**?
- 4) What is the plasma concentration of drug 4 hrs after stopping infusion (infusion was stopped after 24 hrs)?
- 5) What is the infusion rate for a patient weighing 75 kg to maintain a steady state drug level of $10 \mu\text{g/ml}$?
- 6) What is the plasma drug concentration 4 hrs after an IV dose of 1 mg/kg followed by a constant infusion of 0.2 mg/kg/hr ?

Drug elimination from the body

It is irreversible removal of drug from the body by all routes of elimination

» Routes of drug elimination

1) Excretion: mainly renal excretion (kidney) or excretion into sweat, bile, saliva or milk

2) Metabolism (biotransformation): enzymatic conversion of drug into metabolite in liver

Renal excretion of drug

- » It is the route of excretion for non volatile, water soluble, low molecular weight and slowly metabolized drugs
- » Drugs are removed in kidney through one or combination of the following
 - 1) Glomerular filtration
 - 2) Active tubular secretion
 - 3) Renal tubular reabsorption

	Glomerular filtration	Active secretion	Renal Reabsorption
Indicator	Substance excreted by GF only e.g. 1. Inulin 2. Creatinine	1. Para amino hippuric acid (PAH) 2. Iodopyracet	-
Normal	$Cl_{inulin} = Cl_{cr} = GFR$ $= 125 - 130 \text{ ml/min}$	425 – 650 ml/min	-
Mechanism	» Simple diffusion of small ionized or unionized free drug from blood to urine with conc. gradient » Protein bound drugs (become large molecules) are not filtered through glomeruli	» Active transport <u>from blood to urine against conc. gradient (require carrier and energy)</u> » 2 carrier systems are known in kidney for weak acidic and the other for weak basic drugs » Drugs with similar structure may compete for the same carrier system in kidney	» Return unionized (more lipid soluble) drug from urine to blood. » Can be active or passive process » If drug is completely reabsorbed so its Cl is zero e.g. glucose » For weak acids and weak bases; it depends on pH of urine and pKa of the drug » Thus if amphetamine is given with urine alkalinizer; it will be reabsorbed from urine and can not be detected by urine analysis
Rate of renal Excretion = Rate of G. Filtration + Rate of Active secretion – Rate of Reabsorption.			

Clearance

» The volume of the fluid (plasma or blood) containing the drug which is cleared of the drug per unit time, so if clearance of penicillin is 15 ml / min in a patient with $V_d = 12$ liters, this means that 15 ml of the 12 liters are cleared of the drug per minute.

» It may be defined as the rate of drug elimination divided by the plasma concentration.

$$(1) \text{ Total clearance } Cl_T = \frac{dDe/dt}{C_p} = \frac{\mu g / \text{min}}{\mu g / \text{ml}} = \text{ml} / \text{min} \Rightarrow \frac{dDe}{dt} = Cl_T \cdot C_p$$

$$(2) Cl_T = K \cdot V_d = \frac{0.693}{t_{1/2}} \cdot V_d \quad \text{So } \underline{\text{Clearance is inversely proportional to } t_{1/2}}$$

$$\underline{\text{After 2 compartments}} \Rightarrow Cl_T = K \cdot V_p = \frac{0.693}{t_{1/2}} \cdot V_p \text{ and } Cl_r = K_e \cdot V_p \text{ i.e. no of}$$

compartments will not affect drug clearance

$$(3) Cl_T (\text{total clearance}) = Cl_r (\text{renal clearance}) + Cl_h (\text{hepatic clearance})$$

(4) Renal clearance: volume of plasma cleared of drug per unit time through kidney

$Cl_r = K_e \cdot V_d$ where K_e is renal elimination rate constant

Or Renal clearance $Cl_r = \frac{dD_u/dt}{C_p} = \frac{\mu g/min}{\mu g/ml} = ml/min$

Or Renal clearance $Cl_r = \frac{GFR + \text{rate of active tubular secretion} - \text{rate of tubular reabsorption}}{C_p}$

(5) Hepatic clearance $Cl_h = K_m \cdot V_d$ where K_m is hepatic elimination rate constant

(6) $Cl_T = \frac{D_o(\text{IV dose})}{(AUC)_0^\infty} = \frac{f(\text{fraction of drug absorbed}) \times D_o(\text{oral dose})}{(AUC)_0^\infty}$

$Cl_r = \frac{D_u^\infty}{(AUC)_0^\infty}$ and $Cl_h = \frac{M_u^\infty}{(AUC)_0^\infty}$ where M_u^∞ is amount of drug metabolized

(7) $Cl_r = Cl_T \cdot f_u$, where f_u is fraction of drug excreted unchanged in urine

(8) $f_u = \frac{D_u^\infty(\text{total amount of drug excreted unchanged in urine})}{f(\text{fraction of drug absorbed}) \cdot D_o(\text{oral dose})} = \frac{D_u^\infty}{IV \text{ dose}} = \frac{D_u^\infty}{D_u^\infty + M_u^\infty}$

$f_u = \frac{K_e}{K} = \frac{Cl_r}{Cl_T}$ so $Cl_r = Cl_T \cdot \frac{K_e}{K}$

(9) Fraction of drug metabolized = $(1 - f_u)$ or $= \frac{M_u^\infty}{IV \text{ dose}} = \frac{M_u^\infty}{f \cdot \text{oral dose}}$

And % of drug metabolized = $\frac{K_m}{K} \times 100$

(10) $Cl_T = \frac{R}{C_{ss}}$ in steady state of infusion (for both one and two compartments)

Determination of the most probable mechanism of excretion of the drug in kidney

Clearance ratio: Ratio of renal clearance of the drug to clearance of Inulin or creatinine (where $Cl_{inulin} = Cl_{cr} = GFR = 125 - 130 \text{ ml/min}$).

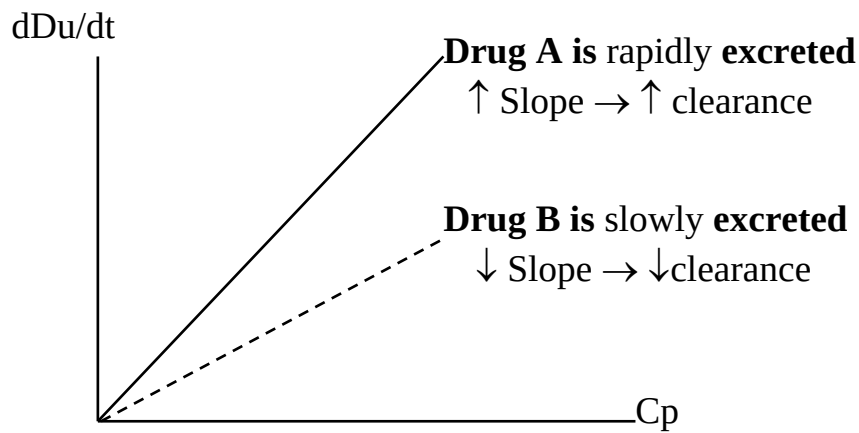
Clearance ratio	Probable mechanism of renal excretion
$\frac{Cl_{\text{drug}}}{Cl_{\text{inulin}}} < 1$	Drug is partially reabsorbed

$\frac{Cl_{\text{drug}}}{Cl_{\text{inulin}}} = 1$	Drug is filtered only (GF only)
$\frac{Cl_{\text{drug}}}{Cl_{\text{inulin}}} > 1$	Drug is actively secreted
$\frac{Cl_{\text{drug}}}{Cl_{\text{inulin}}} = \text{zero}$	Drug is completely reabsorbed e.g. glucose

Determination of renal clearance

(1) Renal clearance $Cl_r = \frac{dDu/dt}{C_p} \Rightarrow \frac{dDu}{dt} = Cl_r \cdot C_p$

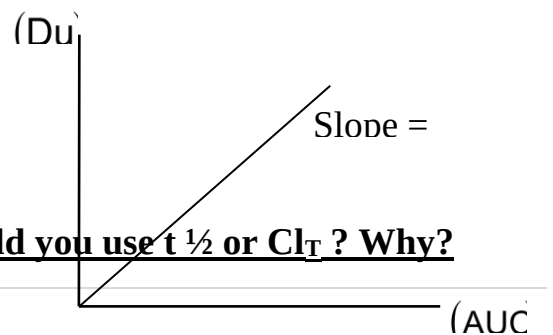
Plot rate of drug excretion in urine (dDu / dt) VS C_p , slope = clearance.



(2) $\frac{dDu}{dt} = Cl_r \cdot C_p \Rightarrow dDu = Cl_r \cdot C_p dt \Rightarrow$ by integration $\int_0^t dDu = Cl_r \cdot C_p \int_0^t dt$

$$[Du]_0^t = Cl_r \cdot [AUC]_0^t$$

Plotting of cumulative amount of drug excreted in urine $(Du)_0^t$ versus $(AUC)_0^t$ of plasma concentration / time curve.



To study removal of a drug from the body, would you use $t_{1/2}$ or Cl_T ? Why?

We use Cl_T because it considers both $t_{1/2}$ and $V_d \Rightarrow Cl_T = K \cdot V_d = \frac{0.693}{t_{1/2}} \cdot V_d$

So in certain pathological conditions e.g. edema & obesity V_d will change, hence Removal of the drug from the body while $t_{1/2}$ may remain constant i.e. doesn't change significantly although removal of the drug from the body has already changed.

Example 15

A new antibiotic is actively secreted by the kidney and V_d is 35 liters in the normal adult, the clearance of the drug is 650 ml / min.

a- What is the usual $t_{1/2}$ of the drug?

b- What would be the new $t_{1/2}$ for this drug in an adult with partial renal failure whose clearance was only 75 ml / min?

Example 16

A single dose (100 mg) of a drug is given orally, the drug is systemically available, total amount of unchanged drug recovered in urine is 60 mg & total amount of metabolite recovered in urine is 30 mg, according to $t_{1/2} = 3.3$ hr, $V_d = 1000$ ml, determine Cl_T ; Cl_r ; Cl_{nr} ??!

Example 17

In order to estimate the renal clearance of a drug in patient, a 2 hr post dose urine sample was collected and found to contain 200 mg of drug. A midpoint plasma sample was taken (1 hr post dose) and the drug concentration in plasma was found to be 2.5 mg%. What is the renal clearance for this drug in this patient?

Example 18

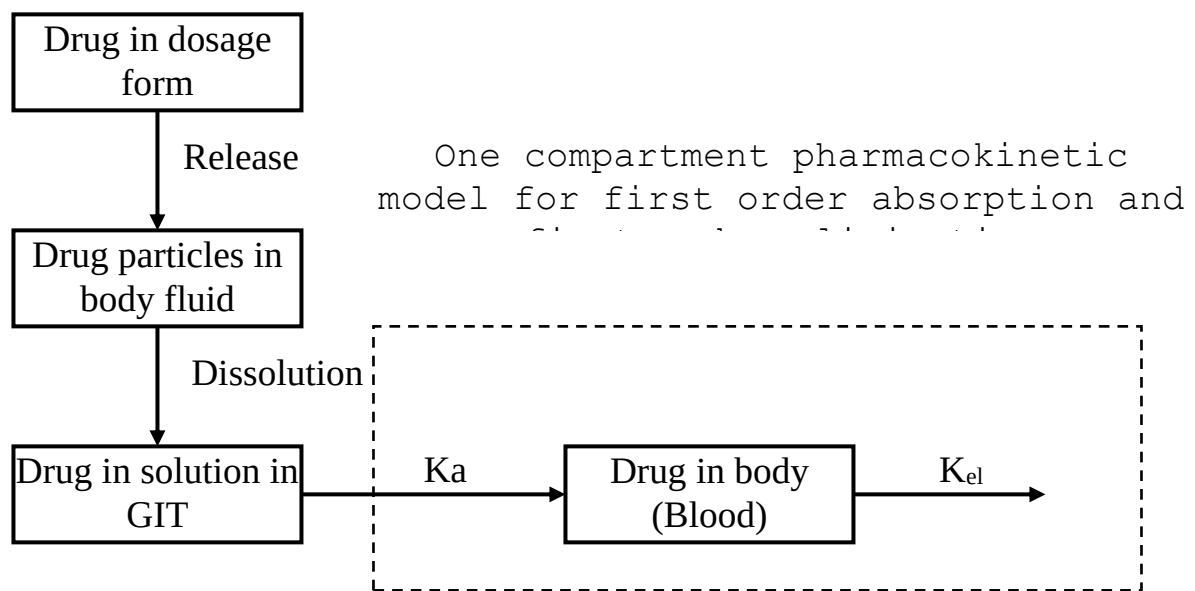
A single 250 mg oral dose of antibiotic is given to a young man (age 32 years, creatinine clearance 122 ml/min, 78 kg). From the literature the drug is known to have an apparent V_d equal to 21 % of the body weight and an elimination half-life of 2 hours. The dose is normally 90 % bioavailable. Urinary excretion of the unchanged drug is equal to 70 % of the absorbed dose. Determine renal, non renal and total clearance

Zero order absorption

- » Drug is absorbed from GI lumen to blood by a constant rate but elimination is first order
- » e.g. Osmotic pump, sustained release tablets
- » Similar to administration of drug by IV infusion

First order one compartment absorption model

- » Absorption is first order for rapidly dissolving dosage forms e.g. immediate release tablets, capsules, suppositories and IM aqueous injections
- » Elimination is first order
- » Drug undergo instantaneous equilibrium and distribution (no distribution phase)



After oral administration, drug dissolve in GI fluid and start to be absorbed according to first order process and rate of disappearance of drug from GIT is given by

$$\frac{dD_G}{dt} = -K_a \cdot F \cdot D_G$$

Where D_G is amount of drug in solution in GIT any time

F is fraction of the dose absorbed (equal 1 for fully absorbed drug and zero for fully unabsorbed drugs)

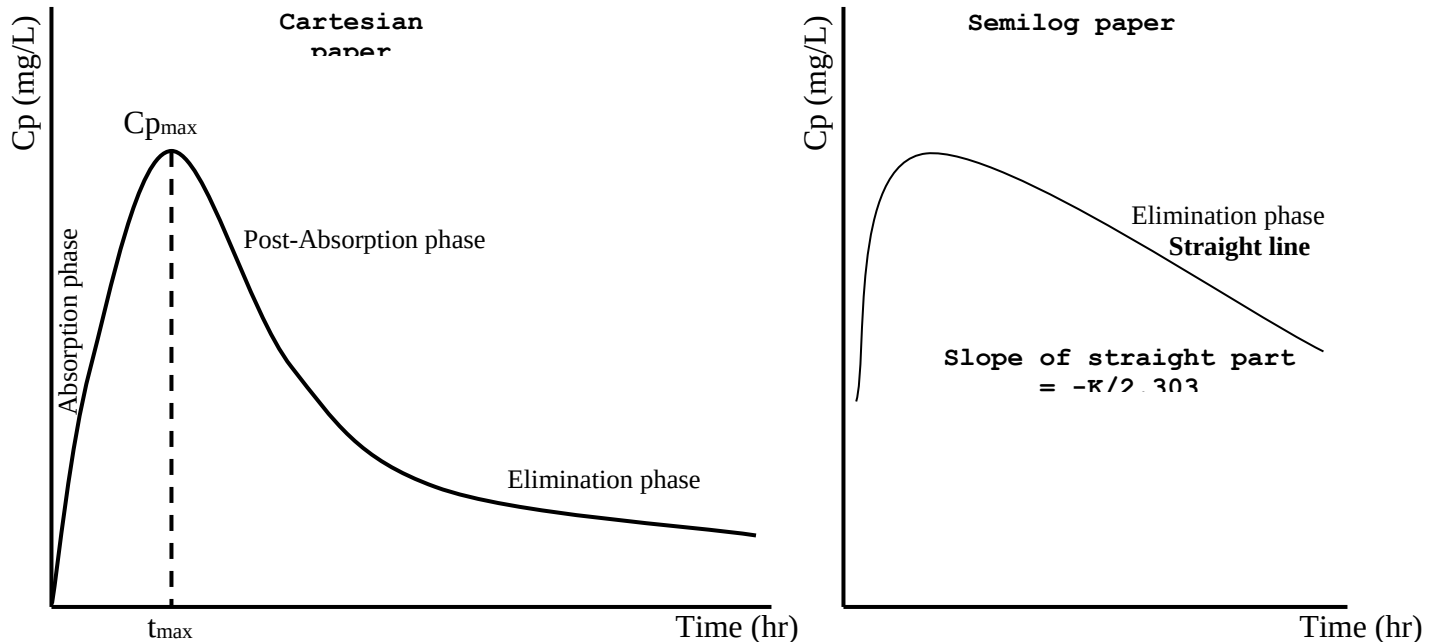
Amount of drug in GIT at any time is given by $D_{GI} = D_0 e^{-K_a \cdot t}$ (where D_0 is oral dose)

Rate of drug change inside the body is given by $\frac{dD_B}{dt} = \text{rate of input} - \text{rate of output}$

$$\frac{dD_B}{dt} = F \cdot K_a \cdot D_{GI} - K \cdot D_B \quad \text{Or} \quad \frac{dD_B}{dt} = F \cdot K_a \cdot D_0 e^{-K_a \cdot t} - K \cdot D_B$$

By integration $\Rightarrow C_p = \frac{F.K_a.D_0}{V_d(K_a - K)}(e^{-K.t} - e^{-K_a.t})$ or $C_p = A(e^{-K.t} - e^{-K_a.t})$

Plot of C_p versus time on both Cartesian and semilog paper give the following curves



T_{max} is independent on dose but depend on absorption and elimination rate constants

according to the following equation $t_{max} = \frac{\ln(K_a / K)}{(K_a - K)}$

N.B.: t_{max} is always calculated from the above equation (more accurate) than from the graph

Determination of K_a from oral absorption data (Method of residual = Feathering model = Stripping = peeling the curve)

Plasma concentration after any time is given by

$$C_p = \frac{F.K_a.D_0}{V_d(K_a - K)}(e^{-K.t} - e^{-K_a.t}) \text{ Or } C_{p1} = A(e^{-K.t} - e^{-K_a.t})$$

Where $A = \frac{F.K_a.D_0}{V_d(K_a - K)} \Rightarrow V_d = \frac{F.K_a.D_0}{A(K_a - K)}$

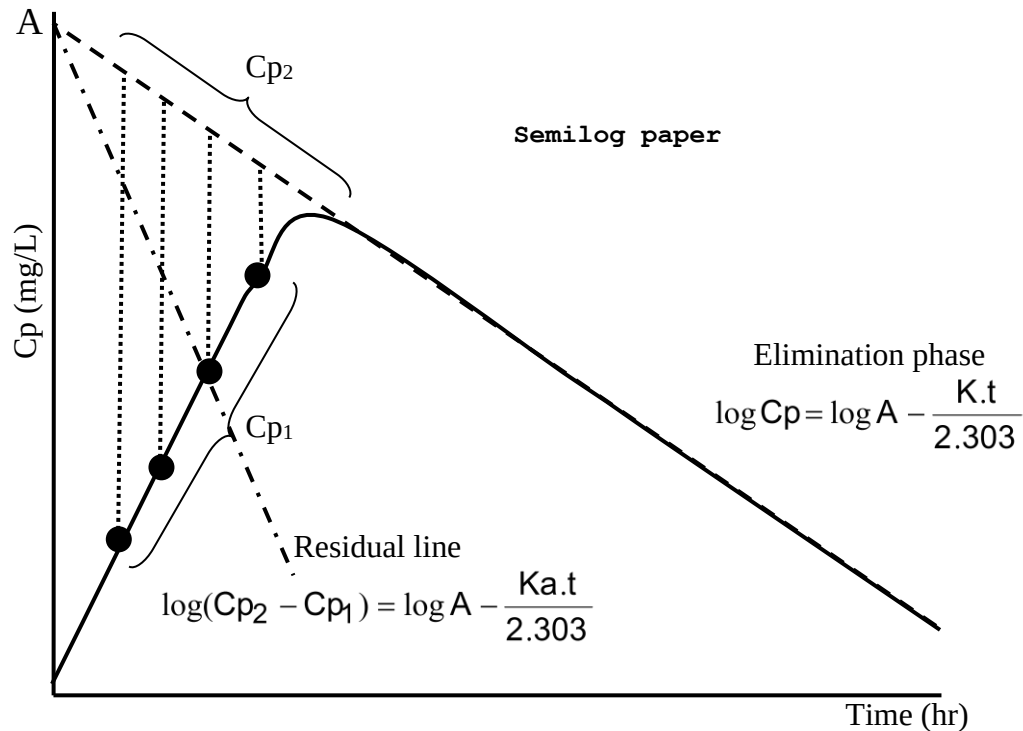
N.B.:

1) For determination of V_d after oral administration accurately; F must be known or better

determine it after IV injection from $V_d = \frac{D_B^0}{C_p^0}$

2) K_a must be greater than $K \Rightarrow$ absorption is faster than elimination thus $K_a.t > K.t$ and in some time after the peak $e^{-K_a.t}$ is zero and the above equation become $Cp_2 = Ae^{-K.t}$ expressing elimination phase only

3) Subtracting Cp_1 from Cp_2 will yield $(Cp_2 - Cp_1) = Ae^{-K_a.t}$ for residual line



If residual line is not straight $\Rightarrow$ absorption is not a first order process

Example 22

The plasma drug concentration in a patient (50 kg) who had received a single **oral** dose of a drug (10 mg / kg) were determined as follow

Time (hr)	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28
Cp (µg/ml)	0	23.7	35.4	39.6	39.6	37.2	33.7	29.7	25.8	22.1	18.7	15.8	13.2	11	9.14

Assuming that drug is **80 %** absorbed, find (using feathering model)

[1] Absorption rate constant **K_a**

[2] Elimination half life **$t_{1/2}$**

[3] **t_{max}**

[4] Volume of distribution of the drug **V_d**

[5] **Cp_{max}**

[6] **Cp** after 11 hours

Wagner — Nelson method (fraction of drug unabsorbed VS Time)

It is a numeric and more general method for determination of K_a from plasma concentration data following extravascular administration

- It assumes that

- (1) Body behaves as single homogenous compartment.
- (2) Drug elimination obey 1st order kinetics (determination of K is very critical)
- (3) No assumptions concerning absorption process.

**** Determination of K_a by plot of fraction of drug unabsorbed versus Time.**

- After single oral dose of a drug.

$$D_o (\text{Total Dose}) = D_G (\text{amount of drug in GIT}) + D_B (\text{amount of drug in body}) + D_u (\text{amount of drug in urine})$$

$$\heartsuit \text{ Ab (amount absorbed at any time t) } = D_B + D_u$$

$$\heartsuit \text{ Ab}^\infty: \text{ amount of drug absorbed at time } \infty (\text{max}^m \text{ amount of drug absorbed})$$

$$\heartsuit \text{ At any time t, fraction of drug absorbed} = \frac{Ab}{Ab^\infty}$$

$$\heartsuit \text{ Fraction of drug unabsorbed} = 1 - \frac{Ab}{Ab^\infty}$$

At any time t

$$D_u = K V_d (AUC)_0^t, D_B = C_p V_d$$

$$\therefore Ab = C_p V_d + K V_d (AUC)_0^t$$

at $t = \infty, C_p = 0$

$$\therefore DB^\infty = 0 \quad Du^\infty = K V_d (AUC)_0^\infty$$

$$\therefore Ab^\infty = 0 + K V_d (AUC)_0^\infty$$

$$\therefore \frac{Ab}{Ab^\infty} = \frac{C_p V_d + K V_d (AUC)_0^t}{K V_d (AUC)_0^\infty} = \frac{C_p + K (AUC)_0^t}{K (AUC)_0^\infty}$$

$\therefore$ Fraction of drug unabsorbed at any time

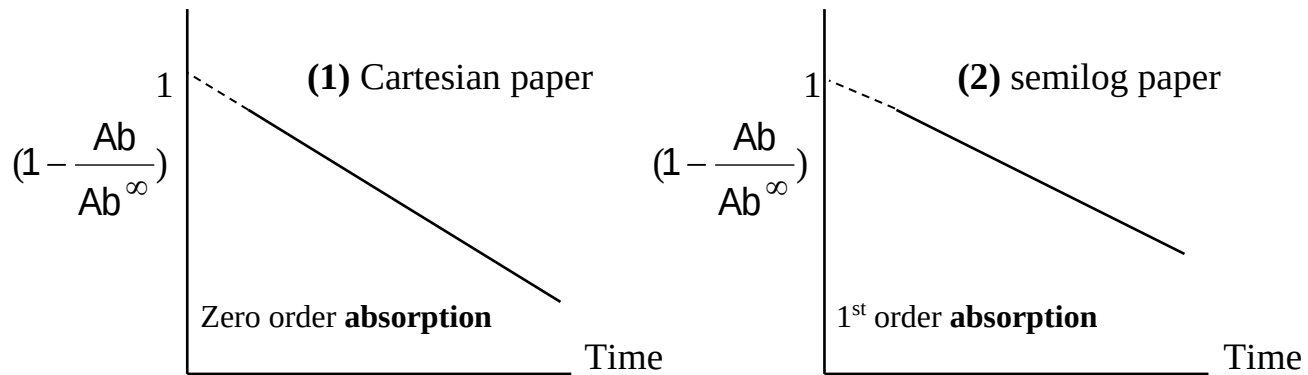
$$\left(1 - \frac{Ab}{Ab^\infty}\right) = \left(1 - \frac{C_p V_d + K V_d (AUC)_0^t}{K V_d (AUC)_0^\infty}\right) = 1 - \frac{C_p + K (AUC)_0^t}{K (AUC)_0^\infty}$$

Steps

- 1) Plot C_p VS time on a semilog paper
- 2) Find K from slope of terminal part of the curve (first order elimination)
- 3) Find $(AUC)_0^t$ from plot of C_p versus time by trapezoidal method
- 4) Multiply K by each $(AUC)_0^t$
- 5) Find $(AUC)_0^\infty$ by summation of all (AUC) from time 0 to time ∞

6) Multiply K by $(AUC)_0^\infty$ 7) Determine $(1 - \frac{Ab}{Ab^\infty})$ at each time.

8) Plot $(1 - \frac{Ab}{Ab^\infty})$ VS time on Cartesian & semilog paper. State whether absorption is zero order or 1st order.



N.B.:

1) Amount of drug remaining at any time following first order absorption is given from

$$D_{GI} = D_0 e^{-K_a \cdot t} \Rightarrow \frac{D_{GI}}{D_0} = e^{-K_a \cdot t} \Rightarrow \text{After log} \Rightarrow \log \frac{D_{GI}}{D_0} = \frac{-K_a \cdot t}{2.303}$$

Thus plot of $\frac{D_{GI}}{D_0}$ (fraction of drug unabsorbed) versus time will give straight line with slope $-K_a/2.303$ (if absorption is first order)

2) Terminal part of the curve $(1 - Ab/Ab^\infty)$ versus time will be scattered and nonlinear; thus the terminal part is excluded and only the initial linear part of the curve is used for calculation of slope $(-K_a/2.303)$ i.e. calculations stop at $C_{p_{max}}$ only

Example 23

A tablet containing 100 mg of drug was administered to a healthy volunteer and plasma drug concentration versus time was obtained, **does absorption follow 1st or zero order pattern?**

Determine K_a ?

Time (hr)	C_p (mg/ml)	$(AUC)_0^t$	$K(AUC)_0^t$	$C_p + K(AUC)_0^t$? (Ab)	$\frac{Ab}{Ab^\infty}$?	$(1 - \frac{Ab}{Ab^\infty})$?
0.25	0.6					
0.5	1.2					
0.75	1.8					
1	2.3					
1.5	3.4					

2	4.3					
3	6					
6	5.6					
12	2.3					
18	0.9					
24	0.4					

Lag time

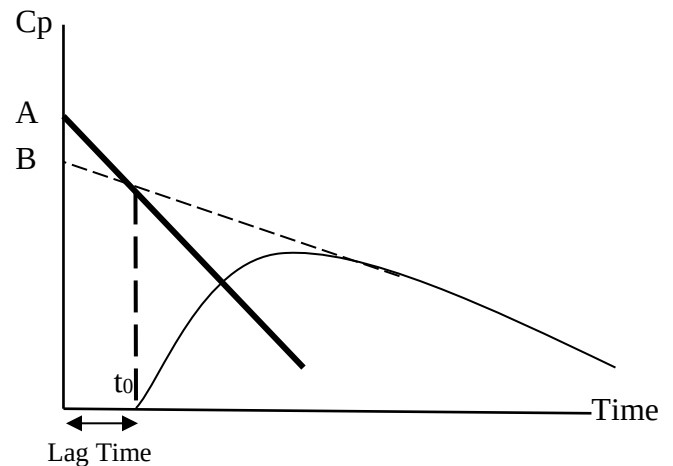
Definition time delay before start (commencement) of 1st order absorption of drugs

Cause

(1) ↓ gastric emptying

(2) ↓ Intestinal motility.

Or any physiological factor, that cause absorption not to start immediately



- observed by presence of 2 point for Y – axis intercept

A: intercept of residual curve.

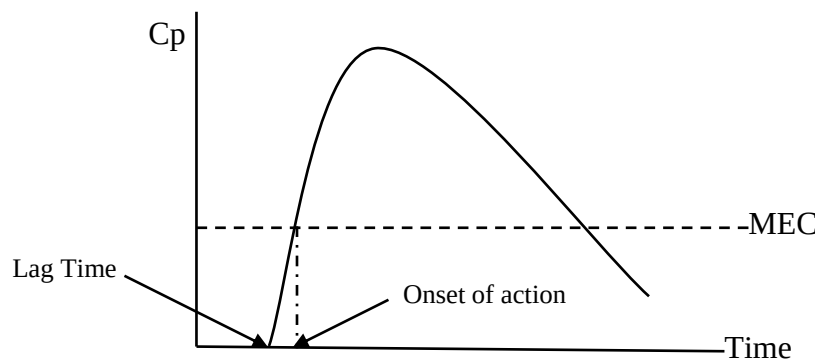
B: intercept of elimination line.

∴ Time t₀: beginning of drug absorption.

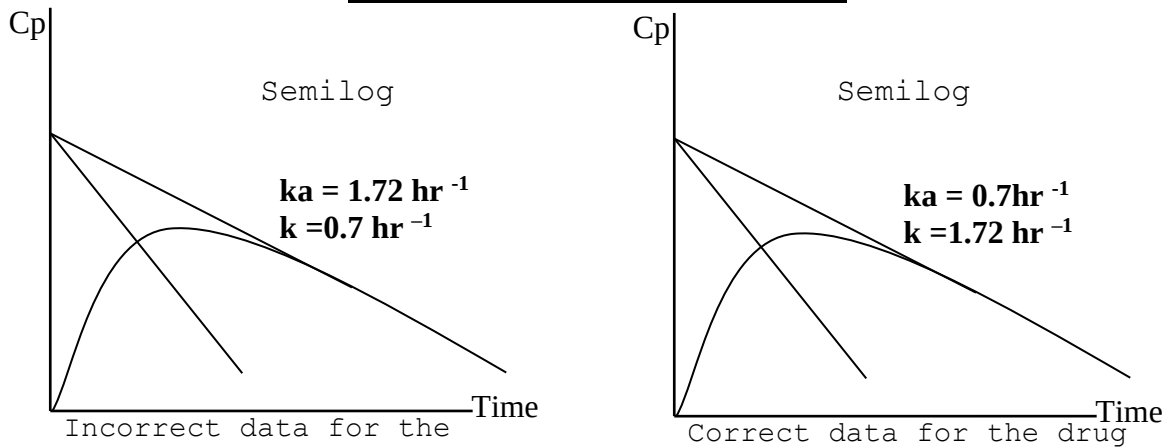
Description of the curve in case of presence of lag time

$$C_p = A [e^{-K(t-t_0)} - e^{-K_a(t-t_0)}]$$

Lag time is not the onset time of pharmacologic activity



Flip – flop of k_a and k



Definition: interchange (reversal) of k_a and k after 1st order absorption e.g. **isoproterenol**

Or it is the phenomena observed when rate of elimination is faster than rate of absorption or $k > k_a$

I.V $k = 1.72 \text{ hr}^{-1}$

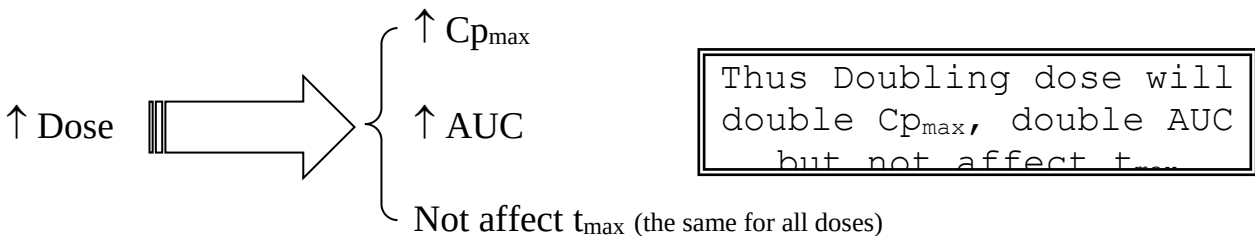
(Feathering) oral $k = 0.7 \text{ hr}^{-1}$ $k_a = 1.72 \text{ hr}^{-1}$ (flip – flop).

Situations where flip-flop situation may be observed

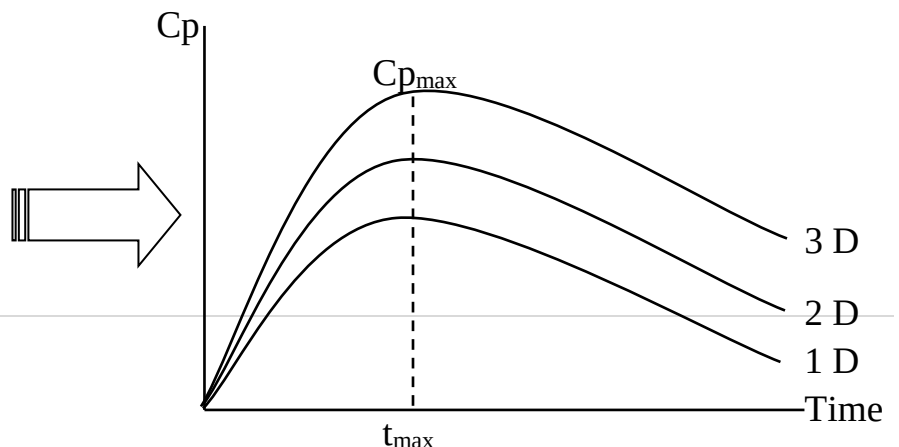
- 1) Drugs which are rapidly eliminated having $k > 0.69 \text{ hr}^{-1}$ have high chance of flip -flop.
- 2) SR oral dosage forms slow absorption process thus may cause k to be greater than k_a thus results in flip-flop phenomena

- So to be sure of the estimate after oral administration, compare k calculated from oral administration with k from I.V data. → I.V data is more reliable.

Effect of Dose, k_a and k on $C_{p_{\max}}$, $t_{\max}$ and AUC

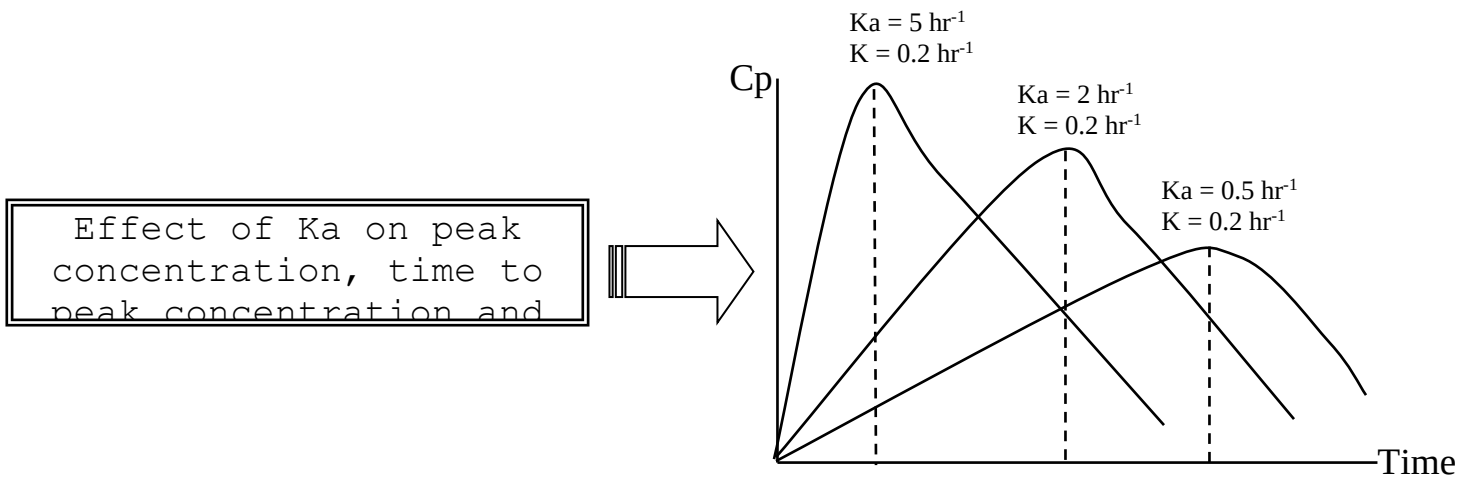
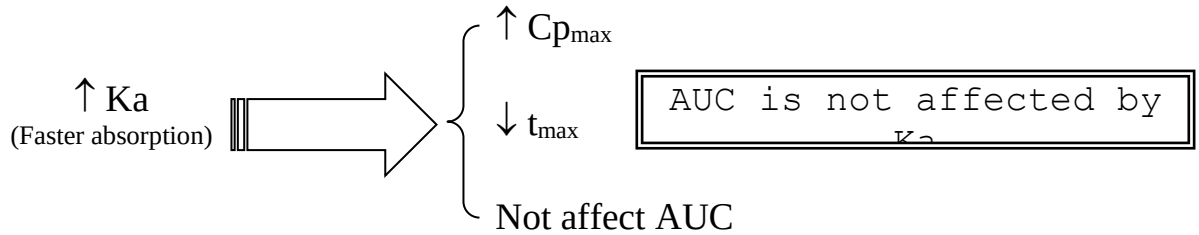


Effect of dose on peak concentration, time to peak concentration and

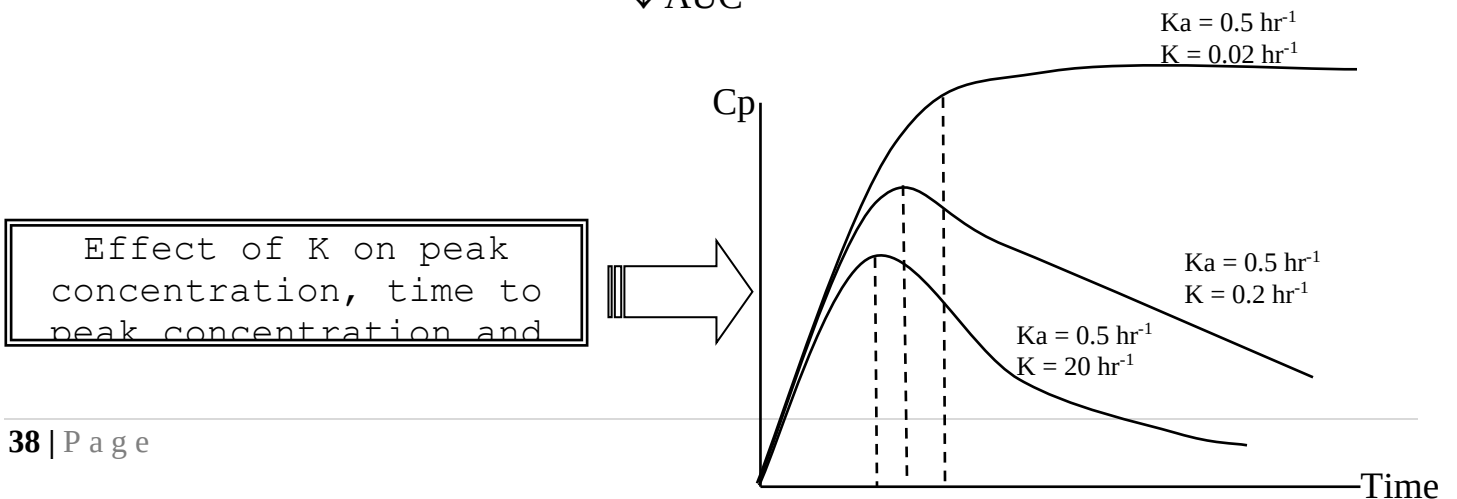
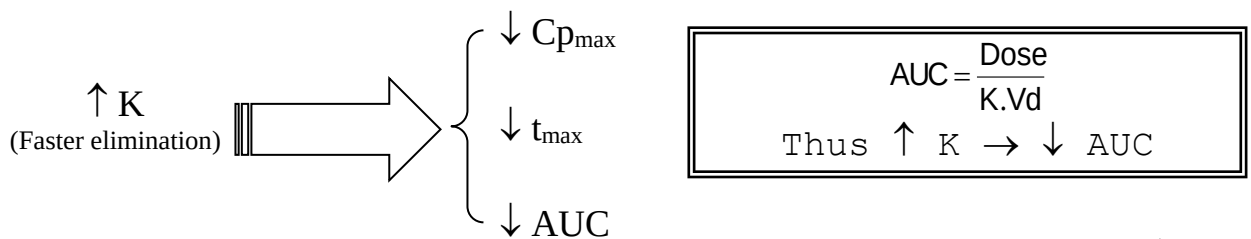


$t_{\max}$ Time to peak concentration depend on K_a and K only

1) Effect of K_a on $t_{\max}$ where K is kept constant (unchanged)

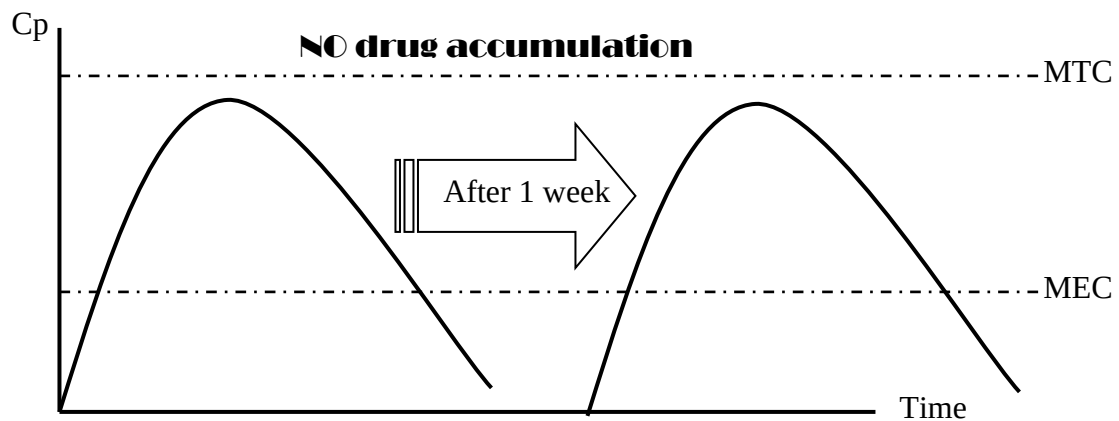
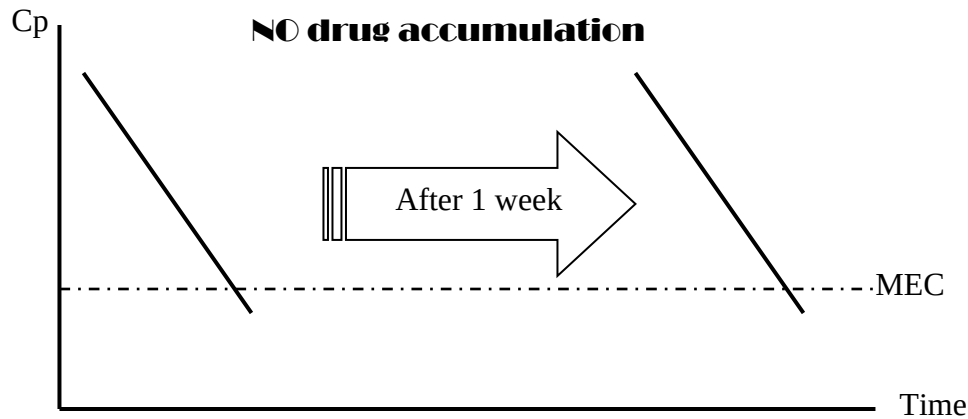


2) Effect of K on $t_{\max}$ where K_a is kept constant (unchanged)

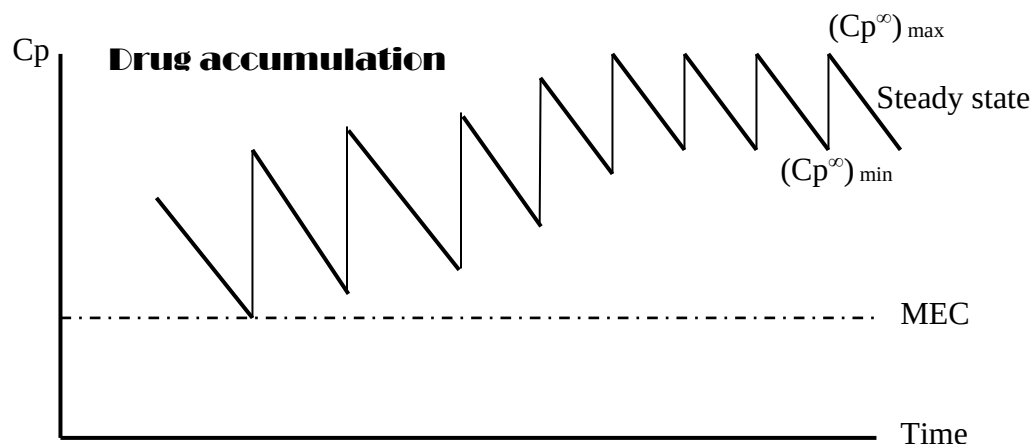


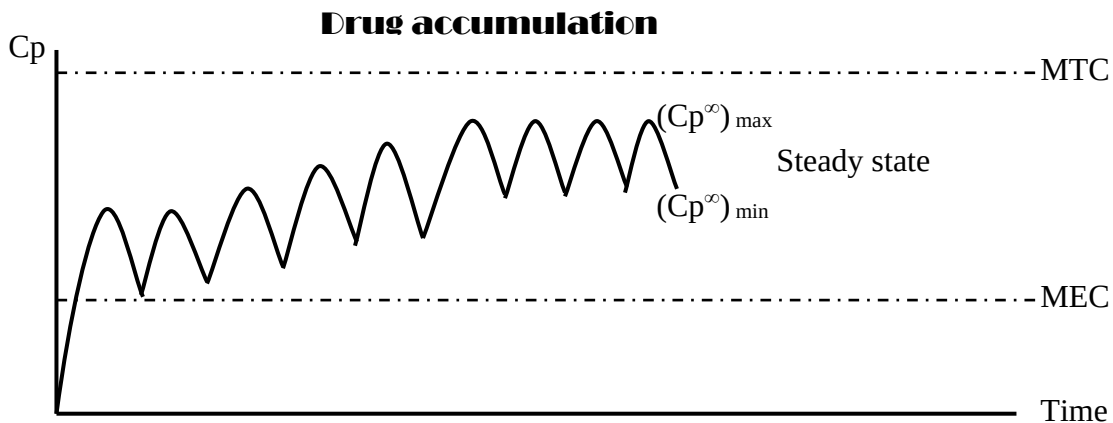
Multiple dosage regimens

After a single dose, plasma drug concentration reaches a maximum level then decline below the MEC of the drug; if a second dose is given after a long time no drug accumulation will occur



For drugs e.g. antibiotics and antirheumatics, dose is given at constant dosage interval leading to drug accumulation inside the body and drug concentration reaches a steady state as follow





- Plasma drug level at steady state resulting from multiple dosage regimen must be adjusted to be within the therapeutic index of the drug (effective and safe range)

► **Adjustment of dosage regimen depend on 2 parameters**

- 1) Size of dose 2) Dosage interval τ (tau)
- At steady state plasma drug level fluctuate between $C_{p\max}^{\infty}$ and $C_{p\min}^{\infty}$
- $C_{p\max}^{\infty}$ Should be always below the MTC (minimum toxic concentration) of the drug to be safe and it is an important indication of drug accumulation (accumulation index R)

$$R = \frac{C_{\max}^{\infty} \text{ (at steady state)}}{(C_{n=1})_{\max} \text{ (after the first dose)}}$$

$R = 1 \Rightarrow$ No accumulation

$$R > 1 \Rightarrow \text{accumulation occur}$$

$$R = \frac{\frac{D^0}{Vd}}{(1 - e^{-k\tau})} \cdot \frac{1}{\frac{D^0}{Vd} \text{ (after the first dose)}} \Rightarrow R = \frac{1}{(1 - e^{-k\tau})}$$

Where $C_p^0 = D^0/V_d$

- So drug accumulation with R-index depends on elimination rate constant K , dosage interval τ and independent on the dose

Accumulation half-life

- The time required to reach one half (50%) of the steady state plasma level

$$\text{Accumulation on } t_{1/2} = t_{1/2} \left[1 + 3.3 \log \frac{K_a}{K_a - K} \right]$$

- ▶▶ In case of IV injection; absorption is very rapid and $K_a \cong \infty$ and K is very small compared to K_a thus K can be removed from the equation Accumulation $t_{1/2} = t_{1/2} [1 + 3.3 \log \frac{K_a}{K}]$
- ▶▶ So accumulation $t_{1/2} =$ elimination $t_{1/2}$ for IV injection
- ▶▶ i.e. the time required to reach 50% of steady state drug level (accumulation $t_{1/2}$) depends only on the elimination $t_{1/2}$ and independent on the dose or the dosage interval
- ▶▶ Time required to reach 95% of steady state is $5 t_{1/2}$ and **time required to reach 99% of steady state is 6.6 (≈ 7) $t_{1/2}$** and time required to reach 99.9% of steady state is $9.96 (\approx 10) t_{1/2}$
- ▶▶ If the drug is given at a dosage interval equal to its $t_{1/2}$ so the number of doses required to reach steady state is 6.6 doses (≈ 7 doses)

Amount of drug remaining in the body at steady state

- ▶▶ After single dosage $D_B = D_B^0 e^{-kt}$ and after multiple dosage $D_B = D_B^0 e^{-k\tau}$
- ▶▶ **Fraction of drug remaining in the body** $f = \frac{D_B}{D^0} = e^{-k\tau}$ (for any given dose; f depend on K and τ only). If τ is large, f will be small as D_B is smaller with time
- ▶▶ The injected dose D^0 can be given from $D^0 = D_{\max}^{\infty} - D_{\min}^{\infty}$

Where $D_{\max}^{\infty}$: is maximum amount of drug in the body at steady state $\Rightarrow D_{\max}^{\infty} = \frac{D^0}{(1-f)}$

And $D_{\min}^{\infty}$: is minimum amount of drug in the body at steady state

- ▶▶ The average amount of the drug in the body at steady state D_{ave}^{∞} is calculated from

$$D_{\text{ave}}^{\infty} = \frac{F \cdot D^0}{k \cdot \tau} \text{ Or } D_{\text{ave}}^{\infty} = \frac{F \cdot D^0}{0.693 \cdot t_{1/2} \cdot \tau} \text{ or } D_{\text{ave}}^{\infty} = \frac{F \cdot D^0 \times 1.44 t_{1/2}}{\tau}$$

- ▶▶ D_{ave}^{∞} is not the arithmetic mean of $D_{\max}^{\infty}$ and $D_{\min}^{\infty}$ because the drug in the body declines exponentially according to 1st order elimination kinetics

Concentration of drug in the body at steady state

- ▶▶ Maximum conc at steady state $C_{p_{\max}}^{\infty} = \frac{D_{\max}^{\infty}}{V_d}$ and $C_{p_{\min}}^{\infty} = \frac{C_{p^0}}{1 - e^{-k\tau}}$
- ▶▶ Minimum conc at steady state $C_{p_{\min}}^{\infty} = \frac{D_{\min}^{\infty}}{V_d}$ and $C_{p_{\min}}^{\infty} = \frac{C_{p^0} e^{-k\tau}}{1 - e^{-k\tau}}$ and $C_{ss\min} = C_{ss\max} \cdot e^{-k\tau}$

► Average conc at steady state $C_{p_{ave}}^{\infty} = \frac{D_{ave}^{\infty}}{V_d}$ and $C_{p_{ave}}^{\infty} = \frac{F \cdot D^0}{V_d \cdot k \cdot \tau}$

► $C_{p_{ave}}^{\infty}$ is not the arithmetic mean of $C_{p_{max}}^{\infty}$ and $C_{p_{min}}^{\infty}$ because the drug in the body declines exponentially according to 1st order elimination kinetics

Example 19

A patient receives 1000 mg every 6 hours by repetitive IV injection of an antibiotic with elimination half-life of 3 hours. Assume the drug is distributed according to one-compartment model and the volume of distribution is 20 L

- Find the maximum, minimum and average amount of drug in the body at steady state
- Determine the maximum, minimum and average plasma concentration of the drug at steady state

Example 20: The pharmacokinetic data of clindamycin were reported by DeHann and co-workers (1972) as follows: $k = 0.247 \text{ hr}^{-1}$ $t_{1/2} = 2.81 \text{ hr}$ $V_d = 43.9 \text{ L}/1.73\text{m}^2$
What is the steady-state concentration of the drug after 150 mg of the drug have been given orally every 6 hours for a week (assume the drug is 100% absorbed)?

Example 21: The elimination half-life of tobramycin was reported by Regamey and associates (1973) to be 2.15 hr; the volume of distribution was reported to be 33.5% of body weight.

- What is the dose for an 80-kg individual if a steady - state level of 2.5 µg/ml is desired? Assume that the drug is given intravenously every 8 hours.
- The manufacturer has suggested that in normal cases tobramycin should be given at a rate of 1 mg/kg every 8 hrs. With this dosage regimen, what the average steady-state level be?

NON-COMPARTMENTAL PHARMACOKINETICS

- » Pharmacokinetic parameters are obtained without fitting the data to any specific compartmental model
- » The data follow linear pharmacokinetics

Statistical moment

- » A mathematical description of **probability density function (PDF)** which describes the true relationship between concentration and time
- » **PDF**: represents the probability of concentration or residence time in the population i.e. it is the true mean of the entire population

Mean residence time (MRT): the average total time molecules of a given drug dose will reside or spend in the body, measured for **instantaneous administration** (IV route only)

Mean transit time (MTT): the average time molecules of a given drug dose will reside or spend in the kinetic system, measured for **non-instantaneous administration** (routes other than IV route e.g. oral route)

Mean absorption time (MAT): the average time for drug molecules spend in site of absorption to enter a kinetic space e.g. blood

; Mean Arrival Time = Mean Absorption Time if the drug is 100 % absorbed but they will be different if some of drug is degraded or lost in GIT (unabsorbed)
; The average time drug molecule spend at site of absorption may differ from average arrival time into systemic circulation e.g. if the drug molecule overcome its absorption site in duodenum and passed down the small intestine
; If $F < 100\% \Rightarrow$ Mean arrival time will be shorter than mean absorption time

$$\text{» } \mathbf{MAT = MRT_{oral (OR m)} - MRT_{iv} \text{ OR } MAT = MTT - MRT_{iv}}$$

$$\text{» } \mathbf{MAT = \frac{1}{K_a}} \text{ Where } K_a \text{ is rate constant of first order absorption}$$

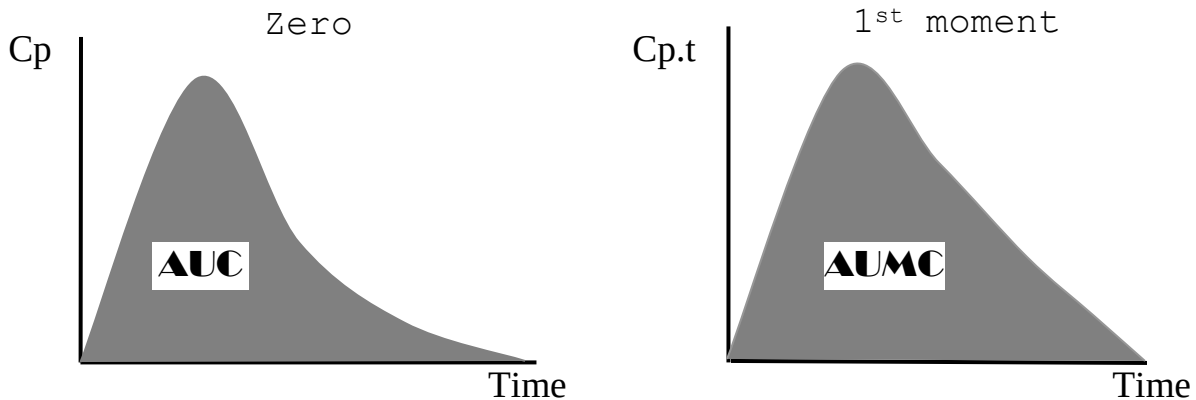
$$\text{Absorption } t_{\frac{1}{2}} = \frac{0.693}{K_a} = 0.693 \text{ MAT}$$

$$\text{» } \mathbf{MRT_{iv} = \frac{1}{K}} \text{ Where } K \text{ is rate constant of first order elimination}$$

$$\text{Elimination } t_{\frac{1}{2}} = \frac{0.693}{K} = 0.693 \text{ MRT}_{iv}$$

Pharmacokinetic moment (PK Moment)

- ▶▶ Product of concentration of drug and time, will give the 1st PK moment = $C_p \times t$ (versus time)
- ▶▶ **AUMC** Area under 1st PK moment curve ($C_p \times t$ versus time) is called 1st PK moment
- ▶▶ **AUC** Area under the curve (C_p versus time) is called zero PK moment



Mean Residence Time (MRT)

- ▶▶ Independent on route administration
- ▶▶ Describes the average time all drug molecules will reside in the body
- ▶▶ It is statistical moment analogous to & directly proportional to elimination $t_{1/2}$
- ▶▶ For IV route MRT reflects the elimination rate process ($\downarrow$ MRT $\rightarrow$ rapid elimination and vice versa)

$$\text{For IV route } \text{MRT} = \frac{(AUMC)_0^\infty}{(AUC)_0^\infty}$$

$$(AUC)_0^\infty = (AUC)_0^{t_n} + (AUC)_{t_n}^\infty \Rightarrow (AUC)_0^\infty = (AUC)_0^{t_n} + \frac{C_{p_n}}{\lambda_z}$$

Where λ_z ($= k$): is $2.303 \times$ slope of the terminal straight exponential part of C_p versus time

$(AUC)_0^{t_n}$: can be calculated by trapezoidal rule

$$(AUMC)_0^\infty = (AUMC)_0^{t_n} + (AUMC)_{t_n}^\infty \Rightarrow (AUMC)_0^\infty = (AUMC)_0^{t_n} + \frac{t_n \cdot C_{p_n}}{\lambda_z} + \frac{C_{p_n}}{\lambda_z^2}$$

$(AUMC)_0^{t_n}$: can be calculated by trapezoidal rule

Volume of distribution at steady state

$$V_{dss} = \frac{Cl}{K} = Cl \cdot \text{MRT} = \frac{D_{iv}^0}{(AUC)} \cdot \frac{AUMC}{(AUC)} = \frac{D_{iv}^0 \cdot (AUMC)}{(AUC)^2}$$

Example

A dose of 300 mg of a drug was administered to a subject via an IV bolus. Blood samples were obtained and the following concentrations obtained at the following times

Time (hr)	Concentration (mg/L)
0.25	8.21
0.5	7.87
1.0	7.23
3.0	5.15
6.0	3.09
12.0	1.11
18.0	0.4

Using non-compartmental methods, calculate the Cl, Vdss and MRT

Answer

Plot Cp versus time on semilog paper

From graph determine $C_p^0 \Rightarrow 8.57 \text{ mg/L}$ and $K (= \lambda_z) = 0.173 \text{ hr}^{-1}$

Calculate $(AUC)_0^\infty$ and $(AUMC)_0^\infty$

Time (hr)	Concentration (mg/L)	AUC (mg.hr/L)	t.Cp (mg/L).hr	AUMC (mg.hr ² /L)
0	8.57	-	0	-
0.25	8.21	2.10	2.05	0.26
0.5	7.87	2.01	3.94	0.75
1.0	7.23	3.78	7.23	2.79
3.0	5.15	12.38	15.45	22.68
6.0	3.09	12.36	18.54	50.99
12.0	1.11	12.6	13.32	95.58
18.0	0.4	4.53	7.2	61.56
Total		49.76		234.61

$$(AUC)_0^\infty = (AUC)_0^{18} + (AUC)_{18}^\infty \Rightarrow (AUC)_0^\infty = (AUC)_0^{t_n} + \frac{C_{p_n}}{\lambda_z} \Rightarrow$$

$$(AUC)_0^\infty = 49.76 + \frac{0.4}{0.173} = 52.07 \text{ mg.hr / L}$$

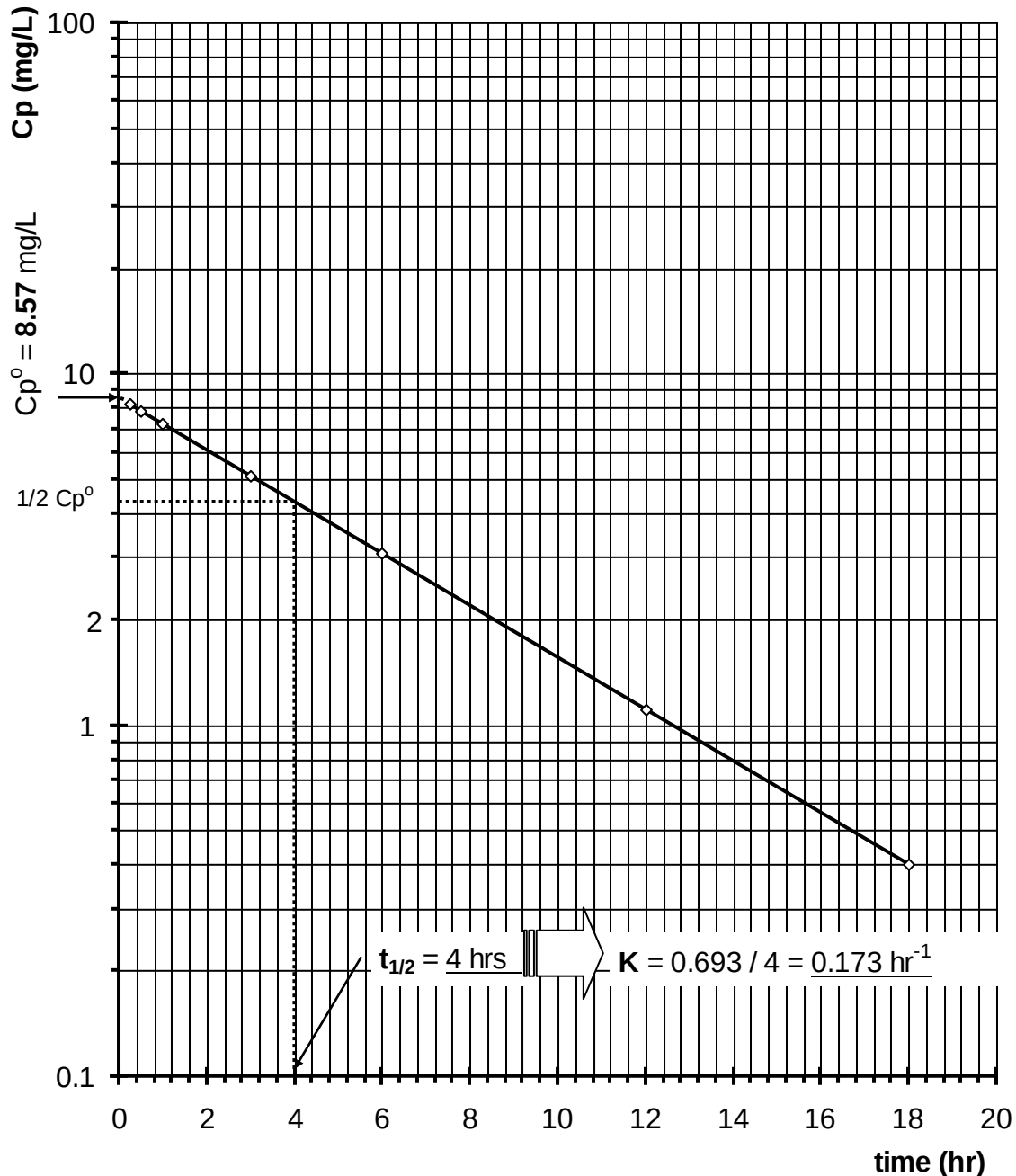
$$(AUMC)_0^\infty = (AUMC)_0^{t_n} + (AUMC)_{t_n}^\infty \Rightarrow (AUMC)_0^\infty = (AUMC)_0^{t_n} + \frac{t.C_{p_n}}{\lambda_z} + \frac{C_{p_n}}{\lambda_z^2}$$

$$\Rightarrow (AUMC)_0^\infty = 234.61 + \frac{18 \times 0.4}{0.173} + \frac{0.4}{(0.173)^2} = 289.59 \text{ mg.hr}^2 / \text{L}$$

$$MRT = \frac{(AUMC)_0^\infty}{(AUC)_0^\infty} \Rightarrow MRT = \frac{289.59 \text{ mg.hr}^2 / \text{L}}{52.07 \text{ mg.hr} / \text{L}} = 5.56 \text{ hrs}$$

$$V_{dss} = \frac{D_{iv}^\circ \cdot (AUMC)_0^\infty}{([AUC]_0^\infty)^2} = \frac{300 \text{ mg} \times 289.59 (\text{mg.hr}^2 / \text{L})}{(52.07 \text{ mg.hr} / \text{L})^2} = 32 \text{ L}$$

$$V_{dss} = \frac{Cl}{K} = Cl \cdot MRT \Rightarrow Cl = \frac{V_{dss}}{MRT} = \frac{32 \text{ L}}{5.56 \text{ hrs}} = 5.76 \text{ L/hr}$$



Advantages of Non-compartmental analysis model

- 1) Need fewer assumptions than compartmental analysis
- 2) Avoid common problem seen with compartmental analysis e.g. procainamide was given IV into 10 subjects after pharmacokinetic analysis; kinetics of the drug in some subjects is best described by 2-compartment model while in other subjects it is best described by 3-compartment model!!

Disadvantages of Non-compartmental analysis model

- 1) No determination of number of compartments to describe **drug disposition**
- 2) Can not show (elucidate) organ specific elimination
- 3) Can not relate derived (statistical moment parameters) to physiological parameters
- 4) Can not show drug – drug or drug – nutrient interaction
- 5) Sensitive to sampling frequency

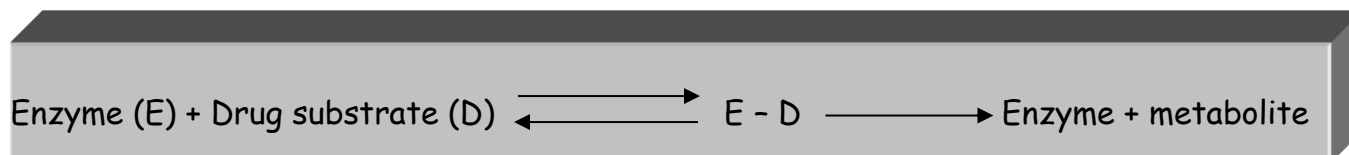
NON-LINEAR PHARMACOKINETICS

Linear pharmacokinetics	Non linear pharmacokinetics
when dose of drug is increased or decreased two fold; the concentration of the drug will either increase or decrease two fold i.e. change in concentration of drug can be expected	concentration of drug in plasma either more or less than expected from change in dose range
Pharmacokinetic parameters e.g. K, Vd and $t_{1/2}$ are independent on dose of the drug (dose – independent kinetics)	Pharmacokinetic parameters e.g. K, Vd and $t_{1/2}$ are dependent on dose of the drug (dose – dependent kinetics)

- » The main cause of non linear pharmacokinetics is that the drug is either metabolized or eliminated by a **saturable enzyme system** inside the body

Drug metabolism

- » Occur through cytochrome P-450 and N-acetyltransferase systems

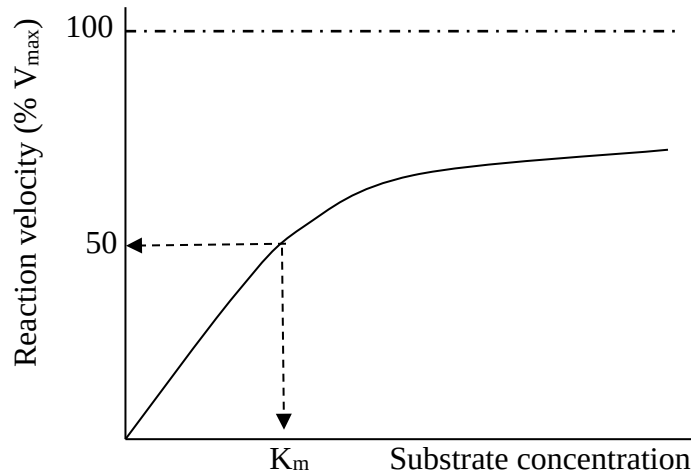


- » Velocity (rate) of enzyme induced reactions is given by **Michaelis-Menten equation**

$$V = -\frac{dC}{dt} = \frac{V_{\max} \cdot C}{K_m + C}$$

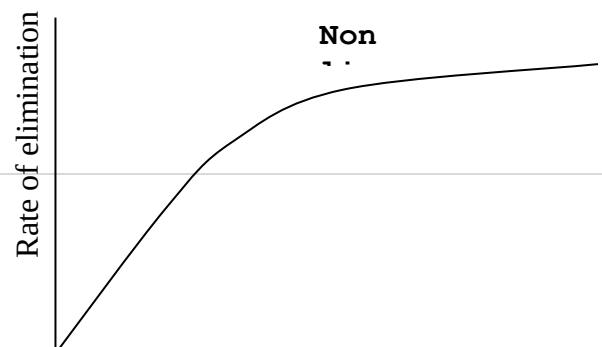
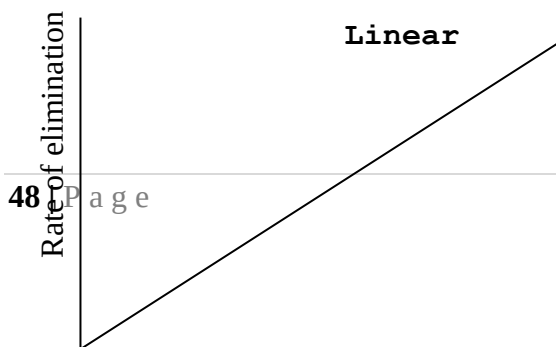
Where **V**: velocity (rate) of the reaction (process)

$V_{\max}$: maximum velocity (rate) of the reaction (process) at very high substrate concentration
 K_m : Michaelis-Menten constant, reflects **affinity** of the drug substrate to enzyme and **it is the concentration of drug at half the maximum velocity** (i.e. when $V = V_{\max} \Rightarrow K_m = C$)



Effect of drug concentration on rate of the reaction

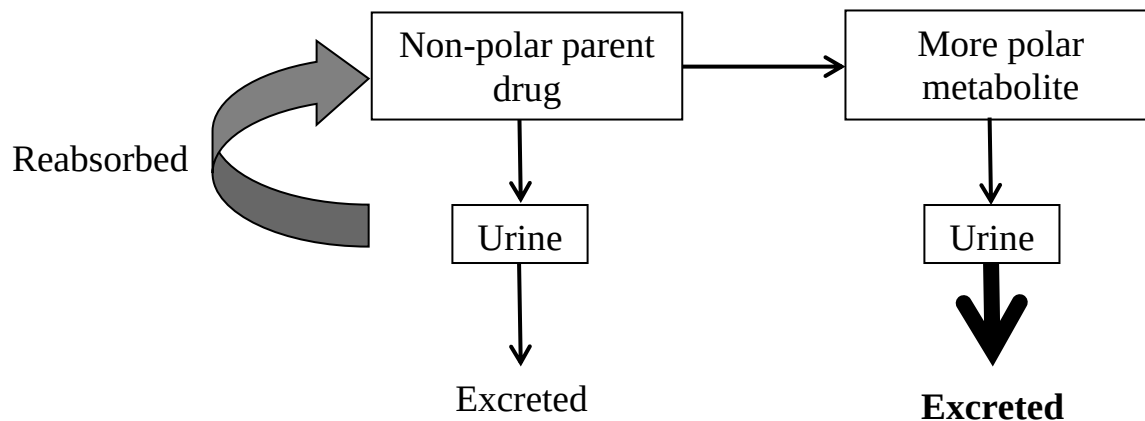
At low concentration	At high concentration
$K_m \gg C \Rightarrow K_m + C \approx K_m$ So $V = -\frac{dC}{dt} = \frac{V_{\max}}{K_m} \cdot C$ and since $\frac{V_{\max}}{K_m}$ is constant $\Rightarrow$ 1st order kinetics	$C \gg K_m \Rightarrow K_m + C \approx C$ So $V = -\frac{dC}{dt} = \frac{V_{\max} \cdot C}{C} = V_{\max}$ and since $V_{\max}$ is constant $\Rightarrow$ zero order kinetics
Maximal rate reaction is not yet reached and rate of reaction will increase till saturation of the enzyme	Maximal rate reaction is reached and can not be exceeded (due to saturation of enzyme)
► Non linear pharmacokinetics is important when high doses of the drug is administered ► At saturation, drug accumulation occur which can not be predicted following a single dose but require multiple drug dosing ► Drugs of non linear pharmacokinetics e.g. phenytoin require dose adjustment and in-vivo bioequivalence studies	



METABOLITE PHARMACOKINETICS

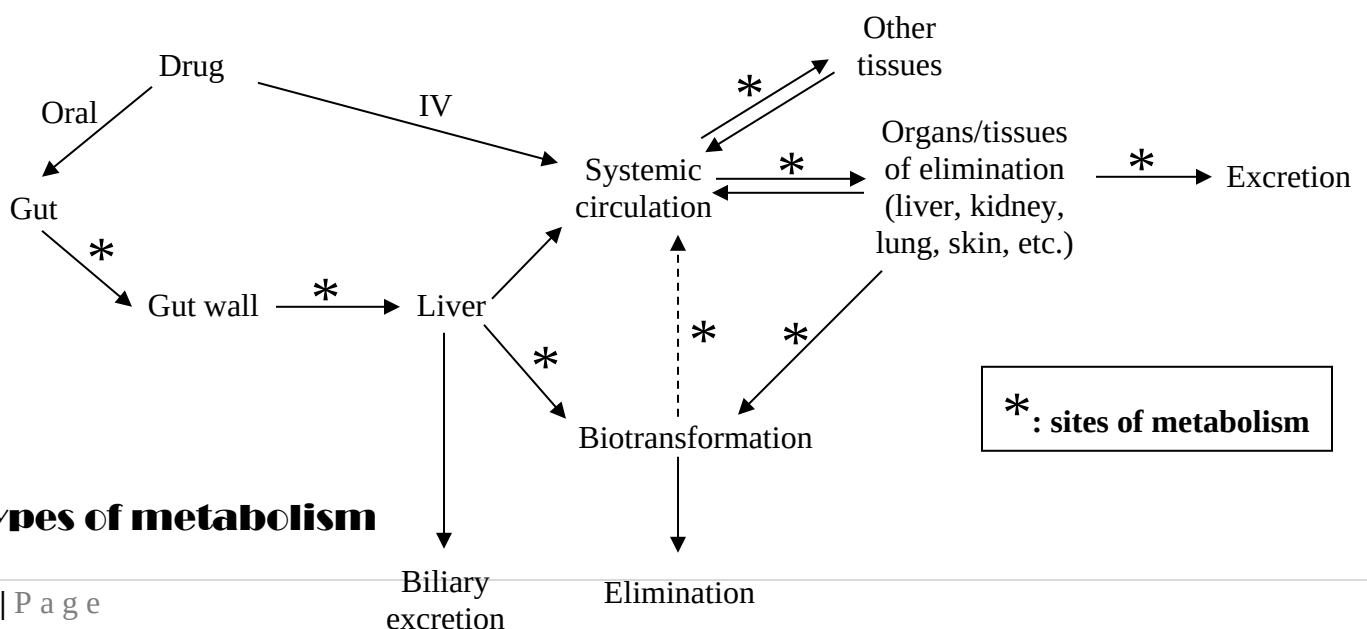
Metabolism (Biotransformation)

Chemical process by which the body **converts** drug often into more polar (water soluble) chemical species (metabolite) thus reduce its volume of distribution (**i.e. become in central compartment**) and enhances excretion and it is also the process by which the body **reduce the activity of the parent drug** (detoxification)



Sites of metabolism

- It can occur in many organs e.g. GI lumen, GI membrane, liver, blood, lungs, skin, kidney, brain, placenta and many other organs
- It is 1st order process



Types of metabolism

Phase I

- ▶▶ Adds polar groups to drug through oxidation/reduction reactions
- ▶▶ Require reducing agent e.g. NADPH, molecular oxygen and microsomal enzyme system
- ▶▶ The terminal oxidizing enzyme is cytochrome P450 (it is so called as its carbon monoxide derivative absorbs UV light at 450 nm)
- ▶▶ Cytochrome P450 system is formed of many isoenzymes which differ in their specificities for drug substrates
- ▶▶ There is more than 70 different P450 genes in various species

Phase II

Conjugation with endogenous compounds to form more polar drug-conjugate

Reasons for concern of metabolite kinetics (why we study metabolite kinetics?)

1) Metabolite can be pharmacologically active e.g.

Imipramine to **desipramine**, Phenacetin to **acetaminophen**, Terfenadine to **fexofenadine** and procainamide to **N-acetyl procainamide**

2) Metabolite can be toxic e.g.

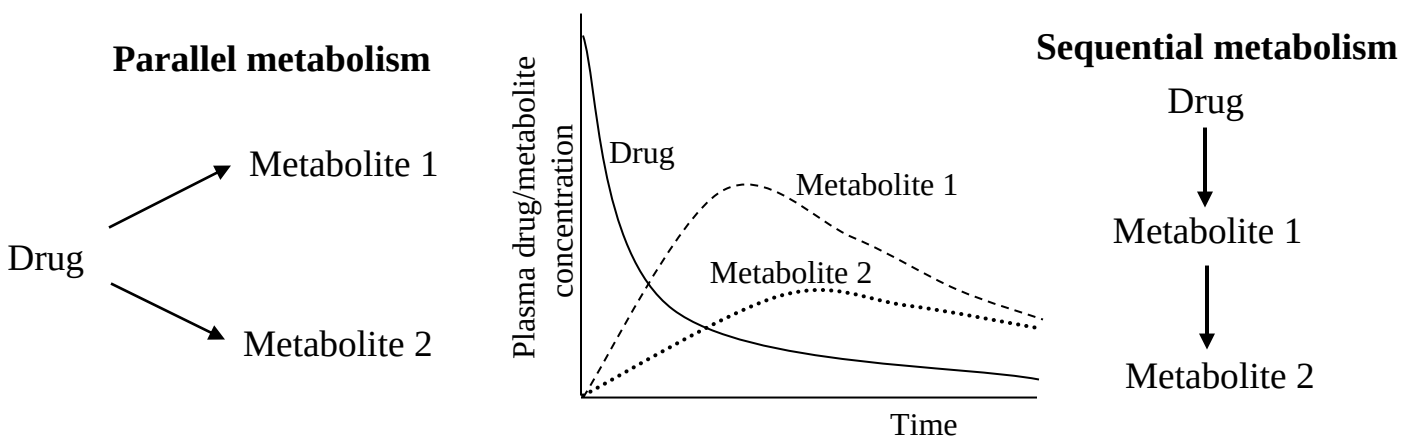
Fluoxetine to **norfluoxetine**, propranolol to **4-OH propranolol**, verapamil to **norverapamil**, diazepam to **desmethyldiazepam** and **toxic metabolites of acetaminophen**

3) Metabolites can interact with the parent drug (and with other drugs) and alter their disposition e.g. it may act as enzyme inducer or inhibitor for P 450 system

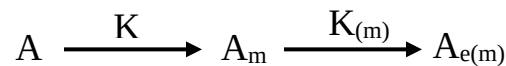
4) Metabolism may vary due to intrinsic genetic factors (e.g. black men are faster acetylators and metabolizers than white men) and extrinsic factors (e.g. smoking and diet may affect metabolism of theophylline)

- ▶▶ Metabolite formation depends on parent drug but it is a separate entity from the parent
- ▶▶ Metabolites have their own pharmacokinetic parameters e.g. Vd and Cl

Metabolite kinetics following an IV single dose of parent compound



Simple model for metabolite kinetics



Where

A: amount of parent drug in the body

K: 1st order rate constant for drug elimination (1st order rate constant for **metabolite formation**)

A_m: amount of metabolite in the body

K_(m): 1st order rate constant for **elimination of metabolite**

A_{e(m)}: amount of metabolite eliminated from the body

Amount of metabolite in plasma at any time (A_m) can be calculated from

$$A_m = \frac{f_m \cdot K \cdot D}{K_m - K} (e^{-Kt} - e^{-K_m \cdot t})$$

Concentration of metabolite in plasma at any time (C_{p(m)}) can be calculated from

$$C_{p(m)} = \frac{A_m}{Vd_{(m)}} \Rightarrow C_{p(m)} = \frac{f_m \cdot K \cdot D}{Vd_{(m)} \cdot (K_m - K)} (e^{-Kt} - e^{-K_m \cdot t})$$

Where D: IV dose of parent drug and f_m: fraction of drug metabolized

$T_{\max(m)}$: time required to reach maximum concentration of metabolite in plasma can be calculated from

$$t_{\max(m)} = \frac{\ln \frac{K_m}{K}}{(K_m - K)} = \frac{2.303 \log \frac{K_m}{K}}{(K_m - K)}$$

At $t_{\max(m)}$: rate of metabolite formation = rate of metabolite elimination i.e. $\frac{dC_{p(m)}}{dt} = \text{zero}$

Maximum concentration of metabolite in plasma can be calculated from

$$C_{p \max(m)} = \frac{f_m \cdot K \cdot D}{V_{d(m)} \cdot (K_m - K)} (e^{-K t_{\max}} - e^{-K_m \cdot t_{\max}})$$

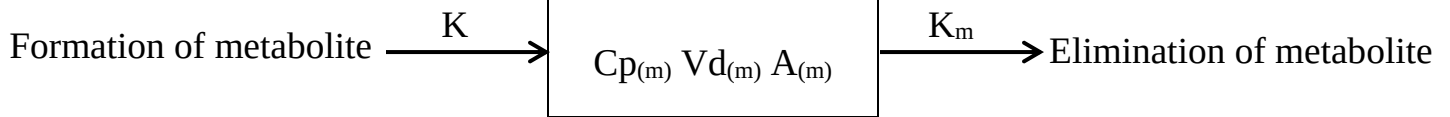
Area under metabolite plasma concentration versus time curve $(AUC)_m$ can be determined from

$$(AUC)_m = \frac{f_m \cdot D}{Cl_m} \text{ but } IV \text{dose} = Cl_T \cdot (AUC) \Rightarrow (AUC)_m = \frac{f_m \cdot Cl_T \cdot (AUC)}{Cl_m}$$

$$\frac{(AUC)_m}{(AUC)} = \frac{f_m \cdot Cl_T}{Cl_m} \Rightarrow f_m = \frac{(AUC)_m}{(AUC)} \cdot \frac{Cl_m}{Cl_T}$$

Volume of distribution of metabolite $[V_{d(m)}]$ can be calculated from

$$V_{d(m)} = \frac{Cl_m}{K_m}$$

One compartment open model following single IV dose for metabolite

Rate limiting step for metabolite kinetics in blood

Elimination – rate limited

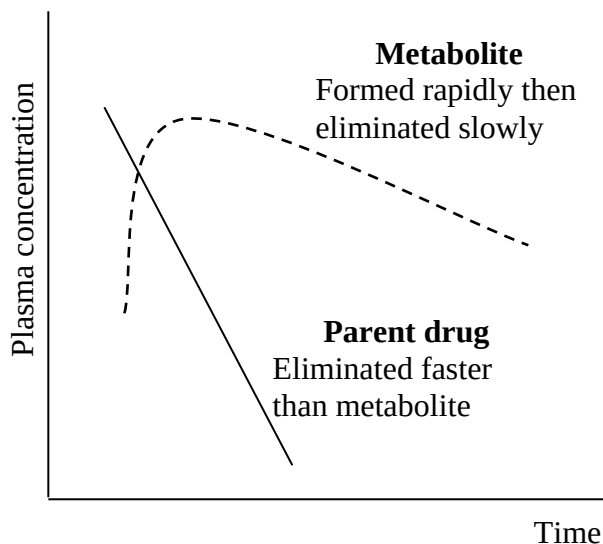
- ▶ Elimination of metabolite is slower than its formation i.e. $K_m < K$ and $T_{1/2}$ of parent drug $< T_{1/2}$ of metabolite
- ▶ So from the equation

$$C_{p_m} = \frac{f_m \cdot K \cdot D}{V_d \cdot (K_m - K)} (e^{-Kt} - e^{-K_m \cdot t})$$

$e^{-Kt} = 1/e^{Kt}$ will approach zero at higher time faster than $e^{-K(m).t}$ and the equation will be

$$C_{p_m} = \frac{f_m \cdot K \cdot D}{V_d \cdot (K_m - K)} e^{-K_m \cdot t}$$

- ▶ **$T_{1/2}$ of metabolite can be calculated from $C_{p(m)}$ versus time curve**
- ▶ Thus decline in $C_{p(m)}$ depends on elimination of metabolite

**Formation – rate limited**

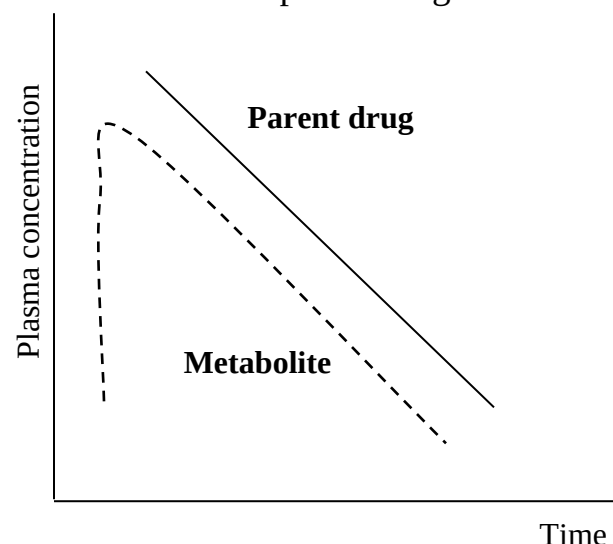
- ▶ Formation of metabolite is slower than its elimination i.e. $K < K_m$ and $T_{1/2}$ of metabolite $< T_{1/2}$ of parent drug
- ▶ So from the equation

$$C_{p_m} = \frac{f_m \cdot K \cdot D}{V_d \cdot (K_m - K)} (e^{-Kt} - e^{-K_m \cdot t})$$

$e^{-K(m).t} = 1/e^{K(m).t}$ will approach zero at higher time faster than $e^{-K.t}$ and the equation will be

$$C_{p_m} = \frac{f_m \cdot K \cdot D}{V_d \cdot (K_m - K)} e^{-K \cdot t}$$

- ▶ **$T_{1/2}$ of metabolite can not be calculated from $C_{p(m)}$ versus time curve as the slope does not reflect rate of metabolite elimination but rate of its elimination and rate of its formation as well and $C_{p(m)}$ depends on K of parent drug not the metabolite**
- ▶ Thus decline in $C_{p(m)}$ will be parallel to the decline of the parent drug



Biopharmaceutics

Drug absorption: drug transfer from site of administration to site of action (blood)

Oral absorption occur in steps

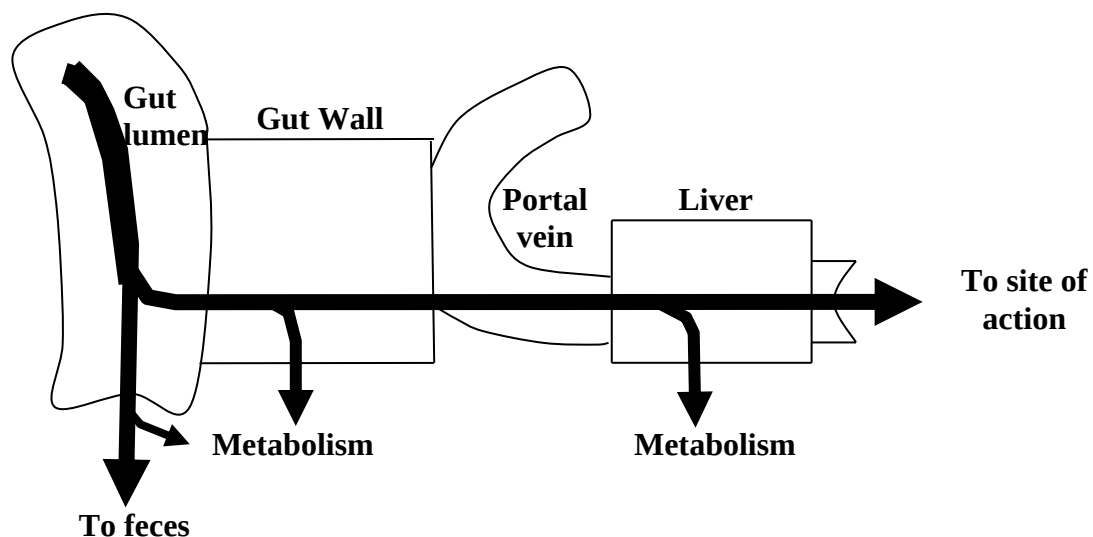
- 1) Crossing the epithelium of GI membrane by transcellular or paracellular pathways and enter blood stream through capillaries
- 2) Passing through hepatoportal system into systemic circulation

Advantages of oral administration

- 1) Convenient, portable, no pain and easy to take e.g. chewable or dispersible tablets for those who can not swallow
- 2) Cheap, no sterilization is required (but must be hygienic), compact, multidose bottles and automated machines for mass production
- 3) Variety fast release, slow release, enteric coated, layered coated tablets, suspensions and mixtures

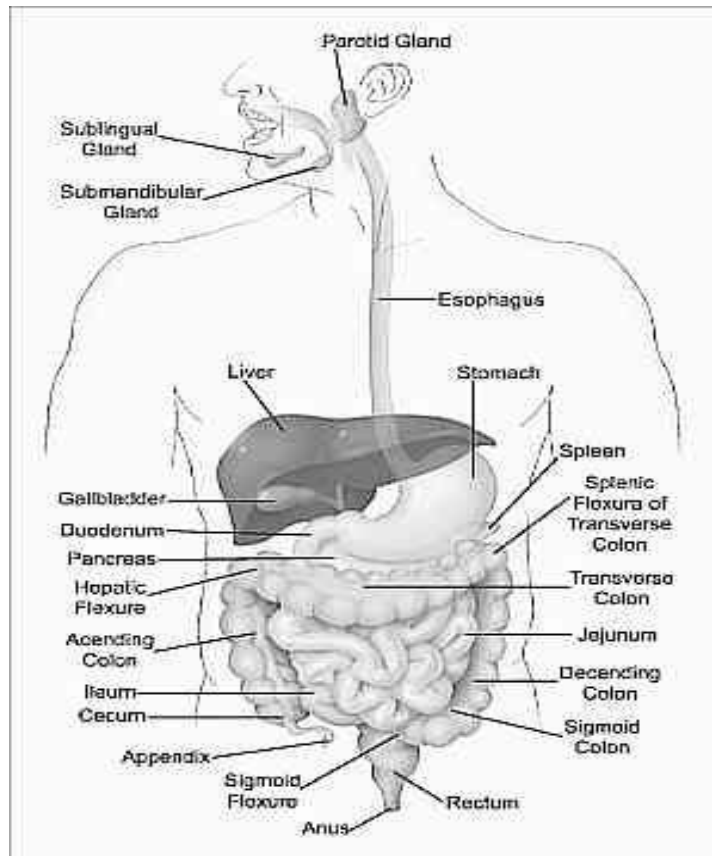
Disadvantages of oral administration

High dose or low solubility of drugs in GI fluids may reduce bioavailability e.g. griseofulvin (it was reformulated in 1970 as micronized powder thus its bioavailability increased and dose was reduced by factor of 2)

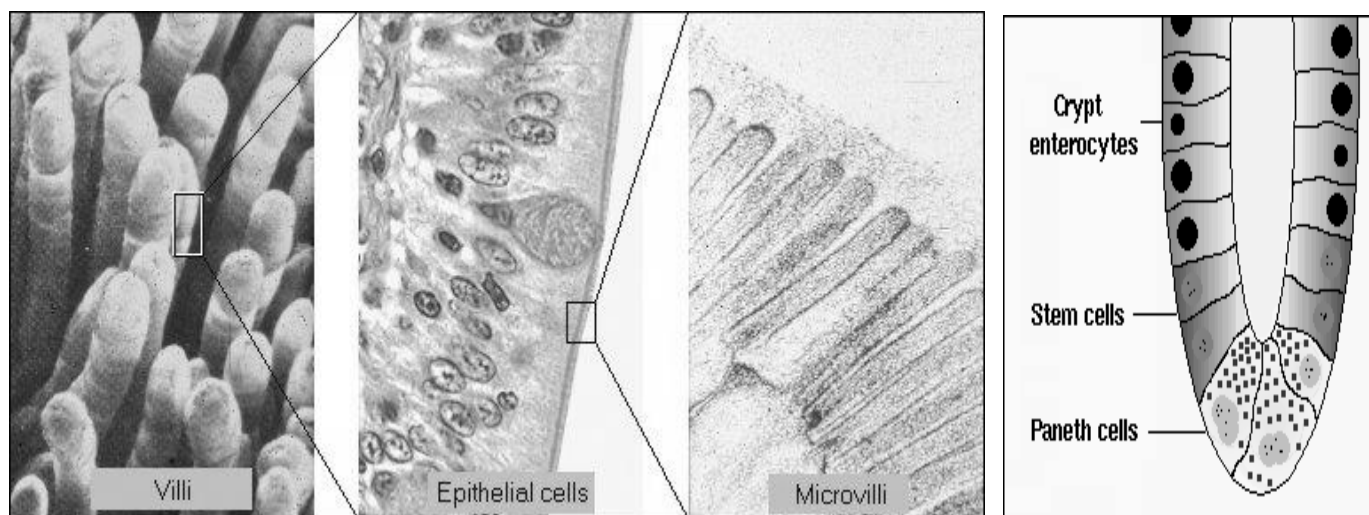


Anatomy and physiology of GIT

- » GIT consists of stomach, small intestine and colon
- » **Small intestine is main site of drug absorption** as its absorption surface area is huge due to its length which may reach 4 – 7 m



- » Small intestine consists of 3 parts
 - 1) Duodenum: 26 cm length, receive secretions from pancreas and liver through pancreatic and bile duct
 - 2) Jejunum: 2.5 m length and represent 40% of small intestine
 - 3) Ileum: 3.5 m length, represent 60% of gut
- » Small intestine is 4 – 5 times longer than large intestine but it is so called as it is thinner
- » **The luminal (cross sectional) area of small intestine is $\frac{1}{2} \text{ m}^2$ but its overall area is about 250 m^2 this is due to**
 - a) Mucosal folds: inner surface is folded (not flat) which aid in mixing of digested food as it acts as baffles
 - b) Villi: projections which protrude into lumen and covered with epithelial cells
 - c) Microvilli: luminal membrane of absorbing epithelia is studded محب with microvilli which resembles a brush thus the intestinal epithelial cells are called the brush border and enzymes embedded in microvilli are called brush border enzyme



►► In addition to the absorptive epithelial cells, gut membrane also contain

- Goblet cells: secretes mucus into intestinal lumen
- Enteroendocrine cells: secretes hormones e.g. Gastrin and cholecystokinin into blood
- Paneth cells: provide defense against microbes in small intestine, they act like **Neutrophils** when exposed to bacteria where they secretes antimicrobial molecules into the gut lumen

- Stem cells: replenish epithelial cells that die

►► Paneth cells are adjacent to stem cells, this suggest that they play a critical role in defending epithelial cell renewal

►► The epithelium of gut is not a monotonous sheet of functionally identical cells i.e. there is a variation in phenotype (appearance) and function of intestinal cells

►► Tight junctions linking epithelial cells vary in their tightness along down the intestine which results of difference in shape of cell and number of transporter molecules (carrier) in the membrane

►► The optimum site for drug absorption after oral administration is the upper portion of the small intestine (due to high perfusion with a network of capillaries that maintain a concentration gradient between gut lumen and blood)

Segment	Function	Surface area	pH	Remarks
Stomach	Digestion of food	3.5 m ²	1 – 3.5	Acidic
Duodenum	Neutralization of acids	2 m ²	4 – 6.5 (Slightly acidic to neutral)	Absorption of most drugs even acidic drugs
Juenum	Absorption of nutrients	180 m ²	5 – 7 (alkaline)	

Ileum	Absorption of nutrients	280 m ²	6 – 8 (alkaline)	
Colon	Absorption of water	1.3 m ²	6 – 8 (alkaline)	Absorption of some drugs

Degradation of the drug may occur due to

- 1) Acid hydrolysis in stomach
- 2) Enzymatic degradation in stomach or intestine
- 3) Metabolism in the brush border of the intestinal wall
- 4) Metabolism by microorganisms in the colon
- 5) Metabolism in liver

Fate of protein drugs after oral administration

- 1) Digestion by **pepsin** (of stomach) and **trypsin, chymotrypsin and carboxypeptidases** (of pancreas) into medium and small peptides (**oligopeptides**)
- 2) Oligopeptides are degraded by **lactase** [its deficiency cause lactose intolerance] and **maltase** peptidase enzymes (integral membrane proteins not a soluble lumen enzymes) into free amino acids and very small peptides
- 3) Normal intestinal enterocytes have **no transporters for proteins and they can not permeate the tight junction**
- 4) **So protein drugs can not be absorbed after oral administration**

Physiological factors affecting drug absorption

- 1) Perfusion of the GIT (blood supply)
 - » Splanchnic circulation (capillary network to GIT) receives 28% of the cardiac output and blood flow increase after meals
 - » ↓ Mesenteric Blood flow → ↓ rate of removal of drug from GIT (absorption) → ↓ rate of bioavailability
 - » Drug may be absorbed through the lymphatic system from GIT not affected by blood supply and bypass hepatic metabolism e.g. absorption of dietary lipids and some lipophilic drugs

2) Gastric emptying

- » Depends on physical form of dosage form e.g. **size and density** (tablets and capsules stay longer than liquids) and **presence of food** in stomach
- » Indigestible solid dosage forms larger than pyloric opening in closed position are prevented from emptying during post prandial period but should wait for interdigestive migrating myoelectric complex (IMMC) which occur at the end of digestive phase to clear food from stomach

» **Cycle of IMMC (1.5 – 2 hrs), consists of 4 phases** (migrate distally from stomach → small intestine → colon)

Phase	Appearance	Duration (min)	Amplitude of contraction	Description
1 (basal)		45 – 60	Low	Nearly empty stomach with no motor activity except few contractions
2 (preburst)		30 – 45	Intermediate	↑ Frequency and amplitude of contraction
3 (burst)	 Housekeeper waves due to its sweeping property	5 – 15	Maximal	Giant peristaltic contractions, occur 3 times per minute, opening of pyloric sphincter and removal of indigestible solids
4		0 – 5	Short transition period of decreasing activity till next cycle begins	

» **Bile secretion starts from phase 2 of IMMC while mucus secretion continue during the 2, 3 and 4 phases**

» **Decreased GER will cause**

↓ Rate and perhaps the extent of drug absorption

↓ Onset of action

↑ Chemical degradation of unstable drugs in acid medium (penicillin)

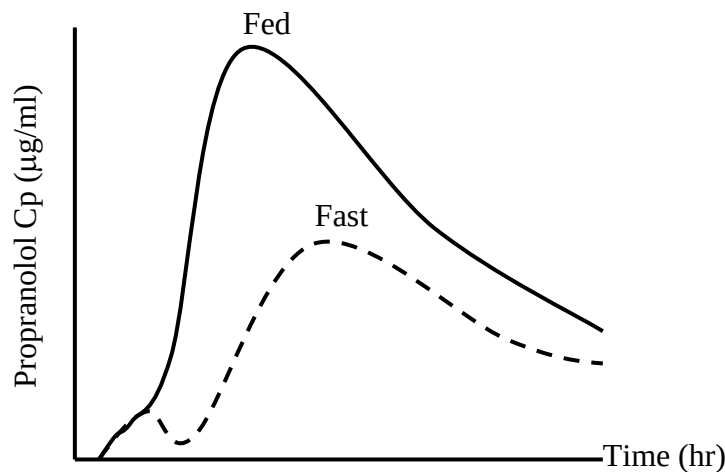
↑ Irritation of gastric mucosa caused by Aspirin due to increased contact time

↑ Residence time of enteric coated tablets in stomach → ↓ onset → ↓ rate of absorption

3) Effect of food on gastric retention and bioavailability

Food may interact with coadministered drug before and after gastrointestinal absorption, during distribution, during metabolism and during elimination

- a) **↑ pH of stomach to 6 or higher** $\Rightarrow$ affects solubility and dissolution of drugs e.g. cola beverage (acidic) $\rightarrow$ ↓ pH of stomach $\rightarrow$ ↑ dissolution of ketoconazole and itraconazole $\rightarrow$ ↑ bioavailability
- b) **↓ GER** $\rightarrow$ ↑ contact time with gastric juice $\rightarrow$ ↑ degradation of acid sensitive drugs e.g. pravastatin (its bioavailability decreased to 31% after fatty meal), erythromycin (gave 28% bioavailability when taken with food), tetracycline and penicillin
- c) **↑ mesenteric blood flow** $\rightarrow$ drug will be absorbed and presented to liver faster $\rightarrow$ ↑ chances of drug to escape liver enzymes $\rightarrow$ ↓ 1st pass effect $\rightarrow$ ↑ bioavailability e.g. propranolol
- d) **↓ Irritation of gastric mucosa** caused by NSAIDs, iron salts and aspirin where rate of absorption may be reduced but extent will be the same but without irritation
- e) **↑ bile secretion (fatty meals)** $\rightarrow$ ↑ emulsification and Solubilization of griseofulvin



4) Small intestine transit	5) Colonic transit
▶▶ Small intestine is the main site of absorption ▶▶ Transit time in intestine is more regular compared to variable gastric transit time ▶▶ Average small intestine transit time is 3 ± 1 hr ▶▶ The total effective time for drug absorption is 3 – 8 hrs thus if we want to make sustained release product its absorption window should be 3 – 8 hrs i.e. we can not make SR over 12 hrs since part of the drug will pass absorption site in small intestine	▶▶ Its surface area is much smaller than the small intestine ▶▶ Transit time in colon is variable 20 – 36 hrs ▶▶ Colon is the absorption site of some drugs which require activation by bacterial cleavage e.g. sulfasalazine
Techniques used to measure transit times of dosage forms in the GIT 1) Radiographic technique ▶▶ Mix dosage form with radio-opaque material e.g. barium for X-ray visualization	

- » Take several radiographs to determine position of the dosage form in different GIT regions
- » Expose volunteers to radiation (during X-ray visualization)

2) Scintigraphic technique

- » Mix dosage form with small amount of radionuclide e.g. ^{99m}Tc (technetium) or ^{113m}In Indium which emits γ -rays
- » Position of the dosage form can be visualized by γ -camera
- » Radionuclides have short half lives and emit small dose of radiation thus it is safer for volunteers

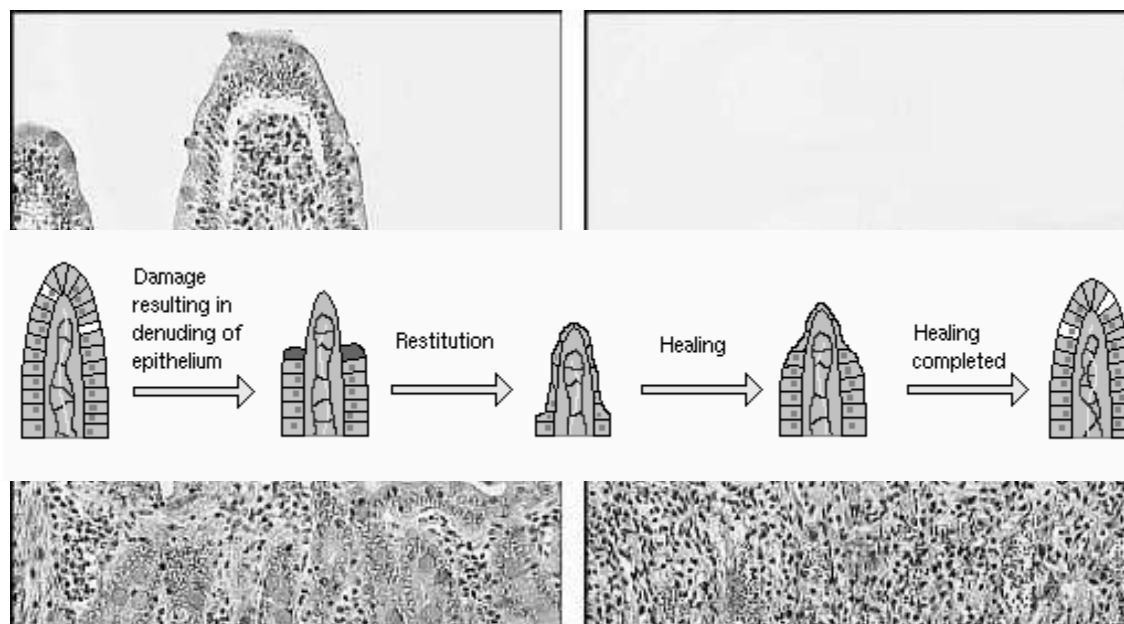
6) Effect of disease states on drug absorption

Drug absorption may be affected by any disease state as follow

Disease state	Effect on drug absorption
CHF	↓ mesenteric blood flow, ↓ rate of absorption and bioavailability
Diarrhea	↑ intestinal motility → ↓ residence time in intestine → ↓ absorption
Parkinson's disease	↓↓↓ GI motility → ↓ absorption
Celiac disease	↑ gastric emptying and ↑ intestinal permeability → ↑ absorption
Achlorhydria	↑ pH of stomach and affects drug solubility
Change of intestinal pH	Affects extent of drug ionization
Crohn's disease and ulcers	Affect permeability of intestinal membrane
Changes in bile secretion (bile stones)	Affect absorption of lipophilic drugs e.g. griseofulvin
Changes in digestive enzyme secretion	Affect stability of some drugs e.g. protein drugs
Alteration of normal GI flora	Affect absorption of some drugs

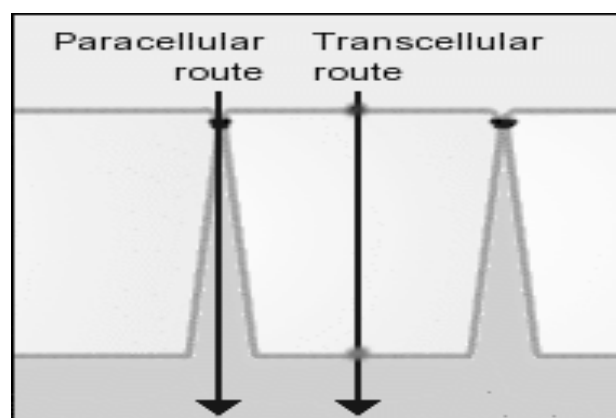
Restitution and healing of GI epithelium after injury

- » Disruption and injury of intestinal mucosa can occur due to local infections by bacteria and viruses, exposure to toxins or physical injury or systemic disease
- » During healing, contraction of the defect villi occur to reduce area of basement membrane that must be covered
- » **Restitution:** occur 1st, it is a mechanism for covering a defect in barrier but does not involve proliferation of epithelial cells
- » **Healing:** epithelial cells on margin of defect proliferate, differentiate and migrate to the damaged area



Transcellular and paracellular absorption

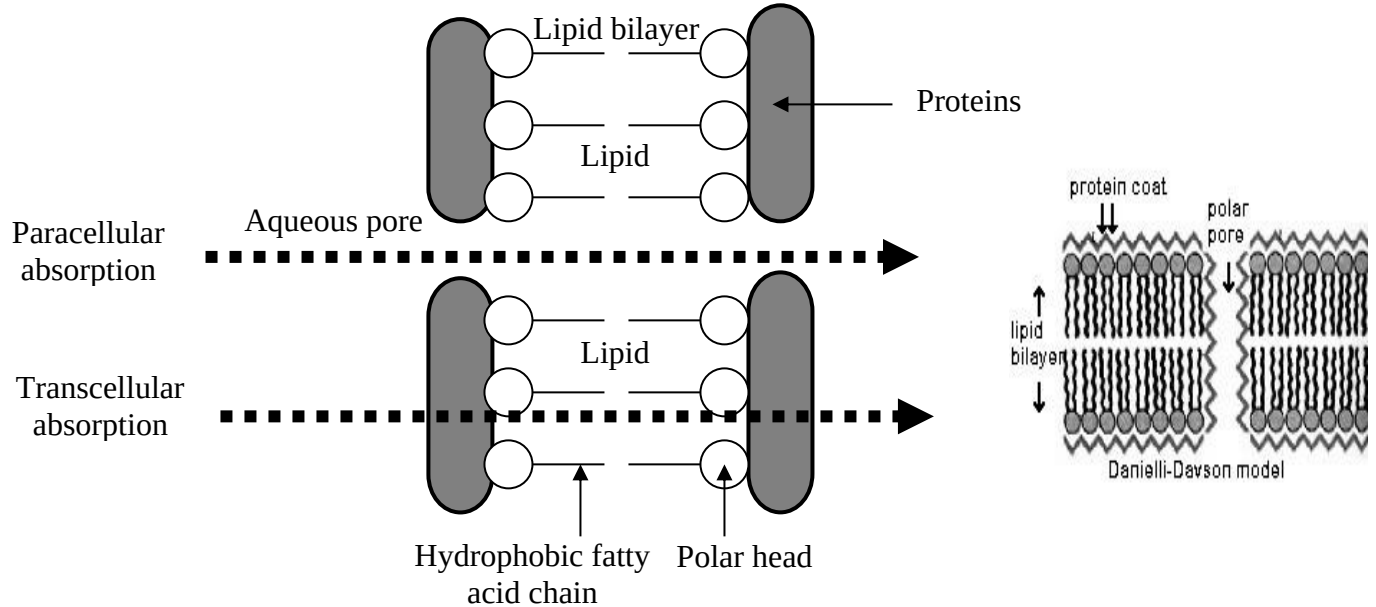
Transcellular absorption	paracellular absorption (through tight the junction)
Drugs pass through lipid sieve (lipid bilayer)	Drugs pass through the aqueous pores present between cell margins
Drug should be unionized, soluble and have certain lipid solubility	Drug should be small in size (molecular size), polar (ionized) e.g. water (can be transported by both routes), urea and certain ions
Highly charged molecules and large molecules e.g. proteins and protein bound drugs can not pass through the GI membrane	



GI cell membranes

►► Semipermeable membranes composed of phospholipids bilayer (proposed by Davson and Danielli in 1952) between 2 surface layers of protein, with the hydrophilic head of PL facing protein layers and hydrophobic tails of PL aligned to inside

- » The membrane is traversed with aqueous pores lined with protein layer
- » **Systemic absorption of drug depends on**
 - a) Physicochemical properties of the drug
 - b) Nature of drug product
 - c) Anatomy and physiologic function at the site of absorption



Mechanisms of drug passage through biological membranes

1) Convective absorption (pore transport)

- » Very small molecules e.g water, urea, ↓ mol - wt sugars e.g glucose & organic electrolytes.
- » They pass through aqueous filled pores (0.4 nm), this pores have never been observed by microscopy but a certain type of protein (transport protein) may form open channel across the lipid membrane through which small molecules can pass rapidly by diffusion
- » Play minor role in absorption of ↑ mol. Wt compounds from G I T but involved in renal excretion and uptake of drug into liver.

2) Passive diffusion

- » The most common method of transcellular passage
- » Drug move from region of high conc. to region of low conc. (with conc. gradient)
- » No external energy (so called passive)

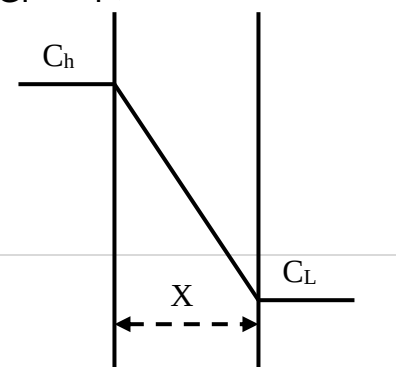
- » Expressed by fick's law of diffusion as follow $\frac{dQ}{dt} = \frac{DAK}{h} (C_{GI} - C_P)$

Where

$\frac{dQ}{dt}$: Rate of diffusion

D: diffusion coefficient (depend on nature of membrane)

K: partition coefficient ($K_{lipid/water}$), depend on nature of drug



A: surface area

h or x: thickness of the membrane

($C_{GI} - C_P$): difference in conc. of drug in GI tract and blood

Diffusion coefficient (D): is the amount of drug which diffuse across membrane of a given unit area per unit time when conc. gradient is unity; its unit is area/time (cm^2/sec)

Thickness of the membrane (h) is constant at all absorption sites except in the capillaries of the brain; it is densely lined with glial fatty cells making diffusion through it into brain quit slow and giving rise to the term BBB (blood brain barrier)

» Since D, A, K and h are constants; so they can be replaced in the above equation by P (permeability coefficient) $P = \frac{DAK}{h}$

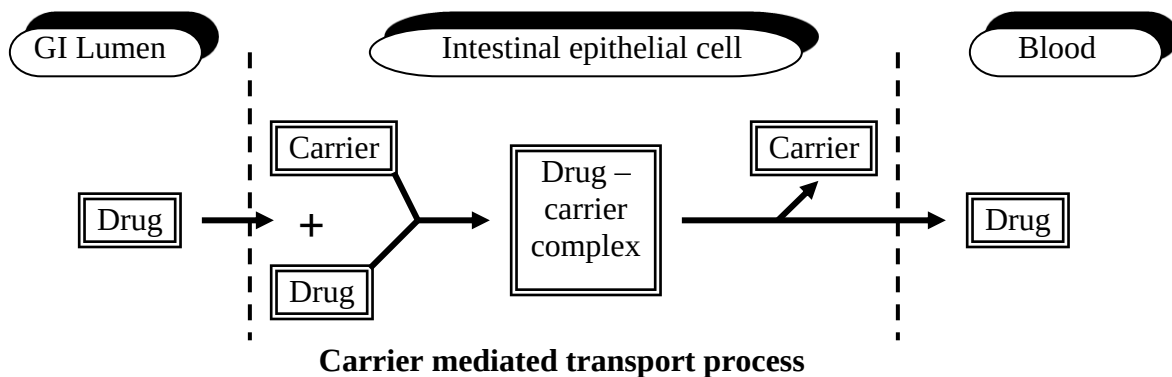
» And since C_p is small compared to C_{GI} the fick's equation become $\frac{dQ}{dt} = p(C_{GI})$

» This equation represents **first order absorption process**

» Drugs to be absorbed by passive diffusion first order process should be **unionized** which will depend on both pK_a of the drug and pH of the medium according to the following equation $\Rightarrow pH = pK_a + \log \frac{\text{ionized}}{\text{unionized}}$

N.B.: permeability of the membrane depend on physicochemical properties of the drug (lipid solubility, degree of unionization and solubility)

3) Carrier mediated transport

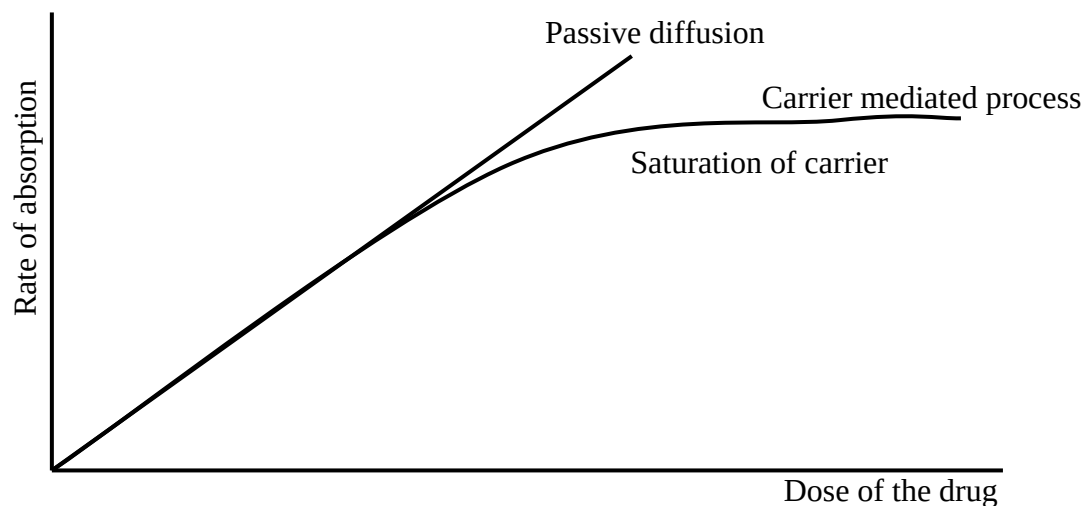


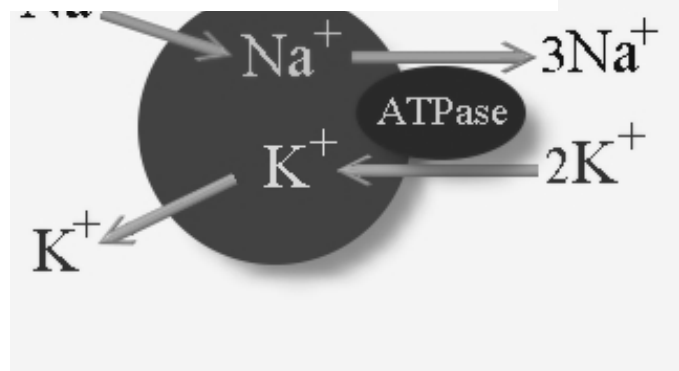
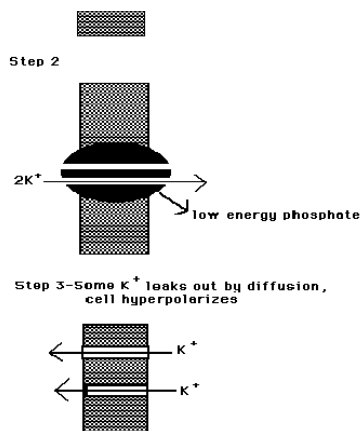
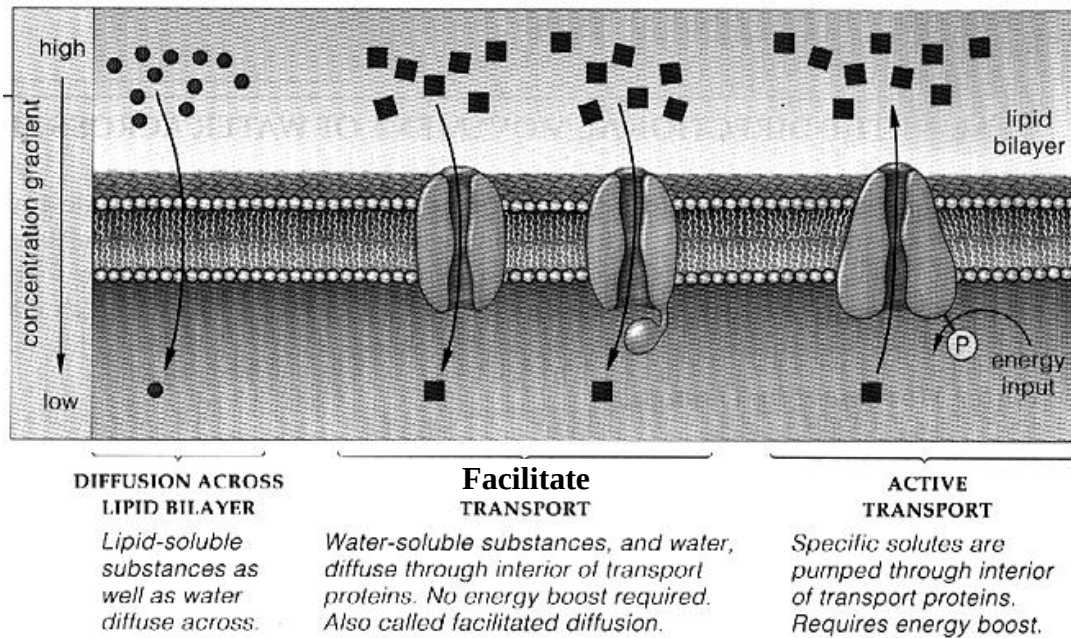
	Active transport	Facilitated diffusion
Direction of flow	Against conc. gradient	With conc. gradient
Carrier	» Required » (glucose and galactose are transported by sodium-glucose co-transporter i.e. it is a sodium dependent hexose transporter known as SGLUT-1)	» Required » Fructose is not co-transported with sodium C.F. glucose and galactose

	▶▶ This carrier transport them together but will not transport either alone) ▶▶ Lactose and starch have no carrier thus should be digested to monosaccharide before absorption	▶▶ Egyptians have a genetic defect for carrier of vitamin B ₁₂ absorption
Energy	Required (supplied by ATP)	Not required
Competition for carrier with similarly structured drugs	Present	Present
Saturation of carrier at high doses and rate of absorption become constant	Present	present
Drugs	Mainly for drugs of endogenous origin e.g. transport of amino acids and glucose and galactose	Play a minor role e.g. reabsorption of drugs from kidney and transport of vitamin B ₁₂ and fructose

Carrier mediated intestinal transport

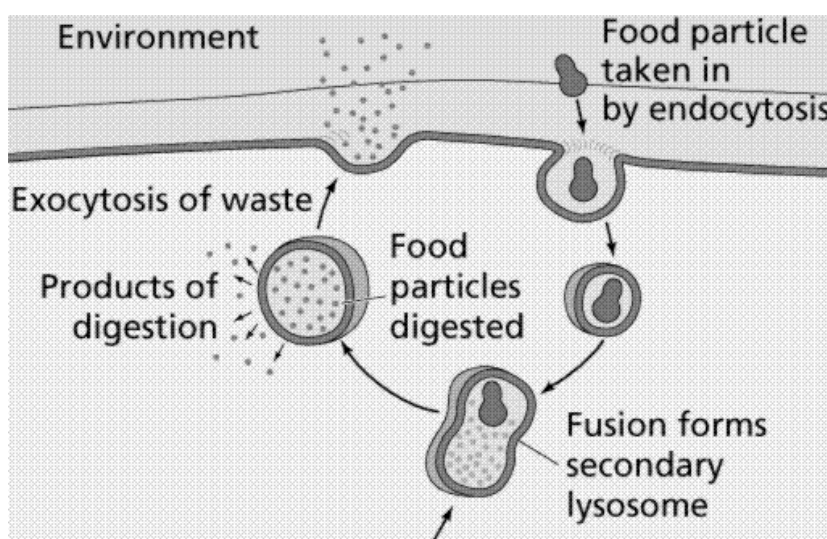
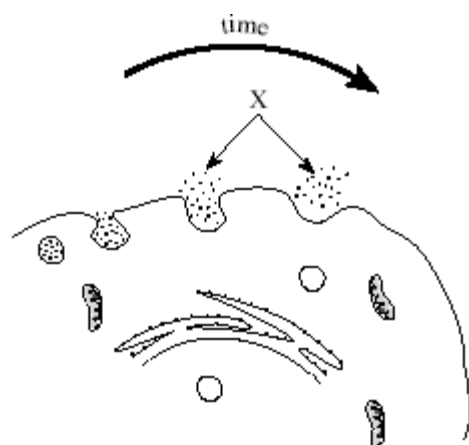
- ▶▶ Intestinal brush border contain many transporters (e.g. P-gp) for nutrients which can transport similarly structured drugs
- ▶▶ P-glycoprotein (P-gp): ↓ permeability of intestinal membrane to various lipophilic and cytotoxic drugs
- ▶▶ Oral cephalosporins are absorbed by another amino acid transporter but cefazolin (a parenteral only cephalosporin) is not absorbed orally by this transporter
- ▶▶ **Inhibition of P-gp → ↑ absorption of some drugs**





4) Vesicular transport

- ▶▶ The process of engulfing particles or dissolved materials by the cell where cell membrane invaginates and surround the material then engulf it to inside of the cell
- ▶▶ Pinocytosis: engulfment of small solutes or fluid (cell drinking)
- ▶▶ Phagocytosis: engulfment of larger particles or macromolecules (cell eating)
- ▶▶ Vesicular transport is the proposed mechanism for absorption of oral sabin polio vaccine and various large proteins
- ▶▶ Endocytosis and exocytosis: processes of moving macromolecules in and out of the cell
- ▶▶ Transport of insulin from insulin producing cells of pancreas to the extracellular space is an example of exocytosis

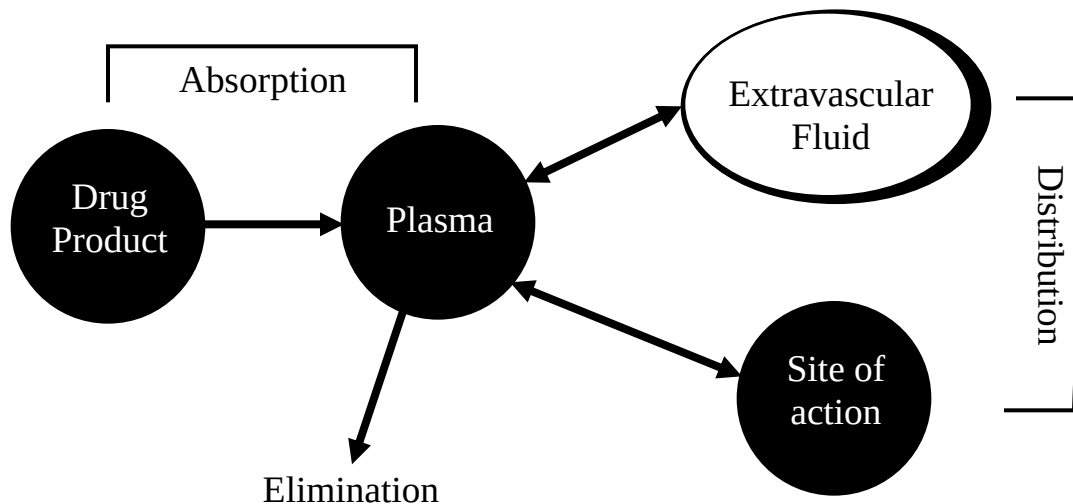


5) Ion pair formation

- ▶▶ This is the proposed mechanism for absorption of drugs which are
 - a) Highly ionized or charged molecules over all pH range of GIT e.g. quaternary nitrogen compounds (tetracyclines) i.e. with extreme pKa (either too high or too low)
 - b) Too large $\Rightarrow$ can not pass by pore transport
 - c) Too lipid insoluble
- ▶▶ They unit with endogenous organic ions of opposite charge $\rightarrow$ absorbable neutral species (**ion - pair**) $\rightarrow$ partition to lipoidal cell membrane $\rightarrow$ **passive diffusion**
- ▶▶ Propranolol (basic drug) form ion pair with oleic acid while quinine (basic drug) form ion pair with hexylsalicylate

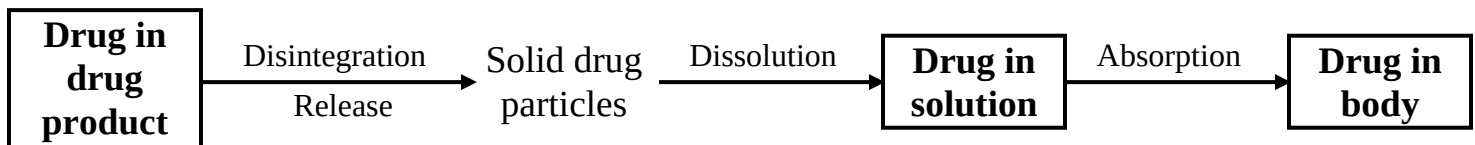
Biopharmaceutic Considerations in Drug Product Design

- ▶▶ Biopharmaceutics is the science that studies the relationship between the physicochemical properties of the drug, the dosage form and the route of administration on the rate and extent of systemic drug absorption.
- ▶▶ Bioavailability captures two essential features, namely rate of absorption and extent of absorption.
- ▶▶ Onset of response is linked to the rate of drug absorption whereas the extent of response is linked to the extent of drug absorption.



Rate Process of Drug Bioavailability

- ▶ After oral administration, absorption process depends on drug **dissociation (release)** from its dosage form, **dissolution** in body fluids, and then **diffusion** across the biological membrane barriers of the gut wall into systemic circulation.
- ▶ The slowest step in a series of kinetic processes is called **rate-limiting step**.



Factors Affecting Bioavailability

- ▶ Bioavailability following oral doses may vary because of **patient-related** or **dosage-form-related factors**.
- ▶ **Patient-related factors** can include the nature and timing of meals, age, disease, genetic traits and gastrointestinal physiology
- ▶ **The dosage form related factors include**
 - 1) **Physicochemical factors** and
 - 2) **Pharmaceutical factors** (nature of dosage form, array of formulation (e.g. non-active ingredients) and manufacturing variables).

The dosage form factors

I. Physicochemical Nature of the Drug

- ▶ The physical and chemical characteristics of a drug and formulation can affect **drug absorption and stability**.
- ▶ A drug must be dissolved to be absorbed; its rate of dissolution from a given product

influences its rate of absorption especially for sparingly soluble drugs.

- ▶▶ **For example, intravenous solutions are difficult to prepare with drugs that have poor aqueous solubility.**
- ▶▶ Drugs that are physically or chemically unstable may require special excipients, coatings or manufacturing processes to protect the drug from degradation.

Physicochemical property	Effect on solubility and stability of the drug
pKa and pH profile	Necessary for optimum stability and solubility of the final product
Particle size	May affect the solubility of drug and therefore the dissolution rate of the product
Polymorphism	▶▶ The ability of a drug to exist in various crystal forms may change the solubility of the drug. ▶▶ Also, the stability of each form is important, because polymorphs may convert from one form to another
Hygroscopicity	Moisture absorption may affect the physical structure as well as stability of the product
Partition Coefficient	▶▶ May give some indication of the relative affinity of the drug for oil and water. ▶▶ A drug has high affinity for oil may have poor released dissolution from the drug product.
Excipient Interaction	▶▶ The compatibility of the excipients with the drug and sometimes trace elements may affect the stability of the product. ▶▶ It is important to have specifications of all raw materials.
pH Stability profile	▶▶ The stability of the Solution is often affected by the pH of the vehicle ▶▶ knowledge of the stability profile would help to avoid or prevent degradation of the product during storage or after administration

A- Solubility, pH, and Drug Absorption

- ▶▶ A basic drug is more soluble in acidic medium forming a soluble salt conversely an acid drug is more soluble in the intestine forming a soluble salt at the more alkaline pH.
- ▶▶ The solubility pH profile gives a rough estimation of the completeness of dissolution for a dose of a drug in the stomach or the small intestine.
- ▶▶ Solubility may be improved with the addition of an **acidic or basic excipient**.
- ▶▶ Solubilization of aspirin may be increased by the addition of an **alkaline buffer**.

pKa and Degree of Ionization

Most of the drugs are either weak acids or weak bases

Henderson – Hasselbalch equation**Weak acids**

$$\text{pH} = \text{pKa} + \log \frac{C_i}{C_u}$$

$$\text{pKa} = \text{pH} + \log \frac{C_u}{C_i}$$

weak base

$$\text{pH} = \text{pKa} + \log \frac{C_u}{C_i}$$

$$\text{pKa} = \text{pH} + \log \frac{C_i}{C_u}$$

N.B.: ↓ pKa of drug → acidic but when ↑ pKa of drug → basic

$\text{pKa} = -\log k_a$ $\text{PH} = -\log [\text{H}^+]$
--

Degree of ionization (α)

►► It gives indication about the extent of ionization of an acid or base at certain pH

$$\alpha = \frac{C_i}{C_i + C_u}$$

►► If α is 1 ⇒ complete ionization (100% ionization; not absorbed), if $\alpha = 0.8 \Rightarrow 80\%$ ionization but if $\alpha = 0.3 \Rightarrow 30\%$ ionization

►► Degree of ionization depends on pH of the medium and pKa of the drug as follow

$$\text{pKa} = \text{pH} + \log \frac{C_u}{C_i} \text{ (acidic drug)}$$

$$\text{pKa} - \text{pH} = \log \frac{C_u}{C_i} \Rightarrow \text{Take shift log to the equation} \Rightarrow \frac{C_u}{C_i} = \text{shift log}(\text{pKa} - \text{pH})$$

$$\frac{C_u}{C_i} = 10^{\text{pKa}-\text{pH}} \Rightarrow C_u = C_i \cdot 10^{\text{pKa}-\text{pH}} \text{ and } C_i = C_u \cdot 10^{\text{pH}-\text{pKa}}$$

$$\text{But } \alpha_a = \frac{C_i}{C_i + C_u} \Rightarrow \alpha_a = \frac{C_i}{C_i + C_i \cdot 10^{\text{pKa}-\text{pH}}} \Rightarrow \alpha_a = \frac{1}{1 + 10^{\text{pKa}-\text{pH}}}$$

$$\text{►► Thus the degree of un-ionization of acidic drug is } 1 - \alpha_a = \frac{C_u}{C_i + C_u} = \frac{1}{1 + 10^{\text{pH}-\text{pKa}}}$$

$$\text{►► And for basic drug} \Rightarrow \alpha_b = \frac{1}{1 + 10^{\text{pH}-\text{pKa}}}$$

$$\text{►► And the degree of un-ionization of basic drug is} \Rightarrow 1 - \alpha_b = \frac{1}{1 + 10^{\text{pKa}-\text{pH}}}$$

$\alpha_a = \frac{1}{1 + 10^{\text{pKa}-\text{pH}}}$ And $1 - \alpha_a = \frac{1}{1 + 10^{\text{pH}-\text{pKa}}}$ $\alpha_b = \frac{1}{1 + 10^{\text{pH}-\text{pKa}}}$ And $1 - \alpha_b = \frac{1}{1 + 10^{\text{pKa}-\text{pH}}}$
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- ▶▶ **These equations indicate that the degree of ionization of weak acids increases with increasing pH, and decreases in the case of weak bases**

The pH-partition hypothesis of drug absorption

The pH-partition theory states that drugs are absorbed from the biological membranes by passive diffusion, depending on the fraction of un-ionized form of the drug at the pH of the fluids close to that biological membrane.

- ▶▶ In solution, organic electrolytes exist in non ionized (lipid-soluble) and ionized (poor lipid soluble) form. The fraction of each species depends on the pH of the solution.
- ▶▶ Since the GI barriers are much more permeable to unchanged, lipid-soluble solutes, a drug may be well-absorbed from one segment of the GIT, where a favorable pH exists but poorly absorbed from another segment, where a less favorable pH is found.
- ▶▶ The gastrointestinal tracts acts as a lipophilic barrier and thus ionized drugs, compared to un-ionized ones, are more hydrophilic and have minimal membrane transport.
- ▶▶ **The absorption of weakly basic drugs such as antihistamines and antidepressants is favored in the small intestine where such drugs exist largely in a non ionized form.**
- ▶▶ On the other hand, the acidic gastric fluids tend to retard the absorption of weak bases but promote the absorption of weakly acidic drugs such as sulphonamides and NSAIDs.
- ▶▶ Changes in the pH of the fluids in a given segment of tract may improve or impede the absorption of a drug. The pH of gastric fluid varies considerably.
- ▶▶ Gastric secretions have a pH of less than 1, but the pH of gastric contents is usually between 1 and 3 because of dilution and diet. The pH of the stomach contents is distinctly but briefly elevated after meals; pH values of 5 are not unusual.
- ▶▶ Fasting tends to decrease the pH of gastric fluids.
- ▶▶ Disease may also influence the pH of the stomach. The average gastric pH is significantly lower in patients with a duodenal ulcer than in healthy individuals.
- ▶▶ Fats and fatty acids in the diet have been found to inhibit gastric secretions.
- ▶▶ A major clinical effect of antispasmodic drugs such as Atropine and H₂-blockers, like cimetidine and ranitidine is a reduction in gastric secretions.
- ▶▶ Some anticholinergic activity, including suppression of gastric secretion, is commonly found with many other drugs. Antacids are widely used for the purpose of neutralizing gastric acidity and elevating the pH of gastric contents.
- ▶▶ **From the pKa value of drug and pH of body fluid at the absorption site, it is possible to determine the relative amounts of ionized and unionized drug in a solution at a particular pH by using Henderson and Hasselbalch equations:**

Weak acids

$$\text{pH} = \text{pK}_a + \log \frac{\text{I}}{\text{U}} \Rightarrow \text{pK}_a - \text{pH} = -\log \frac{\text{I}}{\text{U}} \Rightarrow \text{pK}_a - \text{pH} = \log \frac{\text{U}}{\text{I}} = \log \frac{[\text{HA}]}{[\text{A}^-]}$$

Drug	Example	pKa	PKa – pH	Drug is mainly	Absorption
Weak acid	Phenytoin, ethosiximide and barbiturates	> 8	> 1 at all pH range of GIT	Mainly unionized at all pH of GIT	Rapid and independent on GI pH
Weak acid	Aspirin, Ibuprofen, phenylbutazone and penicillin analogs	2.5 – 7.5	> 1 in stomach	Mainly unionized in stomach	Better absorbed from stomach
			< 1 in intestine	Mainly ionized in intestine	
Strong acid	Cromolyn sodium	< 2.5	< 1 in all pH range of GIT	Mainly ionized at all pH of GIT	Poorly absorbed from all GIT

Weak bases

$$\text{pH} = \text{pK}_a + \log \frac{\text{U}}{\text{I}} \Rightarrow \text{pK}_a - \text{pH} = -\log \frac{\text{U}}{\text{I}} \Rightarrow \text{pK}_a - \text{pH} = \log \frac{\text{I}}{\text{U}} = \log \frac{[\text{HB}^+]}{[\text{B}]}$$

Drug	Example	pKa	PKa – pH	Drug is mainly	Absorption
Weak base	Caffeine, theophylline, diazepam, oxazepam and nitrazepam	< 5	< 1 at all pH range of GIT	Mainly unionized at all pH of GIT	Rapid and independent on GI pH
Weak base	Morphine analogs, chloroquine, imipramine and amitriptyline	5 – 11	> 1 in stomach	Mainly ionized in stomach	Better absorbed from intestine
			< 1 in intestine	Mainly unionized in intestine	
Strong base	Mecamylamine and guanethidine	> 11	> 1 in all pH range of GIT	Mainly ionized at all pH of GIT	Poorly absorbed from all GIT

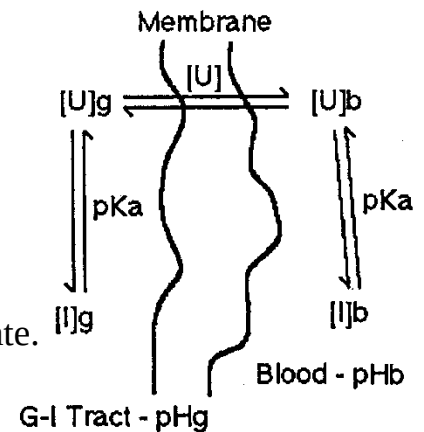
Brodie et al proposed the pH-partition theory to explain the Influence of GI pH and drug pKa on the extent of drug absorption.

►► **Brodie determined the ratio D at equilibrium between blood and GIT**

$$D = \frac{\text{Total concentration in blood}}{\text{Total concentration in GI - tract}} \Rightarrow D = \frac{[U]_b + [I]_b}{[U]_g + [I]_g}$$

►► A large D value will mean that more drug will move from the GI tract to the blood side of the membrane.

►► The larger the D value, the larger the effective concentration gradient, and thus the faster the expected transfer or absorption rate.



Compare D for a weak acid (pKa = 5.4) from the stomach (pH 3.4) or intestine (pH 6.4), with blood pH = 7.4

Stomach

$$[U]/[I] = 10^{pK_a - pH} = 10^{5.4 - 3.4} = 10^2 = 100$$

$$\text{i.e. } 100 [I] = [U] \Rightarrow [I] = 0.01 [U]$$

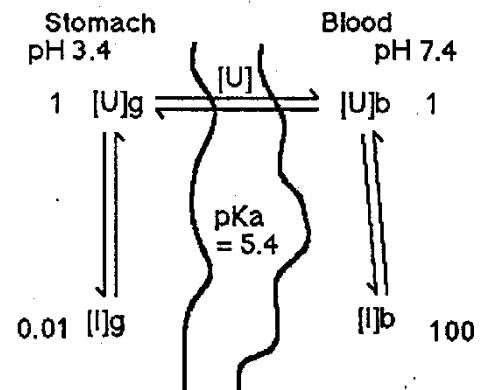
Blood

$$[U]/[I] = 10^{pK_a - pH} = 10^{5.4 - 7.4} = 10^{-2} = 0.01$$

$$\text{i.e. } [I] = 100 [U]$$

Therefore the calculated D value would be

$$D = \frac{\text{Total concentration in blood}}{\text{Total concentration in GI - tract}} \Rightarrow D = \frac{[U]_b + [I]_b}{[U]_g + [I]_g} \Rightarrow D = \frac{[U] + 100[U]}{[U] + 0.01[U]} = \frac{101}{1.01} = 100$$



By comparison in the intestine, pH = 6.4

Intestine

$$[U]/[I] = 10^{pK_a - pH} = 10^{5.4 - 6.4} = 10^{-1} = 0.1$$

$$\text{i.e. } 0.1 [I] = [U] \Rightarrow [I] = 10 [U]$$

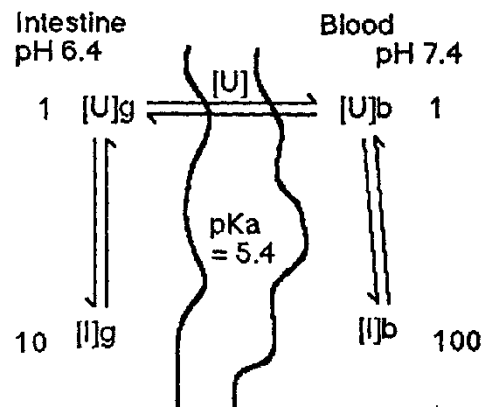
Blood

$$[U]/[I] = 10^{pK_a - pH} = 10^{5.4 - 7.4} = 10^{-2} = 0.01$$

$$\text{i.e. } [I] = 100 [U]$$

Therefore the calculated D value would be

$$D = \frac{\text{Total concentration in blood}}{\text{Total concentration in GI - tract}} \Rightarrow D = \frac{[U]_b + [I]_b}{[U]_g + [I]_g} \Rightarrow D = \frac{[U] + 100[U]}{[U] + 10[U]} = \frac{101}{11} = 9.2$$



∴ Significant absorption of weak acids from the stomach compared with from the intestine.

Limitations of pH-partition hypothesis theory

- 1) Most **weak acids are well-absorbed from the small intestine**, which is contrary to the prediction of the pH-partition hypothesis.
- 2) **Quaternary ammonium compounds are ionized at all pHs**, but are readily absorbed from the GI tract.
- 3) **Barbitone** (pKa 7.8) which is almost totally unionized at gastric pH is only poorly absorbed from the stomach; however **thiopentone**, which has a similar pKa value (pKa 7.6) is much better absorbed from the stomach than barbitone **because the unionized form of thiopentone is more lipid soluble than the unionized form of barbitone**, exhibits a greater affinity for the gastrointestinal membrane and thus better absorbed than barbitone from stomach

These discrepancies arise because pH-partition theory does not take into consideration the following (this deviation can be explained by)

- 1) Large epithelial **surface areas of small intestine** compensates for ionization effects.
- 2) **Long residence time** in the small intestine also compensates for ionization effects.
- 3) Charged drugs, such as quaternary ammonium compounds and tetracyclines, may interact with opposite charged organic ions, resulting in a neutral species, which is absorbable (**ion pair mechanism**)
- 4) Some drugs are absorbed via **active transport**

B-Stability, pH and Drug Absorption

The stability pH profile is a plot of the reaction rate constant for drug degradation versus pH

Erythromycin

Decompose rapidly in acidic medium erythromycin whereas it is relatively stable in neutral or alkaline pH.

To protect it

- a) Erythromycin tablets are enteric coated to protect against acid degradation in the stomach.
- b) Preparation of a less water soluble erythromycin salt that is more stable in the stomach (**prodrugs**) e.g. erythromycin stearate after passing through the stomach undissolved dissolves and dissociates in the intestinal fluid yielding the free base erythromycin which is absorbed.

Penicillin G

Susceptible to acid catalyzed hydrolysis and its bioavailability would be improved if it exhibited minimal dissolution in gastric fluid and rapid dissolution in intestinal fluid

C-Particle Size and Drug Absorption

- ▶▶ Surface area of the dissolving solid affects its dissolution rate and bioavailability
- ▶▶ ↓ Particle size (micronization) → ↑ surface area → ↑ dissolution rate
- ▶▶ The geometric shape of the particle also affects the surface area, and during dissolution, the surface area is constantly changing.
- ▶▶ In dissolution calculations, the solute particle is usually assumed to have retained its geometric shape.
- ▶▶ Many hydrophobic drugs are very active intravenously but are not very effective when given orally due to poor absorption.
- ▶▶ **Griseofulvin, Nitrofurantoin, and many steroids** are drugs with low aqueous solubility; reduction of the particle size decreased by milling to a micronized form has improved the oral absorption of these drugs.
- ▶▶ The reduction in the particle size of **Griseofulvin, Chloramphenicol, and Tetracycline** improve their dissolution.
- ▶▶ **Micronization** also reduces the dose of some drugs, e.g., the dose of Griseofulvin is reduced to one half while the dose of Spironolactone is reduced to 1/20.

Reduction in particle size is not always helpful, rationalize?

The reduction in particle size of drugs e.g. **Aspirin, Phenacetin, Phenobarbital**, by micronization actually decreases their effective surface area and dissolution rate.

The decrease in surface area is attributed to the following factors:

- a) On their surface, hydrophobic drugs **absorb air** and thereby reduce their wettability.
- b) The particles having size below 0.1 micron **reaggregate to form larger particles**.
- c) The particles having certain microsize get **electrical charge**, thereby preventing contact with wetting medium.

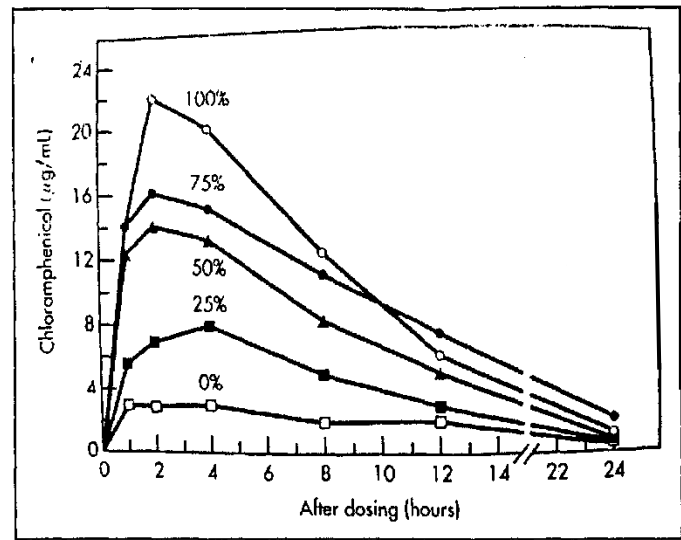
D- Polymorphism, Solvates and Drug Absorption

- ▶▶ **Polymorphism** refers to the arrangement of a drug substance in various crystal forms or polymorphs.
- ▶▶ Amorphous forms are non crystalline forms; Polymorphs have the same chemical structure but different physical properties, such as solubility, density, hardness, and compression characteristics.
- ▶▶ Some **polymorphic crystals** have much lower aqueous solubility than the amorphous forms, causing a product to be incompletely absorbed.
- ▶▶ In general the crystal form that has the lowest free energy is the most stable polymorph. A drug that exists as an amorphous form (non crystalline form) generally dissolves more rapidly than the same drug in a more structurally rigid crystalline form.

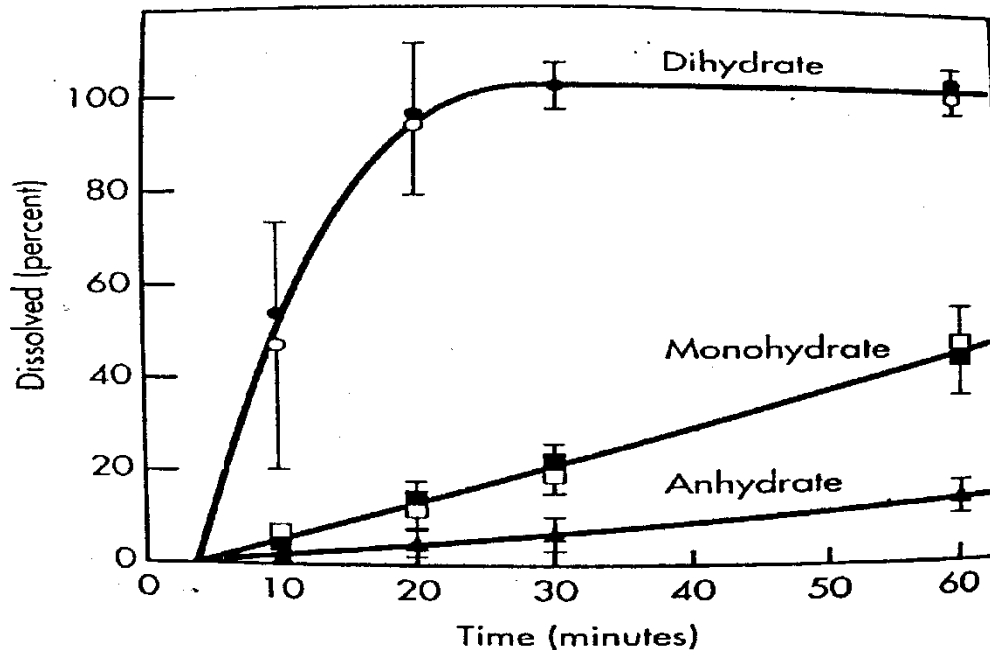
- ▶▶ Some polymorphs are metastable and may convert to a more stable form over time.
- ▶▶ A change in the crystal structure of the drug may cause cracking in a tablet or even prevent a granulation from being compressed into a tablet.

Chloramphenicol

- ▶▶ Has several crystal forms, and when given orally as a suspension the drug concentration in the body was found to be dependant on the percent of β -polymorph in the suspension.
- ▶▶ **The β form is more soluble and better absorbed.**



- ▶▶ Reformulation of a product may be necessary if a new crystal form of a drug is used.
- ▶▶ Some drugs interact with solvent during preparation to form a crystal called a **solvate**.
- ▶▶ Water may form special crystals with drugs called hydrates e.g. **erythromycin hydrates** have quite different solubility compared to the anhydrous form of the drug



Dissolution behavior of erythromycin dehydrate, monohydrate, and anhydrate in phosphate buffer

Dosage form factors affecting drug absorption

- ▶▶ The bioavailability of a drug decrease in the order solution > suspension > capsule > tablet > coated tablet
- ▶▶ ↑ Number of steps a product must undergo before the final absorption step, the slower is the availability and the greater is the potential for bioavailability differences to occur.
- ▶▶ Solutions (elixirs, syrups, or simple solutions) result in faster and more complete absorption of drug, since a dissolution step is not required.
- ▶▶ Enteric-coated tablets do not even begin to release the drug until the tablets empty from the stomach, resulting in poor and erratic bioavailability.
- ▶▶ The pharmaceutical factors that affect dissolution and absorption of a drug from such formulations are as follows:
 - a. The dosage form
 - b. Excipients
 - c. Manufacturing process

a. The Dosage Form

(I) Solutions

- ▶▶ Absorption from an oral solution is rapid and complete, compared with administration in any other oral dosage form.
- ▶▶ The rate limiting step is often the **rate of gastric emptying**.
- ▶▶ When an acidic drug is given in the form of a salt, it may precipitate in the stomach, this precipitate is usually finely divided and is readily re-dissolved and thus causes no absorption problems.

How to use drugs with poor water solubility?

- 1) **Phenytoin** (poorly water soluble) drug is better formulated as **suspension**, of finely divided powder, may have a better bioavailability.
- 2) Some drugs which are poorly soluble in water may be dissolved in **mixed water/alcohol or glycerol solvents**, this is useful for compounds with tight crystal structure, higher melting points that are not ionic. The crystal structure is broken by solution in the mixed solvent.
- 3) **An oily emulsion or soft gelatin capsules** have been used for some compounds to produce improved bioavailability.

(2) Suspensions

- ▶▶ A well formulated suspension is second only to a solution in terms of superior bioavailability.
- ▶▶ Absorption may be dissolution limited; however a suspension of a finely divided powder will maximize the potential for rapid dissolution (↓ **particle size** → ↑ **dissolution**)

rate)

- ▶▶ The addition of a **surfactant** will improve dispersion of a suspension and may improve the absorption of very fine particle size suspensions and prevent caking.

(3) Capsules

- ▶▶ The hard gelatin shell should **disrupt rapidly** and allow the contents to be mixed with the GIT contents.
- ▶▶ The capsule contents **should not be subjected to high compression forces** which would tend to reduce the effective surface area, thus a capsule should perform better than a tablet.
- ▶▶ If a drug is **hydrophobic** a **dispersing agent** should be added to the capsule formulation, these diluents will work to disperse the powder, minimize aggregation and maximize the surface area of the powder.

(4) Tablets

- ▶▶ The tablet is the most commonly used oral dosage form; it is also complex in nature.
- ▶▶ **2 barriers exist between the solid drug and drug in solution**
 - 1) **The reduction in effective surface area** produced during the compression process.
 - 2) **Coating of tablets** to mask an unpleasant taste, to protect the tablet ingredients during storage, or to improve the tablets appearance, this coat must break down quickly or it may hinder a drug's bioavailability.

Sustained release tablets

Benefits

- 1) For short half-life drugs, provide less frequent dosing and thus better compliance.
- 2) Reduce variations in plasma/blood levels for more consistent result.

Problems

- 1) More complicated formulation and may be more erratic in result.
- 2) A failure of the controlled release mechanism result in release of a large toxic dose.
- 3) More expensive technology

b. Excipients (formulation factors)

- ▶▶ They are **non drug components of a formulation**; added to drug for convenient formulation to ensure acceptability, physiochemical stability during the shelf-life, uniformity of composition and dosage and optimum bioavailability of the drug product.
- ▶▶ **The excipients added to the drug formulation** are vehicle, diluents, binders and granulating agents, disintegrants, lubricants, coatings, suspending agents, emulsifiers, surfactants, buffers, complexing agents, colorants, sweeteners, and crystal growth

inhibitors.

- ▶▶ Excipients are known to influence **absorption of drugs**.
- ▶▶ Excipients in the drug product may also affect the **dissolution kinetics of the drug**; either by **altering the medium in which the drug is dissolving** or by **reacting with the drug itself**

1) Suspending agents

Increase the viscosity of the drug vehicle and thereby diminish the rate of drug dissolution from suspensions e.g. Methyl cellulose was found to reduce the rate and extent of absorption of **nitrofurantoin** from aqueous suspensions. In addition, nitrofurantoin may form complexes with methylcellulose which could also tend to reduce the absorption of this drug

2) Tablet lubricants (e.g. magnesium stearate)

May repel water and reduce dissolution when used in large quantities.

3) Coatings particularly shellac

Will crosslink upon ageing and decrease the dissolution rate.

4) Surfactants

May affect drug dissolution in unpredictable fashion

Low concentrations of surfactants: ↓ surface tension and ↑ rate of drug dissolution

Higher surfactants concentrations: tend to form micelles with the drug and thus decrease the dissolution rate.

Polysorbate 80 enhanced gastrointestinal absorption of **phenacetin** in humans resulting from because the surfactant prevent aggregation and thus increasing surface area and dissolution rate of the drug particles in the gastrointestinal fluids.

Some excipients, such as sodium bicarbonate

- ▶▶ May change the pH of the medium surrounding the active drug substance e.g. **Aspirin**, a weak acid when formulated with sodium carbonate, will form water soluble salt in alkaline medium in which the drug rapidly dissolves.
- ▶▶ The term for this process is **dissolution in reactive medium**.
- ▶▶ The solid drug dissolves rapidly in the reactive solvent surrounding the solid particle. However, as the dissolved drug molecules diffuse outward into the bulk solvent, the drug may precipitate out of solution with very fine particle size; these small particles have enormous collective surface area, dispersing and redissolving readily for more rapid absorption upon contact with the mucosal surface.
- ▶▶ Excipients in a formulation may interact directly with the drug to form a water soluble or water insoluble complex e.g. if **tetracycline** is formulated with **Calcium Carbonate**, **an insoluble complex of calcium tetracycline is formed that has a slow rate of**

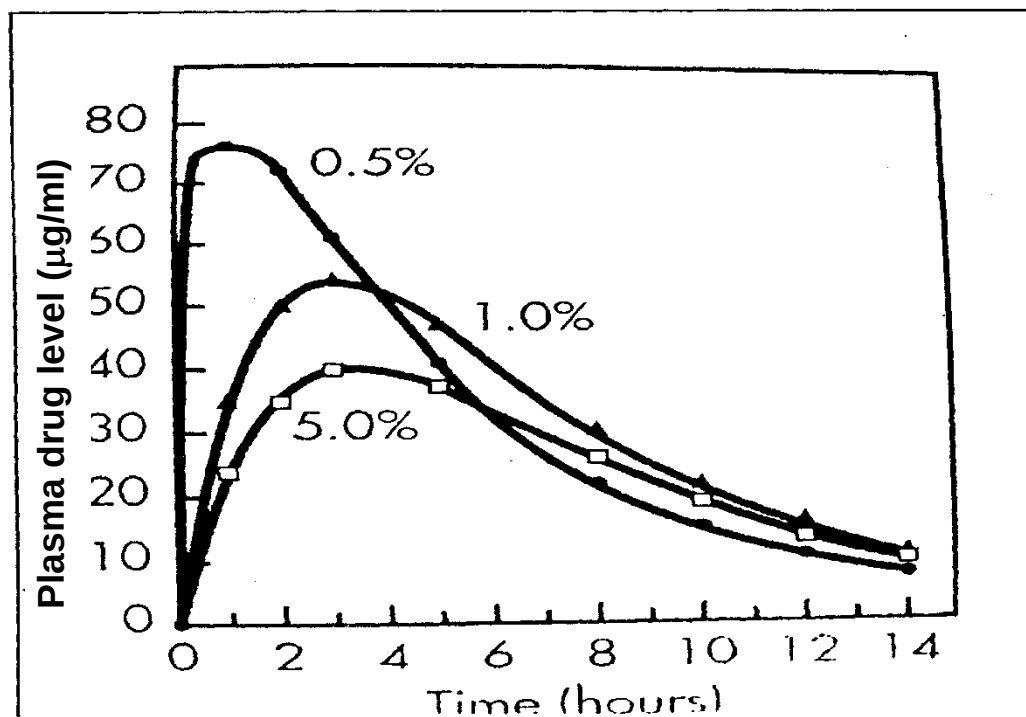
dissolution and poor absorption.

Excipients may be added intentionally to the formulation to enhance the rate and extent of drug absorption e.g.

- 1) Excipients that $\uparrow$ aqueous solubility of the drug generally $\uparrow$ rate of dissolution and drug absorption.
- 2) Excipients may $\uparrow$ retention time of the drug in the gastrointestinal tract and $\uparrow$ total amount of drug absorbed.
- 3) Excipients may act as carriers to $\uparrow$ drug diffusion across the intestinal wall.

Common excipients may change the functionality of the drug substance and the bioavailability of the drug from the dosage form i.e. Excipients may delay or slow the rate of drug absorption e.g.

- An excessive quantity of magnesium stearate (a hydrophobic lubricant) in the formulation may retard drug dissolution and slow the rate of drug absorption
- Talc may be included in tablets as glidant to interfere with the absorption of cyanocobalamin by adsorption of the vitamin on its surface



Effect of lubricant on drug absorption (Percentage of magnesium stearate in formulation, incomplete drug absorption occurs for formulation with 5% mg stearate)

The Australian outbreak of phenytoin intoxication which occurred in sodium phenytoin capsules

- ▶▶ Many epileptic patients who had been previously stabilized with sodium phenytoin capsules containing **Calcium sulphate dihydrate as the diluent**, developed clinical features of phenytoin overdose when given sodium phenytoin capsules containing **lactose as the diluent** even though the quantity of drug in each capsule formulation was identical.
- ▶▶ It was later shown that the excipient calcium sulphate dihydrate had been responsible for decreasing the gastrointestinal absorption of phenytoin, possibly because part of the administered dose of drug formed a poorly absorbable calcium phenytoin complex.
- ▶▶ Hence when the calcium sulphate dihydrate was replaced by lactose, without any alteration in the quantity of drug in each capsule or in the frequency of administration of such capsules, the accompanying improved bioavailability resulted in higher plasma levels of phenytoin.
- ▶▶ In many patients, the higher plasma levels exceeded the maximum safe concentration of phenytoin and produced toxic side effects.

c. Effect of Manufacturing Process

In the tablet manufacture, the following three processes are important:

- 1) Method of granulation
- 2) Compression force
- 3) Intensity of packing capsule contents.

1) Method of granulation

- ▶▶ Granulation is the most common technique followed in the manufacture of tablets.
- ▶▶ This technique enhances the dissolution rate of poorly soluble drugs.
- ▶▶ **The wet granulation process** is the most conventional technique in the manufacture of tablets.
- ▶▶ **The agglomerative phase of comminution technique** is better and gives a product that is stronger and dissolves faster than those made by other granulation methods.

2) Compression force

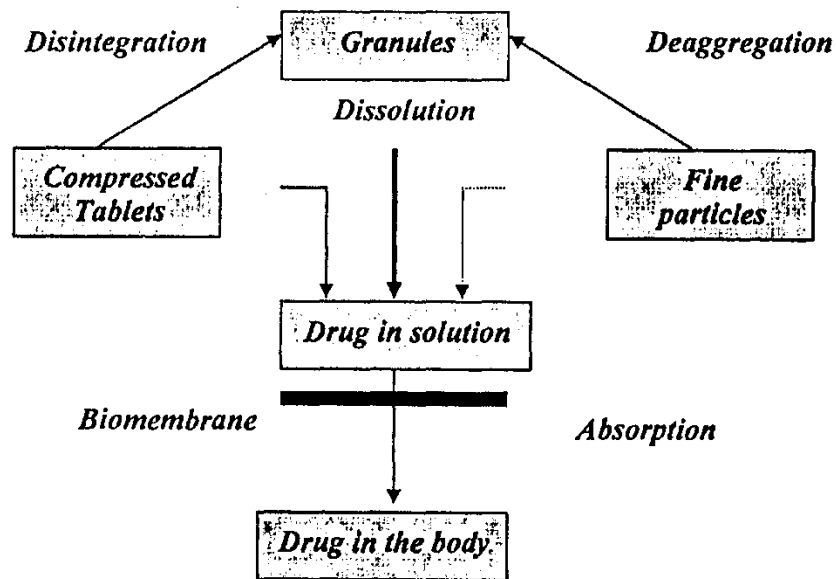
- ▶▶ The compression force used in tablet process also alters density, porosity, hardness, disintegration time and dissolution of tablets.
- ▶▶ ↑ Compression force → ↑ density and hardness of tablet and ↓ porosity, ↓ penetrability of the solvent into the tablet and ↓ wettability by forming a firmer layer of the lubricant and in many cases tends to promote strong bonding between particles.

3) Intensity of packing of capsule contents

- ▶▶ ↑ Packing density of the capsule dosage may either inhibit or promote dissolution rate. The diffusion of GI fluids into the tightly filled capsules gives rise to high pressure within the capsule, thereby causing rapid bursting and dissolution of contents. Opposite of this may

also be possible

- ▶▶ The bioavailability of a drug from the various dosage forms is of the following order:
- ▶▶ Solution > emulsion > suspension > soft gelatin capsule > hard gelatin capsule > tablets
- ▶▶ Among tablets, the order is: uncoated tablets > Coated tablets > Enteric coated > SR/CR products.



BIOAVAILABILITY

Multi-source drug product

Drug products containing the same active ingredients in the same dosage form but marketed by more than one pharmaceutical manufacturer (with different trade names)

Single-source drug product

Patent drug product which is exclusive to only one manufacturer

The orange book

A book published annually by FDA identifying drug products approved on basis of safety and effectiveness and contains therapeutic equivalence evaluations for approved Multisource prescription drug products

Bioavailability = physiological availability = biological availability: the rate and extent of drug absorption from dosage form and become available at the site of action. For drug products intended not to reach blood; their rate and extent will be measured by assessing effect of drug at site of action e.g. antacids (measured by their ANC)

Concept of bioavailability was introduced in 1945 by Oser et al during study of relative absorption of vitamins from pharmaceutical products

Bioavailability concept entered the political arena in the late 1960's as a result of

- 1) Increasing number of prescriptions being written generically e.g. 500 mg paracetamol every 6 hours (how can we ensure safety of such dose?)
- 2) Pronouncements that drug will be purchased on the basis of price i.e. the higher price the better the drug (wrong theory) e.g. Panadol and Paramol (difference in price may be due to difference in rate of absorption or difference in expiration date but the same rate and extent of absorption)
- 3) Pharmacist selects the product
- 4) The probability that formulary systems would be established in an increasing number of situations e.g. BNF (British National Formulary) containing drugs by trade name that are frequently used in England

In 1961 FDA said that bioavailability is the same as biological availability and physiological availability

Definitions

- **Drug substance:** active pharmaceutical ingredient (API) or component of drug product responsible for pharmacologic activity
- **Drug product:** the finished dosage form that contain active ingredient with inactive ingredients
- **Brand Name:** Trade name of the drug.

- **Chemical Name:** Name used by the organic chemist to the material (not the product) to indicate the chemical structure of the drug.
- **Generic Name:** The nonproprietary name مش ملك أى واحد, the name assigned to the compound during early investigations e.g. acetaminophen.
- **Proprietary Name:** The trade name given to the product (Innovator product) by the mother company.
- **Non-proprietary name:** generic name assigned to the compound during early investigations e.g. acetaminophen.
- **cGMP:** current good manufacturing practice
- **GLP:** good laboratory practice
- **GSP:** good storage practice
- **GCP:** good clinical practice
- **GPP:** good pharmacy practice
- **FIFO:** first in first out, materials that enter factory first must be used first
- **Pharmaceutic equivalents:** drug products contain

The same	But may differ in
1) Dosage form 2) Active ingredient (A.I) 3) Conc. or strength f A.I 4) Route of administration	1) Shape 2) Scoring configuration 3) Release mechanism 4) Packaging 5) Excipients e.g. color, flavor or preservative 6) expiration date 7) Within certain limits labeling

- **Pharmaceutic Alternatives** : drug products contain the same therapeutic moiety or its precursor (as the same salt or ester) but not necessarily in the same amount or dosage form e.g. drug products containing tetracycline PO₄ or tetracycline HCl equivalent to 250 mg tetracycline base (as both will dissociate producing tetracycline) or Panadol tablet and syrup
- **Pharmaceutic substitution**: (require physician's approval) dispensing of pharmaceutic alternatives for the prescribed drug product e.g. Ampicillin suspension instead of Ampicillin capsule or tetracycline HCl instead of tetracycline phosphate.
- **Therapeutic equivalents**: must be pharmaceutic equivalents, when given to the same individuals in the same dosage regimen will give same clinical efficacy and safety profile
- **Therapeutic Alternatives**: drug products contain different active ingredients (belong to the same pharmacologic class) used for the same therapeutic or clinical objective. They are expected to have the same therapeutic effect in patients
- **Therapeutic substitution**: (require physician's approval) dispensing of therapeutic alternatives for the prescribed drug product e.g. amoxicillin instead of Ampicillin or ibuprofen instead of naproxen.
- **Bioequivalent drug products**: pharmaceutic equivalents or pharmaceutic alternatives which when given to the same individual in the same dosage regimen (the same experimental conditions) will give the same rate and extent of drug absorption (comparable bioavailability).
- **Interchangeable drug products**: pharmaceutic or Bioequivalents accepted as therapeutic equivalents and the pharmacist has the right to dispense any of them without need for physician's approval

Purpose of bioavailability study

- (1) Determine safety and efficacy of drug product.
- (2) Compare bioavailability of drug products (bioequivalence study)

- (3) Done for approved drug ingredients and those not yet approved for marketing by FDA.
- (4) To ensure that drug meet the required standards of identity, strength, quality and purity.
- (5) Determine differences between drug products from different companies.
- (6) Determine effect of changing physicochemical properties of drug and drug product on its pharmacokinetics
- (7) The Egyptian authorities (MHP Ministry of Health and Population) apply the same requirement of bioequivalence study for registration of a new drug product

Fundamental concepts for determining bioavailability

Availability of drug or its active metabolite to the end-organ (site of action) is controlled by

- 1) The rate and extent of release of drug from dosage form and its subsequent absorption from solution state
 - 2) The 1st pass effect (which may occur through journey of drug till it reach blood i.e. metabolism in saliva, degradation in GI lumen or intestinal wall or in liver)
 - 3) The combined processes of protein binding, distribution to various body fluids, metabolism and urinary, fecal or lung elimination
- In bioavailability studies we compare magnitude of 1 and 2 but in comparative bioavailability studies we compare 1 only (rate and extent of absorption)

Absolute and Relative bioavailability

Absolute Bioavailability (AB = F = F *)

‣ Bioavailability of oral drug product as compared to I.V administration

$$\text{‣ } AUC = \frac{F \cdot \text{dose}}{K \cdot V_d} \Rightarrow F = \frac{AUC}{\text{Dose}} \cdot K \cdot V_d$$

- ▶ **AUC (plasma data) and Du^{∞} (urine data) are used to measure the total amount of unaltered drug that reaches blood**
- ▶ **Done by comparing AUC after oral and IV administration provided that K and Vd remain constant**

Plasma data:

$$\text{Absolute bioavailability} = F = \frac{[AUC]_{PO} / \text{Dose}_{PO}}{[AUC]_{IV} / \text{Dose}_{IV}}$$

$$\% \text{ Absolute bioavailability} = \frac{[AUC]_{PO} / \text{Dose}_{PO}}{[AUC]_{IV} / \text{Dose}_{IV}} \times 100$$

Urine data:

$$\text{Absolute bioavailability} = F = \frac{[Du^{\infty}]_{PO} / \text{Dose}_{PO}}{[Du^{\infty}]_{IV} / \text{Dose}_{IV}}$$

$$\% \text{ Absolute bioavailability} = \frac{[Du^{\infty}]_{PO} / \text{Dose}_{PO}}{[Du^{\infty}]_{IV} / \text{Dose}_{IV}} \times 100$$

A.B = F (fraction of dose that is bioavailable)

- 1) For I.V route (vascular route) = 1
- 2) For oral route (extravascular route) $F = 0 - 1$ (0 for completely unavailable drugs and 1 for completely absorbed drugs)

Relative (Apparent) Bioavailability

Bioavailability of drug product compared to a recognized standard (reference standard usually innovator drug = drug of mother company)

Plasma data

$$\text{Relative bioavailability} = \frac{[AUC]_{\text{test}} / \text{Dose}_{\text{test}}}{[AUC]_{RS} / \text{Dose}_{RS}}$$

$$\% \text{ Relative bioavailability} = \frac{[AUC]_{\text{test}} / \text{Dose}_{\text{test}}}{[AUC]_{\text{RS}} / \text{Dose}_{\text{RS}}} \times 100$$

Urine data

$$\text{Relative bioavailability} = \frac{[Du^{\infty}]_{\text{test}} / \text{Dose}_{\text{test}}}{[Du^{\infty}]_{\text{RS}} / \text{Dose}_{\text{RS}}}$$

$$\% \text{ Relative bioavailability} = \frac{[Du^{\infty}]_{\text{test}} / \text{Dose}_{\text{test}}}{[Du^{\infty}]_{\text{RS}} / \text{Dose}_{\text{RS}}} \times 100$$

Determinations (Assessment) of 1st pass effect

- ▶▶ First pass effect is important for drugs having high clearances
- ▶▶ F: bioavailable fraction of drug after oral administration
- ▶▶ (1 – F): Fractions of dose lost by 1st pass effect

$$\text{Clearance} = \frac{\text{Dose}_{\text{IV}}}{[AUC]_{\text{IV}}} \quad \text{And} \quad \text{Clearance} = \frac{F \cdot \text{Dose}_{\text{oral}}}{[AUC]_{\text{oral}}}$$

$$\frac{\text{Dose}_{\text{IV}}}{[AUC]_{\text{IV}}} = \frac{F \cdot \text{Dose}_{\text{oral}}}{[AUC]_{\text{oral}}} \Rightarrow F = \frac{[AUC]_{\text{oral}} \cdot \text{Dose}_{\text{IV}}}{[AUC]_{\text{IV}} \cdot \text{Dose}_{\text{oral}}} \Rightarrow F = \frac{[AUC]_{\text{oral}} / \text{Dose}_{\text{oral}}}{[AUC]_{\text{IV}} / \text{Dose}_{\text{IV}}}$$

Assessment of bioavailability

5 methods

(1) Plasma data: t_{max} , $C_{p\text{max}}$, AUC and $t_{1/2}$

(2) Urine data: Du^{∞} , $\frac{dDu}{dt}$, t^{∞}

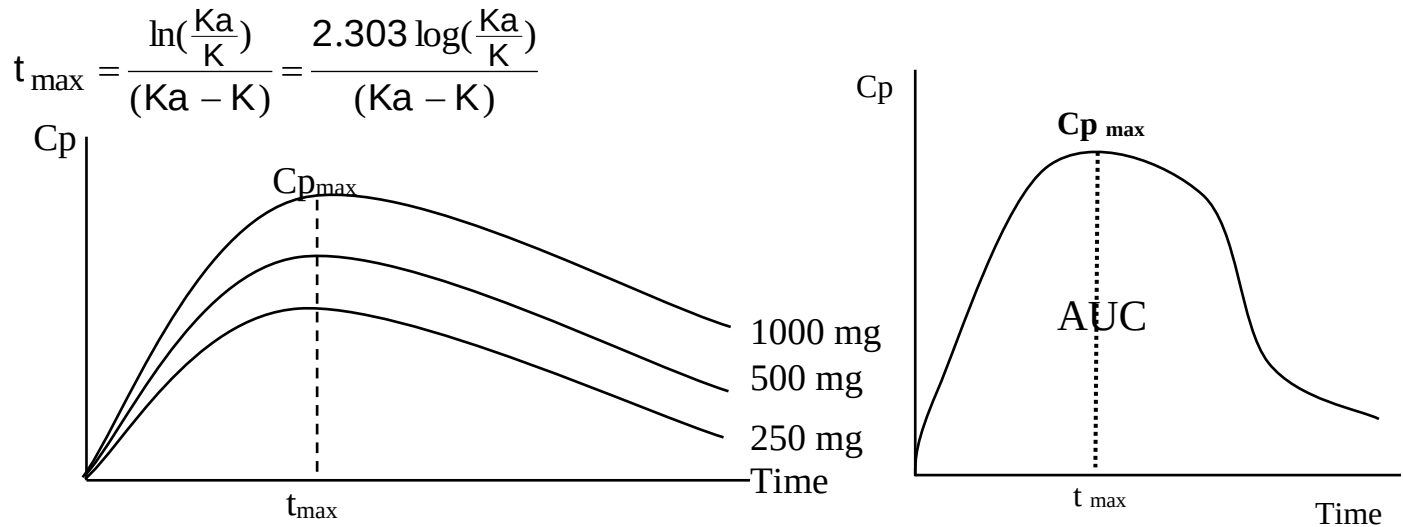
(3) Acute pharmacologic effect.

(4) Clinical observation

(5) In-vitro studies (drug dissolution)

(1) Plasma data

$t_{\max}$: determine rate of drug absorption, The faster drug absorption $\rightarrow \downarrow t_{\max}$



$C_{p\max}$: maximum plasma drug concentration, indicate that drug is sufficiently absorbed to give therapeutic response and provide warning of possibly toxic levels of the drug

AUC (Area under curve): measure extent of drug absorption (total amount of drug that reaches blood).

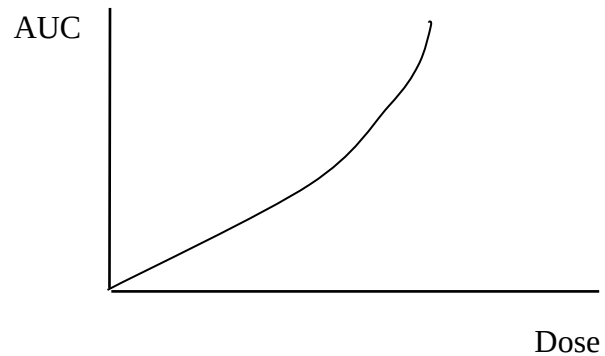
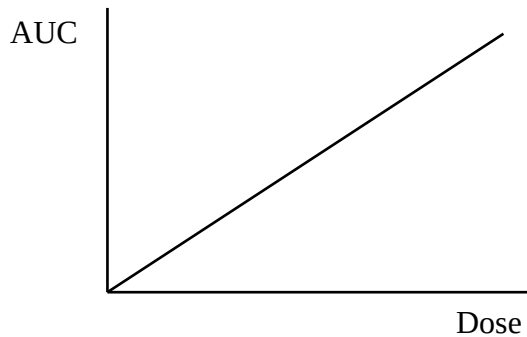
$$[AUC]_0^\infty = \frac{IV \text{ dose}}{K.V_d} \quad \text{Or} \quad [AUC]_0^\infty = \frac{F.\text{oral dose}}{K.V_d}$$

AUC is determined by (1) trapezoidal method (2) Using planimeter

» Unit of AUC is concentration. Time ($\mu\text{g}.\text{hr}/\text{ml}$)

» $\uparrow$ Dose $\rightarrow \uparrow$ AUC (either linear or non-linear)

» Non linear increase of AUC following increase of dose indicate saturation of one of the metabolic pathways through which drug was consumed



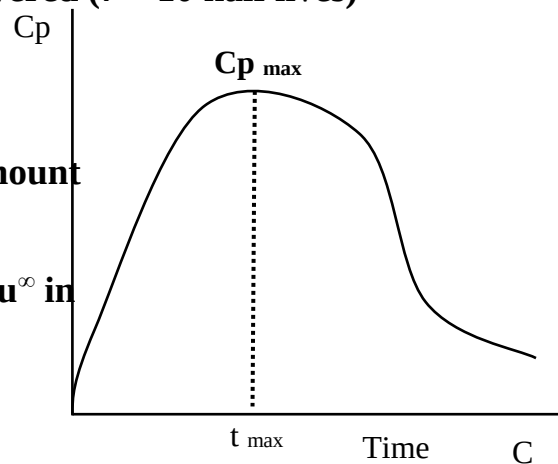
(2) Urine data

To obtain a valid estimation of bioavailability using urine data

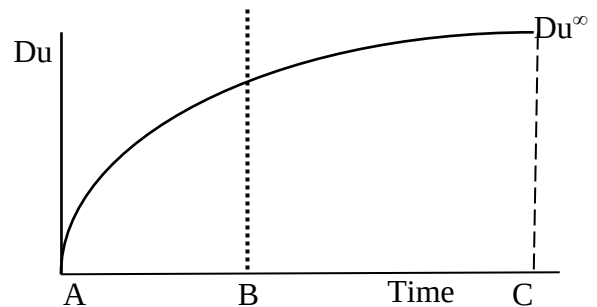
- a) Drug must be excreted in urine in significant amount
- b) Collection of urine samples till all drug is recovered (7 – 10 half lives)

» Du^{∞} : Total cumulative amount of drug excreted in urine, It is directly proportional to amount of drug absorbed

» When drug is almost completely eliminated, Du^{∞} in Urine is reached (i.e. when C_p approaches zero)



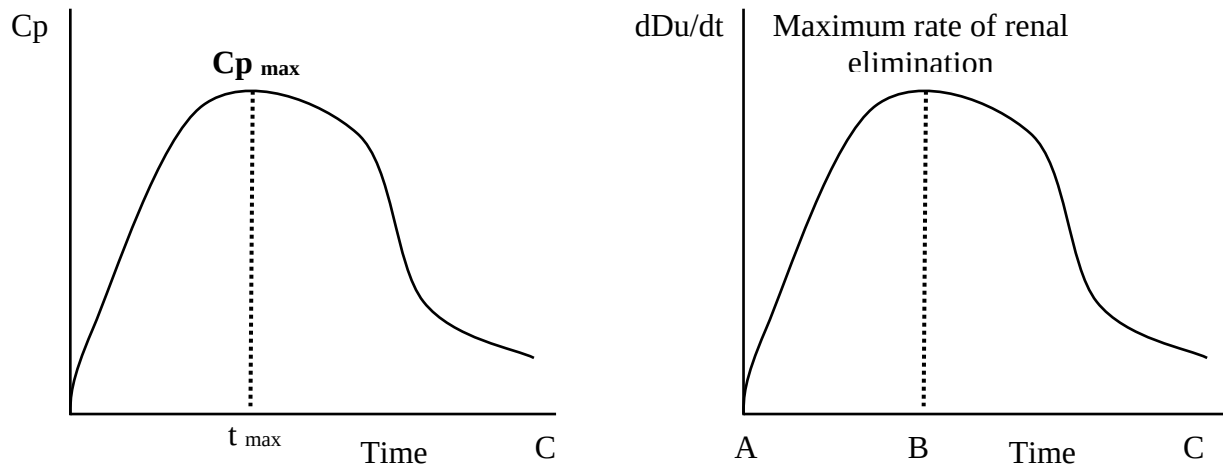
» $Du^{\infty} \propto DB$ (amount of drug in blood)



» Rate of urinary drug elimination $\frac{dDu}{dt}$

» Depends on K and C_p as follow $\frac{dDu}{dt} = K \cdot V_d \cdot C_p$

►► A graph of rate of urinary excretion versus time should be identical to graph of C_p versus time



►► t^∞ : time for maximum urinary excretion (total time for the drug to be completely excreted and eliminated)

In the above graphs

- 1) Point C: represent complete elimination of the drug where C_p approaches zero and D_u is maximum (D_u^∞)
- 2) The segment A-B of the curves of urine data is related to rate of drug absorption (i.e. similar to $t_{\ max}$ of plasma curve)

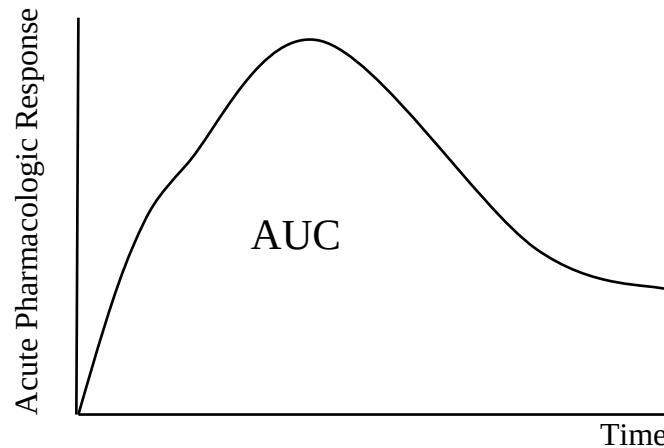
(3) Acute pharmacological effect

►► Used when quantitative measurement of the drug is not accurate, not available or not reproducible e.g. topical corticosteroids

►► Plot the pharmacologic effect (bleaching effect of skin by corticosteroids) versus time; then measure AUC under that curve to be the extent of bioavailability of that drug

►► e.g. (1) pupil diameter (2) heart rate (3) B . P.

►► The pharmacologic effect should be measured for at least 3 half lives of the drug



(4) Clinical response

» Drug is given to patients, certain clinical response is determined as follow:-

- (1) Lack of response (**therapeutic failure**) → drug is unavailable
- (2) Good **clinical (therapeutic)** response → drug is reasonably available
- (3) Toxicity → drug is highly available.

» This is the least sensitive, least accurate and least reproducible for determining bioavailability because drugs may have similar bioavailability but variable clinical response among individuals which might be due to difference in pharmacodynamics of the drug (relationship between drug and receptor site may be due to)

- (1) ↓ Receptor sensitivity to the drug.
- (2) Drug interaction
- (3) Age
- (4) Drug tolerance
- (5) Unknown pathophysiologic factor.

» This method is used for topical acne preparations and topical antifungal preparations e.g. ketoconazole

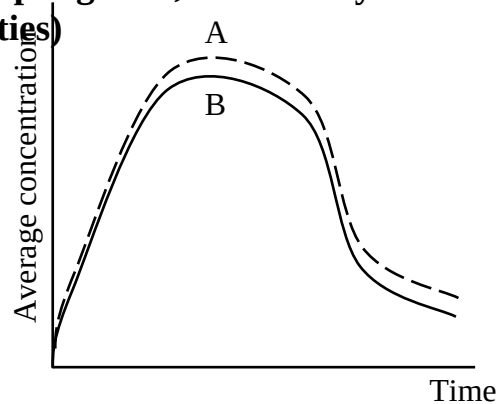
(5) In-vitro studies (dissolution testing)

► It is the cheapest method to study bioavailability of the drug as it is done in-vitro using certain equipments (dissolution tester) under certain conditions without volunteers or patients

► The test formulation that gives the most rapid rate of drug dissolution in-vitro will generally have the most rapid rate of drug bioavailability in-vivo

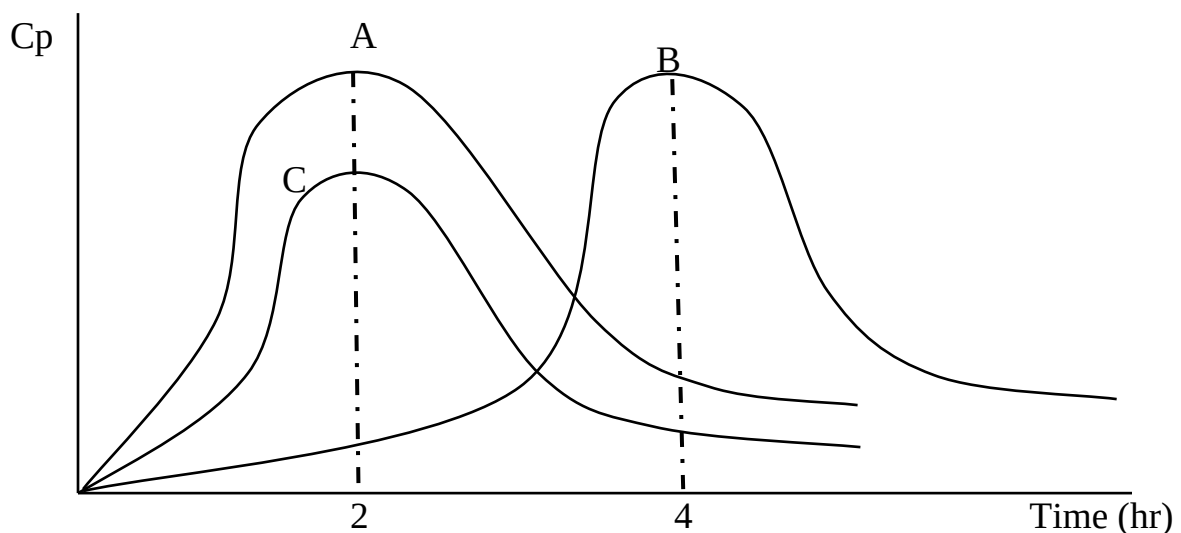
The superimposition principle

States that, if the difference between average blood concentration following administration of 2 or more formulations of the drug containing the same dose doesn't differ significantly at sampling time, then it may be stated that they are Bioequivalents (have equal bioavailabilities)



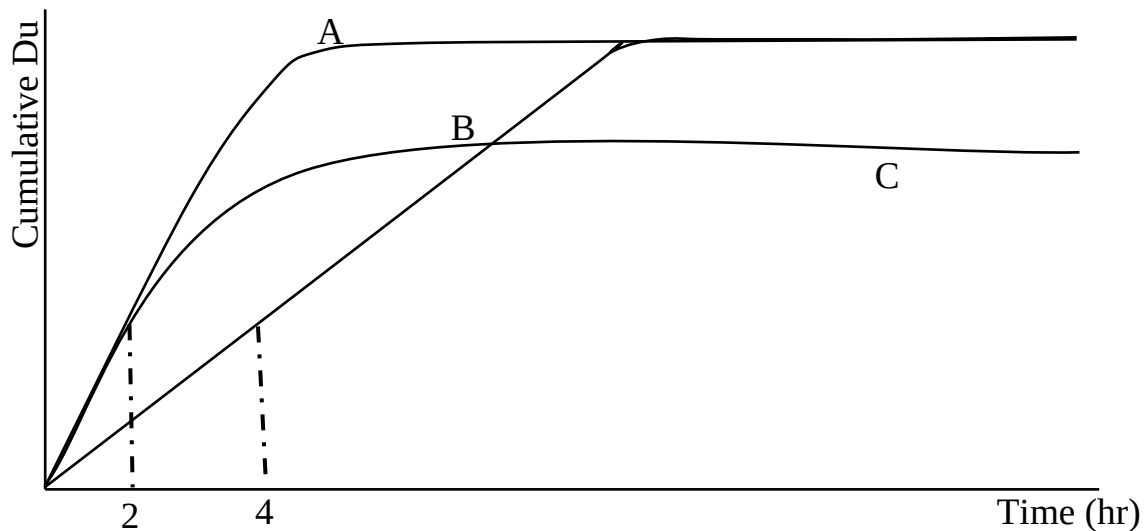
Example

A, B and C are different formulations of the same drug and the same dosage form. When given to 12 volunteers and plasma data were collected the following curves were obtained



- ▶▶ A and B have the same extent of bioavailability as (AUC) of A = (AUC) of B
- ▶▶ A has higher rate of bioavailability than B as $t_{\max}$ of A = 2 hours while $t_{\max}$ of B = 4 hours
- ▶▶ (AUC) of C = $\frac{1}{2}$ (AUC) of A so C is half the extent of bioavailability of A
- ▶▶ A and C have the same rate of bioavailability as they have the same $t_{\max}$ of 2 hours
- ▶▶ C has twice the rate but have half the extent of bioavailability of B

Urine data curve that fit the above plasma data is as follow



Bioequivalence studies

Bioequivalence: when comparing total amount absorbed from a test and recognized reference standard dosage form containing equal doses of the same drug they were found equivalent in their rate and extent of drug absorption i.e. Bioequivalents .

Reference standard

- contain drug in the same quantity , in most bioavailable form (solution or suspension)
- Same route of administration, but may be through other routes to answer specific requirement.
- Usually it is Innovator's product: accepted, approve formulation for its clinical safety and efficacy by FDA i.e. it is the reference listed drug RLD (listed in orange book)
- Should contain the drug at least 5% of the whole product

Bases for determining bioequivalence

According to FDA, drug is considered bioavailable if

(1) Rate & extent of drug product don't differ significantly from reference standard.

(2) Drug product differ from R.S in rate but not in extent as difference in rate may be

(1) Intentional as reflected by labeling e.g. Panadol and Panadol act-fast or sustained

release products.

(2) Not determining to actual activity, safety of the drug.

Design and evaluation of bioequivalence study

- ▶▶ The bioavailability of the generic drug product (new test product) is compared to the brand-name drug product (reference standard)
- ▶▶ Statistical analysis is used to detect difference in bioavailability that are not due to volunteer's variability but due to difference in bioavailability of the examined products
- ▶▶ The main reason for bioequivalence study is that in the past products that were thought to be pharmaceutical equivalents did not give similar therapeutic effects in patients

1) Design

The basic design of bioequivalence study is determined by

1- The scientific question to be answered (to prove bioequivalence or bio-nonequivalence or to compare rate and extent or rate only)

2- The nature of the reference material and the dosage form to be tested

3- The availability and sensitivity of the analytical methods

4- Benefit-risk and ethical considerations with regard to testing in humans

- ▶▶ Both reference and test products should have the same dose, dosage form and route of administration
- ▶▶ Bioequivalence study may be done as single dose or multiple dose study (steady state) or both
- ▶▶ Bioequivalence study is done on normal, healthy male and female volunteers who have signed consent to be in the study (consent required only in human studies)

- Weight of volunteers 54 – 91 kg (affect Vd), age 20 – 50 years, number of volunteers depends on the expected intersubject (difference in metabolism) and intrasubject (difference in body temperature) variability

2) Analytical method

Analytical methods used to measure active ingredient, its active metabolite or the acute pharmacologic effect should be accurate, precise (specific) and of sufficient sensitivity

Types of bioequivalence study

Bioequivalence studies are usually evaluated by a single dose, two treatments

Two periods , two sequences (Ts and ST), open label, randomized crossover design comparing equal doses of test and reference standard products on adult healthy subjects fasted overnight. This study required for both immediate and modified release oral dosage forms, then take plasma samples at different time intervals

1) Single dose bioequivalence study

Aim: reduce variation of data by proper choice of volunteers.

Volunteers: normal, healthy human

Age: 20 – 50 years old.

Weight: 54 – 91 kg.

Sex: male or female (affect drugs e.g hormones and steroids).

Reference standard: innovator drug (mother company), the same dosage form.

Administration: (1) fasting over night and 2 - 4 hrs after medication.

(2) No other medication taken 1 week before test.

(3) Via Latin square cross over design.

Assay: highly sensitive method (H P L C).

Parameters obtained: CP_{max} , t_{max} , AUC, $t_{1/2}$, K.

Analysis of data: ANOVA test (analysis of variance) may be one way or two way ANOVA

Food intervention (food effect)

- Food effect studies on drug bioavailability are done using meal conditions which are expected to give the greatest effect on GI physiology so drug availability is maximized

► These study designs use single dose, randomized, two treatment, two periods, crossover study comparing equal doses of test and standard products following fasting overnight and volunteers are given the recommended meal 30 minutes before dosing

2) Multiple dose regimen bioequivalence study

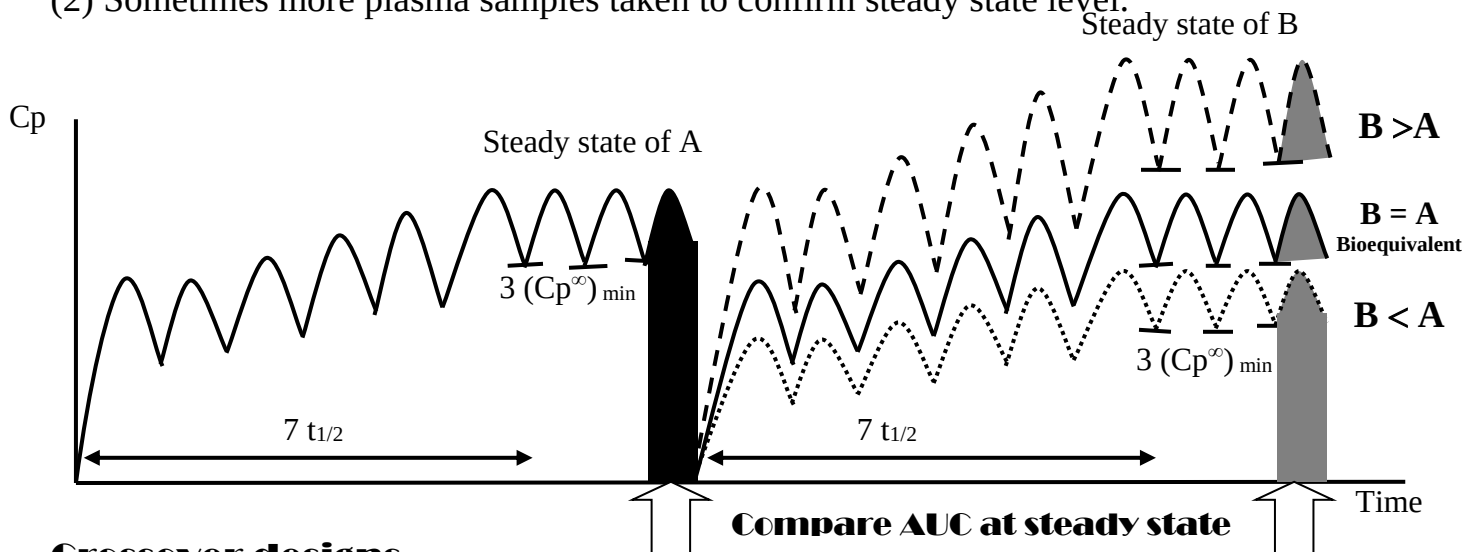
- Used for drugs used for treatment of chronic diseases e.g anticonvulsants, cardiotonics & hypoglycemic.
- **Parameters measured:** AUC, Cp^{∞}_{ave} , determined only after steady state is reached
(demonstrated by measurement of 3 consecutive trough concentrations C_{min} at 3 consecutive days and they are the same)

Advantages

- (1) Determine drug induced changes in metabolism, Cl_T and absorption of the drug
“Can’t be detected in single dose study”
- (2) Plasma contains more drug, so assay is more accurate.

Disadvantage

- (1) Study takes more time.
- (2) Sometimes more plasma samples taken to confirm steady state level.



Crossover designs

Latin square crossover design for a bioequivalence study of 3 drug products in 6 human volunteers

Subjects	Drug product		
	Study period 1	Study period 2	Study period 3
1	A	B	C
2	B	C	A
3	C	A	B
4	A	C	B

5	C	B	A
6	B	A	C

- » Each subject receives each drug product only for once
- » Each subject acts as his own control thus reduce subject to subject variation
- » After each subject receives drug product plasma samples are withdrawn at suitable time intervals for valid plasma data, AUC and absorption and elimination phases are described
- » **Period** means the time interval in which a study is performed (2 period study means it is performed in 2 different days separated by **washout period of 10 $t_{1/2}$**)
- » **Sequence** means number of different orders in treatment groups in a study

Replicated Crossover designs

Used for

- 1) Determination of individual bioequivalence
- 2) Estimate within subject variance for both test and reference drug products
- 3) Estimate the subject-by-formulation interaction variance

FDA recommends it is always done as 4 periods, 2 sequences and 2 formulation design

	Period 1	Period 2	Period 3	Period 4
Sequence 1	T	R	T	R
Sequence 2	R	T	R	T

Evaluation of the data

1) Analytical method

- » Should be accurate, precise and sensitive
- » Only one analytical method should be used as use of more than one analytical method may give different values

2) Pharmacokinetic evaluation

Pharmacokinetic parameters e.g. $[AUC]_0^\infty$, t_{max} , C_{pmax} , K and $t_{1/2}$ are determined

3) Statistical evaluation

- » Data are presented as tabulated and graphical form for evaluation
 - » ANOVA (Analysis Of Variance) is used to determine statistical difference between the pharmacokinetic parameters, **HOW?**
- 1) The difference between pharmacokinetic parameters is significant if the **probability p** that these differences is **due to chance only** is less than 1 in 20 i.e. $P \leq 0.05$ thus if $p > 0.05$ this means that these difference may be due to chance i.e. not significant
 - 2) **Confidence level of 95%** is required means that confidence that the difference is significant is $> 95\%$
 - 3) **Significance level of 5% [0.05]**

**Criteria for establishing a bioequivalence requirement
= what are conditions in which in – vivo bioequivalence
study must be done::**

(1) Evidence from well controlled clinical trials, controlled observation on patients that various drug products don't give the same therapeutic effect.

(2) Evidence from well – controlled bioequivalence studies that drug products are not Bioequivalents

(3) Evidence that drug products are potent ones (narrow therapeutic window & low MEC in blood), so safe and effective use of such drug products require careful dosage titration and patient observation.

(4) Medical evidence that lack of bioequivalence will cause serious adverse effect in treatment or prevention of a serious disease.

(5) Physicochemical evidence of the following

(a) Drug has ↓ aqueous solubility < 5 mg / ml.

(b) Dissolution rate is slow < 50 % in 30 min.

(c) Particle size and / or surface area of drug are critical for its bioavailability.

(d) Certain structural forms of the A.I (polymorphs, solvates, and complexes) dissolve poorly, this affect absorption.

(e) High ratio of excipients / A.I > 5/ 1 e.g potent drugs.

(f) Presence of inactive ingredients (hydrophilic or hydrophobic excipients & lubricants) required to ↑ absorption of drug or may interfere with its absorption

(6) Pharmacokinetic evidence of the following

(1) A.I is absorbed from a particular (localized) site in GIT.

(2) Degree of absorption of A.I is poor < 50 % compared to I.V route even when taken in pure form (e.g solution) i.e. $F < 0.5$

(3) Drug suffer (1st pass effect) 1st order metabolism, thus affect rate of absorption which is important in therapeutic effect and / or toxicity of the drug

(4) Drug is rapidly metabolized or excreted, so rapid dissolution & absorption are required for effectiveness.

(5) Drug is unstable in certain part of GIT, so require special coating or formulation e.g buffers, enteric coating or film coating to ensure drug absorption

(6) Drug product undergo dose – dependent kinetics (*non linear kinetics*) in therapeutic range and rate and extent of absorption are important to bioequivalence

Criteria for waiver إعفاء of in vivo bioavailability study

- (1) Drug product is **I.V solution** (bioavailability is **self evident** i.e. 100 % bioavailable).
- (2) Drug product is **used topically** e.g creams, ointments and gels.
- (3) Drug product is used orally for **local action in GIT** e.g antacids and radiopaque medium (BaSO₄).
- (4) Drug product is used by **inhalation** (gas or vapour) e.g inhalation anaesthetics.
- (5) Drug product meets all of the following:-
 - 1- It is an **oral solution, elixir, syrup**.
 - 2- Contains **no inactive ingredients known to affect absorption of A. I.**

Biopharmaceutics Classification System (BCS)

- ▶▶ BCS is a theoretical means for correlating in-vitro drug dissolution with in-vivo bioavailability by Amidon et al in 1985
- ▶▶ It is a scientific way for classifying drug substances based on their aqueous solubility and intestinal permeability
- ▶▶ It classifies active ingredients from immediate release solid oral dosage forms based on
 - a) Solubility
 - b) Intestinal permeability
 - c) Dissolution

Biopharmaceutics Classification System (BCS)			
Class	Solubility	Permeability	Comments
Class 1	High	High	▶▶ Drug dissolves rapidly and is well absorbed ▶▶ No bioavailability problem for immediate release drug products ▶▶ Not need in-vivo bioequivalence study
Class 2	Low	High	▶▶ Drug is dissolution limited and well absorbed ▶▶ Bioavailability is controlled by the dosage form and rate of release of the drug substance ▶▶ Not require in-vivo bioequivalence study
Class 3	High	Low	▶▶ Drug is permeability limited ▶▶ Bioavailability may be incomplete if drug is not released and dissolved within the absorption window ▶▶ Require in-vivo bioequivalence study
Class 4	Low	Low	▶▶ Drug is dissolution limited and absorption limited

			▶▶ Poor formulation that will not deliver good drug availability ▶▶ An alternate route of administration may be needed
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FDA classifies active ingredients as follow

Active ingredient	Comment
Highly soluble	The highest dose strength is soluble in < 250 ml water over a pH range of 1 – 7.5 (pH range of GIT)
Highly permeable	The extent of absorption in human > 90% of the administered dose based on mass-balance (AUC) or in comparison to IV reference dose i.e. if F > 0.9
Rapidly dissolving	When > 85% of the labeled amount dissolves within 30 min using USP apparatus I or II in a volume ≤ 900 ml buffer solution

DISSOLUTION

- ▶▶ The process by which a solid drug substance becomes dissolved in a solvent
- ▶▶ Dissolution testing is an official test used by pharmacopeias for evaluating drug release of solid and semisolid dosage forms.
- ▶▶ They were first developed to quantify the amount and extent of drug release from solid oral dosage forms including immediate/sustained release tablets and capsules.
- ▶▶ Novel dosage forms present unique problems in the development of *in vitro* release technologies simply because of the physicochemical properties of the formulations and the unique physiological environment in which they should release their content.
- ▶▶ Dissolution testing is routinely used in Quality Control (QC) and Research & Development (R&D).

Dissolution testing in QC	Dissolution testing R & D
▶▶ focus of is <u>batch to batch</u> consistency and detection of manufacturing deviations ▶▶ the test should be designed to demonstrate that the dosage forms were manufactured according to specifications	focus of dissolution testing is shifted to provide some <u>predictive estimation of the drug release in respect to the <i>in vivo</i> performance of a drug product</u>

In-vitro dissolution is an important tool in biopharmaceutics

- 1- *In vitro* dissolution properties are supportive for choosing between different alternative formulation candidates for further development and for evaluation of active ingredients/drug substances.
- 2- *In vitro* dissolution data are supportive in the evaluation and interpretation of possible risks, especially in the case of controlled/modified-release dosage forms e.g. as regards dose dumping, food effects on bioavailability or interaction with other drugs, which influence gastrointestinal environmental conditions.
- 3- Biopharmaceutical aspects are as important for stability concerns as they are for batch release after production, *in vitro* dissolution being of high relevance in quality control and quality assurance.
- 4- *In vitro* dissolution data will be of great importance when assessing changes in production site, manufacturing process or formulation and assist in decisions concerning the need for bioavailability studies. Dissolution tests may be used to predict bioavailability.

►► The overall rate of drug dissolution may be described by the **Noyes-Whitney equation**

$$\text{Where } \frac{dC}{dt} \text{ is the rate of drug dissolution at time } t \quad \frac{dC}{dt} = \frac{DA}{h}(C_s - C)$$

D is the diffusion rate constant; **A** is the surface area of the particle

C_s is the concentration of the drug (drug solubility) in the stagnant layer

C is the concentration of the drug in the bulk solvent; **h** is thickness of the stagnant layer.

Dissolution Conditions

Conditions related to the dissolution apparatus and medium

1- Size and shape of dissolution vessel:

Dissolution vessels range in size from milliliters to several liters.

The shape maybe round bottomed (USP) or flat (BP).

2- The volume of the dissolution medium

may range from 500 to 1000 mL and poorly soluble drugs may require up to 2000mL dissolution medium to ensure sink conditions.

3- The extent of agitation

Affects the hydrodynamics of the system and thus affects the dissolution rate.

4- The temperature of the dissolution medium

Most dissolution tests are performed at **37°C**, whereas transdermal drug products are performed at **32°C**.

5- The nature of the dissolution medium

Water or aqueous buffers with pH values simulating GI tract are used as dissolution media.

Compendial Methods of Dissolution

►► USP apparatus 4 and apparatus 7 and modifications of the official apparatuses have shown value for in vitro release for novel dosage forms.

►► Selection of a particular method is specified in the monograph for that drug product.

Apparatus	Name	Drug product
Apparatus 1	Rotating basket	Tablets (including uncoated, buccal or sublingual)
Apparatus 2	Paddle	Tablets, capsules, modified drug products, suspensions
Apparatus 3	Reciprocating cylinder	Extended release products
Apparatus 4	Flow through cell	Drug products containing low water soluble drugs
Apparatus 5	Paddle over disc	Transdermal drug products
Apparatus 6	Cylinder	Transdermal drug products
Apparatus 7	Reciprocating disc	Extended release drug products
Rotating bottle	Non-USP-NF	Extended release drug products (beads)

Diffusion cell (Franz cell)	Non-USP-NF	Ointments, creams, transdermal drug products
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►► Buccal and sublingual tablets are tested applying the uncoated tablet procedure.

►► For apparatus 1 and 2, low rotational speeds affect the reproducibility of the hydrodynamics; whereas at high rotational speeds, turbulence may occur.

(1) Rotating Basket Method (Apparatus 1)

; Consists of a cylindrical basket held by a motor shaft

; The basket holds the sample and rotates in a round flask containing the dissolution medium.
The entire flask is immersed in a constant temperature bath set at 37°C.

; The most common rotating speed for the basket method is 100 rpm.

; Apparatus I is generally preferred for capsules and for dosage forms that tend to float or disintegrate slowly.

Dissolution calibration standards (Dissolution calibrators)

1) Disintegrating tablets of prednisone and lactose; used for apparatus testing disintegrating tablets

2) Non disintegrating tablets of salicylic acid; used as a standard for non-disintegrating tablets.

(2) Paddle Method (Apparatus 2)

►► Consists of a special, coated paddle that minimizes turbulence during stirring

►► The paddle is attached vertically to a variable-speed motor that rotates at a controlled speed.

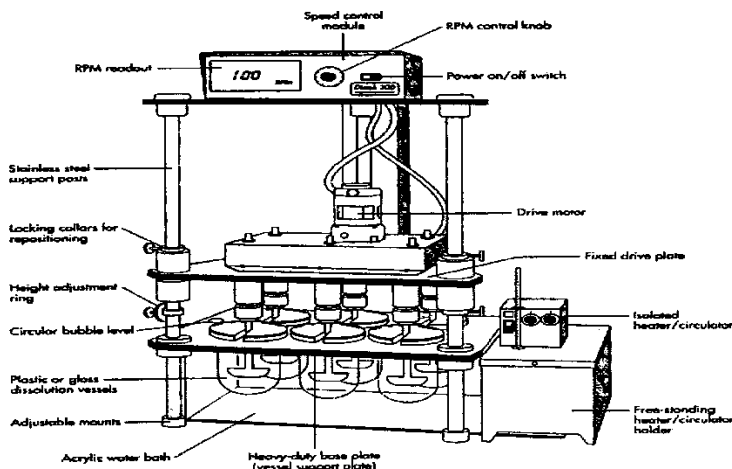
►► The tablet or capsule is placed into the round-bottom dissolution flask, which minimizes turbulence.

►► The apparatus is housed in a constant-temperature water bath maintained at 37°C

►► The position and alignment of the paddle are specified in the USP.

►► The paddle method is very sensitive to tilting.

►► Improper alignment may drastically affect the dissolution results with some drug products.



- ▶▶ The most common operating speeds for Apparatus 2 are 50 rpm for solid oral dosage forms and 25 rpm for suspensions.
- ▶▶ Apparatus 2 is generally preferred for **tablets**.
- ▶▶ A **sinker**, such as a few turns of platinum wire, may be used to
 - 1) Prevent a capsule or tablet from floating.
 - 2) Film coated tablets that stick to the vessel walls or to help position the tablet or capsule under the paddle.

(3) Reciprocating Cylinder Method (Apparatus 3)

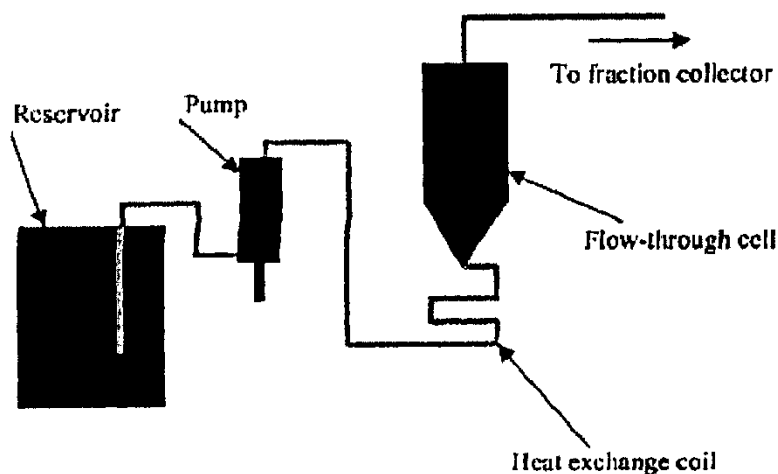
- ▶▶ Consists of a set of cylindrical, flat-bottomed glass vessels equipped with reciprocating cylinders for dissolution testing of extended-release products, particularly **bead-type** modified-release dosage forms.
- ▶▶ Six units are tested, and the dissolution medium is maintained at 37°C.

(4) Flow-through-Cell Method (Apparatus 4)

- ▶▶ Consists of a reservoir for dissolution medium and a pump that forces dissolution medium through the cell holding the test sample
- ▶▶ Flow rate ranges from 4 to 16 mL/min (usually maintained between 10 and 100 mL/min)
- ▶▶ Apparatus 4 may be used for modified-release dosage forms that contain active ingredients having very limited solubility.
- ▶▶ The sample is held in a fixed position while the dissolution medium is pumped through the sample holder, thus dissolving the drug.
- ▶▶ Laminar flow of the medium is achieved by using a pulseless pump.

Advantages

- 1) The dissolution medium may be fresh (open system) or recirculated (closed system). In the case of fresh medium, the dissolution rate at any moment may be obtained
- 2) Easy maintenance of a **sink condition** for dissolution.
- 3) A large volume of dissolution medium may also be used
- 4) Operation is easily automated



5) Paddle-over-Disk Method (Apparatus 5)

- ▶▶ The USP-NF also lists a paddle-over-disk method for testing the release of drugs from transdermal products.
- ▶▶ Consists of a sample holder or disk assembly that holds the product
- ▶▶ The entire preparation is placed in a dissolution flask filled with specified medium maintained at 32°C.
- ▶▶ The paddle is placed directly over the disk assembly. Samples are drawn midway between the surface of the dissolution medium and the top of the paddle blade at specified times.
- ▶▶ Six units are tested during each run.



6) Cylinder Method (Apparatus 6)

- ▶▶ The cylinder method (6) for testing transdermal preparation is modified from the basket method (Apparatus I).
- ▶▶ In place of the basket, a stainless steel cylinder is used to hold the sample.
- ▶▶ The sample is mounted onto an inert porous cellulosic material and the entire system adheres to the cylinder.
- ▶▶ Testing is maintained at 32°C. Samples are drawn midway between the surface of the dissolution medium and the top of the rotating cylinder for analysis.



Methods for Testing Enteric-Coated Products

- ▶▶ The dissolution test is performed in the apparatus specified in the drug monograph (usually Apparatus 2 or Apparatus 1).
- ▶▶ The product is first tested with 0.1 N HCl for 2 hours and then changed to pH 6.8 buffer medium for 45 minutes or for the times specified in the monograph.
- ▶▶ The objective is that no significant dissolution occurs in the acid phase (less than 10% for any sample unit), and a specified percentage of drug must be released in the buffer phase.

Meeting Dissolution Requirements

- ▶▶ The amount of drug to dissolve within a given time period (**Q**) is expressed as a percentage of label content.

- ▶▶ Three stages (S_1 , S_2 and S_3) of testing are allowed by USP-NF.

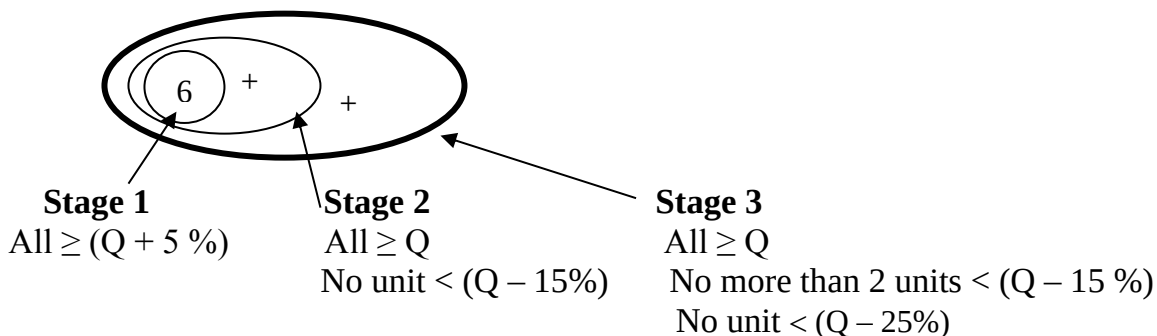
1) Stage 1 $\Rightarrow$ six tablets or capsules are tested for the dissolution test. If the dissolution test fails to meet the criteria for S_1 then six more units are tested.

2) Dissolution testing continues until the dissolution criteria are met or until the three stages are exhausted (Table below).

- ▶▶ For many products the passing value for Q is set at 75% in 45 minutes.

Some products require a Q of 85% in 30 minutes, others 75% in 60 minutes.

stage	Number of units tested	requirement
S_1	6	Each unit is not less (or $\geq$) than $(Q + 5\%)$
S_2	6	▶▶ Average of 12 unit ($S_1 + S_2$), each unit $\geq Q$ ▶▶ No unit $< (Q - 15\%)$
S_3	12	▶▶ Average of 24 units ($S_1 + S_2 + S_3$), each unit $\geq Q$ ▶▶ Not more than 2 units $< (Q - 15\%)$ ▶▶ No unit $< (Q - 25\%)$



Alternative Methods of Dissolution Testing

Rotating Bottle Method

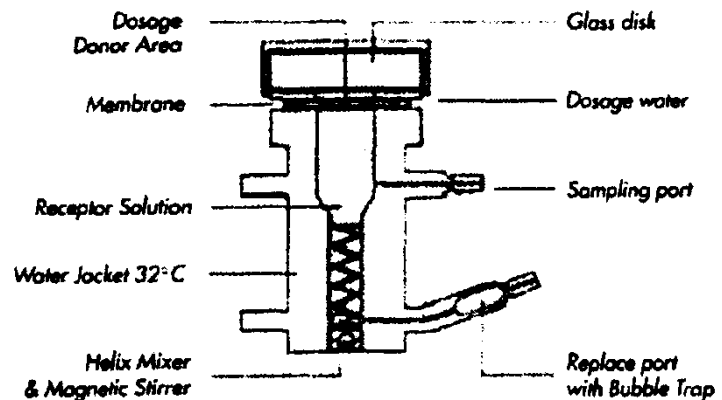
- ▶▶ The rotating bottle method was used mainly for controlled-release beads.
- ▶▶ The dissolution medium may be easily changed, such as from artificial gastric juice to artificial intestinal juice.
- ▶▶ The equipment consists of a rotating rack that holds the sample drug products in bottles. The bottles are capped tightly and rotated in a 37°C temperature bath.
- ▶▶ At various times, the samples are removed from the bottle, decanted through a 40-mesh screen, and the residues are assayed. To the remaining drug residues within the bottles are added an equal volume of fresh medium and the dissolution test is continued.
- ▶▶ A dissolution test with pH 1.2 medium for 1 hour, pH 2.5 medium for the next 1 hour, followed by pH 4.5 medium for 1.5 hours, pH 7 medium for 1.5 hours, and pH 7.5 medium for 2 hours was recommended to simulate condition of the gastrointestinal tract.
- ▶▶ Disadvantage is that this procedure is manual and tedious.

Intrinsic dissolution Method

- ▶▶ Used for screening of a new drug or substance without the effect of excipients
- ▶▶ The dissolution of a drug powder by maintaining a constant surface is called intrinsic dissolution.
- ▶▶ Intrinsic dissolution is usually expressed as $\text{mg}/\text{cm}^2/\text{min}$.

Diffusion Cells

Franz diffusion cell



- ▶▶ The Franz diffusion cell is a static diffusion system that is used for characterizing drug permeation through a skin model.
- ▶▶ The source of skin may be human cadaver ^{جثة} skin or animal skin (hairless mouse skin).
- ▶▶ The skin is mounted on the Franz diffusion cell system.
- ▶▶ The drug permeates across the skin into a receptor fluid compartment that may be sampled at various times.
- ▶▶ The Franz diffusion cell system is useful for comparing in-vitro drug release profiles and skin permeation characteristics to aid in selecting an appropriate formulation that has optimum drug delivery.

Problems of variable control in dissolution testing

Variations of 25% or more may occur with the same type of equipment and procedure.

Variations may be due to

- a) **Central alignment** of the paddle in the paddle method
- b) **Turbulence** can create increased agitation, resulting in a higher dissolution rate
- c) The basket method is more sensitive to **clogging due to gummy materials**. Pieces of small particles can also clog up the basket screen and create a local non-sink condition for dissolution
- d) **Dissolved gas** in the medium may form air bubbles on the surface of the dosage form unit and can affect dissolution in both the basket and paddle methods.
 - ▶▶ A low-density preparation may be poorly wetted in the basket method.
 - ▶▶ A floating dosage form (e.g., suppository) may be placed in a stainless steel coil (sinker) so that the dosage forms at the bottom of the dissolution flask.

In-vitro – In-vivo correlation (IVIVC)

- ▶▶ A relationship between a **biological property of the drug** (such as pharmacodynamic effect, plasma drug concentration or rate of absorption) and a physicochemical property of the drug product containing the drug substance, such as **dissolution rate**
- ▶▶ In order to have an IVIVC, some **property of the drug release from the drug product in vitro** under specified conditions, must **relate to in-vivo drug performance**.
- ▶▶ An IVIVC should be evaluated to demonstrate that predictability of *in-vivo* performance of a drug product from its *in-vitro* dissolution characteristics is maintained over a range of *in-vitro* dissolution release rates and manufacturing changes.
- ▶▶ The *in-vitro* dissolution characteristics are dependent on the physical properties of the active pharmaceutical ingredient (API), the drug formulation, the hydrodynamics of the dissolution apparatus, and dissolution medium.

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"Applied Biopharmaceutics & Pharmacokinetics"; Leon Shargel and Andrew B.C.Yu, 8th Ed., McGraw Hill Medical, USA, 2022.